L02 — study notes

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Class 2 — Introduction to statistical learning · Map of the notes

One claim. Under squared loss the target is the regression function $E[Y \mid X]$, and every learned estimate of it misses a new Y by **squared bias + variance + irreducible noise**. **Flexibility trades the first two**, so the test error is U-shaped while the training error only falls — which is why **the training error can never choose the model** and cross-validation (hold-out, LOOCV, k-fold) estimates the test error instead. **Classification is the same problem with 0–1 loss:** logistic regression models $P(y = 1 \mid x)$ directly, discriminant analysis builds it from $p(x \mid y)$ by Bayes' theorem, and **the cut-off is a statement about costs, not a constant 1/2. Counts get a log link:** Poisson regression reads coefficients as rate ratios and ties the variance to the mean — **a tie that must be checked.**

Three layers per topic:

Layer	What it means here
U — intuition	prediction vs inference; why a straight line fails on a 0/1 response; why counts are not Gaussian; the shape of the bias–variance curves

Layer	What it means here
C — course (examinable)	the decomposition (21) of note 01 with its proof; the coefficient readings of note 02; hold-out, LOOCV with its one-fit shortcut, k-fold and the bootstrap band; the logit likelihood, score, Hessian and IRLS; the Bayes and ML discriminant rules, LDA/QDA, misallocation probabilities, unequal costs, ROC; the Poisson GLM, rate ratios and overdispersion — **all with the decks' own numbers**
G — graduate	the training-error optimism $2\sigma^2(d+1)/n$ (proved, as the root of C_p/AIC); separation and non-existence of the logit MLE; smoothing splines, trees, the one-standard-error rule, nested CV, Fisher's canonical directions, the GLM template. **Named or proved, not examinable**

Reading order

Part I — regression as prediction

1. L02-01-prediction-error-bias-variance — Fundamentos slides 6–30; ISLP §2.1–2.2 (pp. 15–34); AME §1.4–1.5 (pp. 9–23). Settles: $E[Y \mid X]$ **minimises the expected prediction error** (6); the split into reducible and irreducible error (15); **bias² + variance +** σ^2 (21) proved line by line; the tanh world computed **exactly** for polynomial OLS — **optimum** $d^* = 5$**, test error 19.81 against a floor of 16**; and why training error is optimistic by exactly $2\sigma^2(d+1)/n$ (33).
2. L02-02-linear-log-linear-regression — multiplelinearregression.pdf slides 17, 29, 44; W §2.4 (pp. 36–40), §6.2 (pp. 171–178). Settles: "linear" means linear in β; level, log-level, log-log and level-log readings (W Table 2.3); 100β **is an approximation** — $e^{0.306} - 1 = 35.8\%$**, not 30.6%**; interactions and quadratics as slopes that move; the Credit, houseprice and Mroz printouts checked by hand.

Part II — estimating the test error

3. L02-03-cross-validation-motorcycle — the motorcycle deck (11 slides) and Rmd; ISLP §5.1–5.2 (pp. 202–214), §7.1 (pp. 290–291); AME §1.5.1 (pp. 14–21). Settles: adjusted R^2 and BIC pick **order 12**; one hold-out split is a random answer (**63 of 100** splits pick 12) while **10-fold picks it 99 times**; **LOOCV in one fit**, $CV_{(n)} = \frac{1}{n}\sum(e_i/(1-h_{ii}))^2$, derived by Sherman–Morrison; the pairs-bootstrap band and what it leaves out.

Part III — classification

4. L02-04-logistic-regression — logistic part, slides 4–38; graduate-school-admission; ISLP §4.3 (pp. 139–144); W §17.1 (pp. 525–531). Settles: odds, logit, odds ratio and the partial effect $\beta\pi(1-\pi)$; likelihood, score $X'(y-\pi)$, Hessian $-X'WX$; **Newton–Raphson is IRLS**, worked to convergence on the student table (**−3.50413, 0.40489**); the `default` probabilities **0.58% / 58.6%** and cut-off **1936.6**; confounding by balance; the spam confusion matrix (**FDR 9.06%**).

5. L02-05-discriminant-analysis — discriminant part, slides 40–65; ISLP §4.4–4.5 (pp. 146–163); AME Ch. 7, §8.1.4, §9.1. Settles: the Bayes rule minimises the error probability; $h(x)$ and its thresholds; **equal variances → one midpoint cut-off, unequal → two (1.680 and 10.001)**, total error 0.2515 vs 0.3694; LDA by Mahalanobis distance on iris; QDA; admissibility; **unequal costs move the threshold to** $\log\{\pi_2 c_{12}/(\pi_1 c_{21})\}$; ROC curves (AUC 0.802 vs 0.681).

Part IV — counts

6. L02-06-gaussian-vs-poisson-regression — Gaussian-vs-Poisson-regression.html; ISLP §4.6 (pp. 167–173); W §17.3 (pp. 543–546). Settles: on 28 days of blog visits both models give the **same means (109 / 83.33)** and different variances; $\hat{\alpha} = 4.69135$, $\hat{\beta} = -0.26850$ in closed form; **rate ratio 0.7645 (−23.5%)**; overdispersion $\hat{\sigma}^2 = 1.444$ widens the SE by 20%.

Support

7. L02-07-formulas — the exam specification mapped item by item; every numbered result with its note; every number and worked case.
8. L02-08-pitfalls — fifteen confusions with minimal counterexamples, and an original 30-item True/False self-test.
9. L02-09-python — the reference implementation (`L02_python_checks.py`): exact bias–variance, CV by hand, IRLS logit and Poisson against `statsmodels`, discriminant cut-offs, all asserted against the sources' numbers.

The one result to hold

$$E[(Y - \hat{\psi}(X))^2] = \underbrace{\int \text{Viés}^2[\hat{\psi}(x)]\, dF_X(x)}_{\text{falls with flexibility}} + \underbrace{\int \text{Var}[\hat{\psi}(x)]\, dF_X(x)}_{\text{rises with flexibility}} + \underbrace{\sigma^2}_{\text{untouchable}}$$

Every other note is a way to estimate this, or to trade its terms. Note 03 estimates the left side from held-out data; LDA vs QDA (note 05 §8) is a bias–variance choice; and the classification version replaces squared loss by 0–1 loss, whose minimiser is the **Bayes rule** $\arg\max_k \pi_k p_k(x)$ — the classification analogue of $E[Y \mid X]$.

Numbers and cases to have memorized

Case	Number	Meaning
tanh world, $n = 30$, $\sigma = 4$, polynomial OLS	$d^* = 5$, test error **19.81** $(0.94 + 2.87 + 16)$	the U-curve's bottom, 3.81 above the noise floor
training optimism at $d = 5$	$2\sigma^2(d+1)/n = 2(16)(6)/30 =$ **6.4**	why C_p/AIC add a penalty
motorcycle	order **12**; hold-out picks it **63/100**, 10-fold **99/100**	one split is a random answer
log-level, $\beta = 0.306$	**+35.8%** (up), **−26.4%** (down), 30.6% approx.	100β is a first-order reading
`default`, $-10.6513 + 0.0055x$	**0.58%** at 1000, **58.6%** at 2000, 50% cut-off **1936.6**	a cut-off on $\hat{\pi}$ is a cut-off on x

Case	Number	Meaning
student only / with balance	$+0.40489$ (OR 1.499) / -0.7149 (OR 0.489); **5.4% vs 10.5%** at 1500	confounding flips the sign
spam `glm`	error 7.52%, FPR 5.69%, FNR 10.38%, **FDR 9.06%**	FDR ≠ FPR
Gaussian DA, $\mu = (1, 5)$, $\sigma = (6, 2.5)$	cut-offs **1.680, 10.001**; errors 0.3881 / 0.1148; total **0.2515** (vs **0.3694** at equal σ)	quadratic rule, class 1 wins both tails
blog visits	$\hat{\alpha} = 4.69135$, $\hat{\beta} = -0.26850$; RR **0.7645**; $\hat{\sigma}^2 = \mathbf{1.444}$	same means, different variances

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Figures

Built by `figures/make_figures.py` (run: `python Notas/L02/figures/make_figures.py`; every figure ends in `assert` checks on its plotted numbers).

Figure	Shows	Lives in
L02-tanh-regression-function	$\psi(x) = 45\tanh(x/1.9 - 7) + 57$, 30 seeded points, the line, a degree-6 polynomial and the interpolant	L02-01-prediction-error-bias-variance §5
L02-bias-variance-curves	exact avg bias², variance, test and training error vs degree; optimum $d = 5$	L02-01-prediction-error-bias-variance §5
L02-log-linear-readings	$100(e^z - 1)$ vs $100z$ at W's coefficients; the log-log analogue at elasticity 0.257	L02-02-linear-log-linear-regression §4.1
L02-cv-curves	training, hold-out, LOOCV, 10-fold vs order; winners over 100 splits; bootstrap vs formula band (**stand-in data**)	L02-03-cross-validation-motorcycle §8
L02-loocv-shortcut	leverages h_{ii} for $d = 3, 12$; LOOCV by n refits = LOOCV by one fit	L02-03-cross-validation-motorcycle §8
L02-logistic-default	logit vs linear probability on `balance`; student / non-student curves and their cut-offs	L02-04-logistic-regression §1
L02-gaussian-discriminant-regions	$\pi_k p_k(x)$ with the class-2 band $(1.680, 10.001)$; $h(x)$ quadratic vs linear, with the cost threshold	L02-05-discriminant-analysis §5.2
L02-roc	exact ROC curves, AUC 0.802 vs 0.681, ML operating points	L02-05-discriminant-analysis §11

Figure	Shows	Lives in
L02-gaussian-vs-poisson-fit	the 28 counts, common fitted step, Gaussian vs Poisson bands; group (mean, variance) vs each variance function	L02-06-gaussian-vs-poisson-regression §5

Interactive companions

- The Flexibility Dial (`companions/companion-bias-variance-polynomial.html`) — move the polynomial degree d, the training size n and the noise σ on the tanh model; read the exact avg bias^2, variance, training and test error and the optimal degree (19.81 at $d = 5$ by default). Linked from notes 01 and 03.
- The Two-Gaussian Classifier (`companions/companion-gaussian-discriminant.html`) — move the two means and SDs, the prior π_1 and the cost ratio c_{21}/c_{12}; watch the allocation regions, the posterior threshold and the total misclassification probability (0.2515 by default; one cut-off at the midpoint when $\sigma_1 = \sigma_2$). Linked from note 05.

Both pages are verified by `companions/check_companions.py` (the pages' own JavaScript under `node` against a Python recomputation at both ends of every range and at the defaults).

Scope and ceiling

- **Examinable (C):** the results of notes 01–06 as derived there — no code in the final exam (Aula/COURSE: *"You will NOT be asked to write R script or similar"*), so expect the derivations, the hand arithmetic of the printouts, and the readings.
- **Proved but above the line (G):** the optimism $2\sigma^2(d+1)/n$; separation; the GLM template (note 06 §6); the closed-form equal-variance AUC.
- **Named only:** smoothing splines (the Fundamentos learner), trees, boosting and random forests, the classification bias–variance and k-NN of Fundamentos slides 36–50, the one-standard-error rule, nested CV, Fisher's canonical discriminants (LD1/LD2).
- **Belongs to later classes:** ridge, lasso and elastic net (slides 26–27, 32–35 of the logistic deck; `cv.glmnet`) and every Bayesian part — the Bayesian logit (slides 23–25), the admission script's SIR posterior, the blog page's grid posterior, the Mroz page's Bayes factors — are **Classes 3+4**; probit and multinomial models are **Class 7**.
- **Out:** unsupervised learning (clustering, PCA).

Course-wide position

Class	Relation to Class 2
1 — Review	**MSE = variance + bias²** for one parameter becomes (21) pointwise in x; OLS, the hat matrix, R^2, AIC/BIC and overfitting are the tools note 03 reuses; the Bernoulli and Poisson likelihoods become the logit and Poisson GLMs; the bootstrap and its failure for $x_{(n)}$ precede the bootstrap band

Class	Relation to Class 2
2 — Statistical learning	**prediction error, CV, logistic, discriminant, Poisson**
3+4 — Bayesian learning	the posterior predictive (21) of note 04 replaces the plug-in; Bayesian logit, Poisson and model comparison (the Mroz Bayes factors); ridge/lasso as posterior modes; Bayes rules and loss functions generalise note 05 §10
5+6 — Causal inference	the inference column of note 01 §1: which controls are legitimate (the Mroz "return to education" that vanishes), confounding (the student sign flip) made formal
7 — More econometric models	GLMs again; **probit**, ordered and multinomial probit beside the logit; Tobit; heteroskedastic errors (the mcycle noise)
8 — Univariate time series	model choice by information criteria and out-of-sample error, now with time order (no random folds)
9 — Multivariate time series	shrinkage (large BVAR) as the bias–variance trade-off at scale

Rules pack: rules/06 (logistic regression, confusion matrix, ROC/AUC, training and test samples, CV, bias–variance).

Errata found in the sources

Each is documented, with the check, in the note named.

Source, place	Slip	Correct	Note
Fundamentos.pdf slide 16	the cross term of the expansion is written with +2	it is $-2E[(\hat{\psi}(x)-\psi(x))\varepsilon]$ — zero either way	L02-01-prediction-error-bias-variance §3, (13)–(14)
logistic deck slide 9	WLS objective printed as $(z-X\beta)'(z-X\beta)$	$(z-X\beta)'W(z-X\beta)$ — without W the minimiser is not (16)	L02-04-logistic-regression §4
logistic deck slide 10	$\chi^2=\sum r_i$ and $D=\sum d_i$	$\sum r_i^2$ and $\sum d_i^2$	L02-04-logistic-regression §7
logistic deck slide 15	second line labelled "$x=1000$"	the arithmetic is for $x=2000$	L02-04-logistic-regression §5.1
logistic deck slide 18	student-only probabilities 0.43% and 0.29%	**4.31%** and **2.92%** (127/2944, 206/7056; ISLP p. 142)	L02-04-logistic-regression §4.1
logistic deck slide 22	cut-off 1873 attached to students, 1998 to non-students	students **1998**, non-students **1873** ($\hat{\beta}_{\text{student}}<0$)	L02-04-logistic-regression §5.3

Source, place	Slip	Correct	Note
logistic deck slide 38	"false.discovery" column is $FP/(FP+TP+FN)$	FDR $= FP/(FP+TP)$: **9.06%** for `glm`, not 8.20%	L02-04-logistic-regression §9
graduate-school-admission.txt	comment "2/3 admitted and 1/3 not admitted"	about **one third** admitted (the plotted rate ≈ 0.32)	L02-04-logistic-regression §10
discriminant deck slide 41	regions defined with $\cap$	a partition needs $\bigcup_k R_k = \mathbb{R}^p$	L02-05-discriminant-analysis §2
discriminant deck slide 44	multinomial ML rule "class 1 iff $\sum_k x_k \log(\alpha_k/\beta_k) < 0$"	> 0 (check: the Bernoulli case)	L02-05-discriminant-analysis §4
motorcycle Rmd	LOOCV and 10-fold curves plotted under "Root MSE"	the code accumulates SSE and MSE respectively; **the argmin is unaffected**	L02-03-cross-validation-motorcycle §6

Bibliography actually used in these notes

Book / source	Where
The class decks (`Aula/class-02/materials/`): Fundamentos.pdf (Marques), the motorcycle deck and Rmd, the logistic/discriminant deck (Lopes & Marques), multiplelinearregression.pdf	everywhere; the spine of each note
ISLP (2023) §2.1–2.2, §4.3–4.6, §5.1–5.2, §7.1	notes 01, 03–06
Wooldridge 6e §2.4, §6.2, §17.1, §17.3	notes 02, 04, 06
Izbicki & Mendonça (AME, 2020) §1.4–1.5, Ch. 7, §8.1, §9.1	notes 01, 03, 05 (Portuguese cross-check)
Site code: `bias-variance-R.txt`, `capm-regression-R.txt`, `regressaomultipla-return-to-education.html`, `salaries-for-professors.txt`, `graduate-school-admission.txt/.pdf`, `Gaussian-vs-Poisson-regression.html`	notes 01, 02, 04, 06

Related project material: the site's Class 2 list and class note on the motorcycle CV (Aula/COURSE); readings, practice and Python per class (Map/class-plan); exercise numbers by book (Map/exercises-index); the topic rules (rules/06). Practice for this class (statements in the books): ISLP 2.1, 2.3, 4.1, 4.9, 4.12–4.14, 5.3, 5.5, 5.8, 7.6, 7.9; W 17.C1, 17.C4, 17.C5 — each note's `## Links` gives the pages.

Fundamentos slides 6–30 · ISLP §2.1–2.2 · The best predictor is $E[Y \mid X]$, and every estimate of it pays bias², variance and noise

> **Claim.** Under squared loss the best possible predictor of Y from X is the regression function $\psi(x) = E[Y \mid X = x]$ — a two-line calculus result. Any *learned* $\hat{\psi}$ then misses Y by three separate amounts: **noise σ^2 that no method can remove, squared bias that falls as the learner gets more flexible, and variance that rises with it.** The expected test error is U-shaped in flexibility, the training error is not — so the training error can never choose the flexibility. Notes 02–03 build on this: 02 reads the coefficients of the linear learners, 03 estimates the test error by cross-validation. Prev: L01-11-python · Next: L02-02-linear-log-linear-regression · Map: L02-00-map

Notation.

- $X \in \mathbb{R}^p$ predictors, $Y \in \mathbb{R}$ response, joint distribution $F_{X,Y}$ (unknown), marginal F_X.
- $\psi(x) = E[Y \mid X = x]$ the **regression function** (Marques's ψ; ISLP writes f, AME writes r).
- $\varphi : \mathbb{R}^p \to \mathbb{R}$ any candidate predictor; $t = \varphi(x)$ its value at one point.
- $\mathcal{D} = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ i.i.d. training data; $\hat{\psi} = \mathcal{L}[\mathcal{D}]$ the learned function, $\mathcal{L}$ the learning method.
- ε the additive error: $E[\varepsilon] = 0$, $\mathrm{Var}[\varepsilon] = \sigma^2$, independent of X and of $\mathcal{D}$.
- (X, Y) written without index is always a **new** observation, independent of $\mathcal{D}$.
- $\mathrm{EQM}[\hat{\psi}(x)] = E[(\hat{\psi}(x) - \psi(x))^2]$ (*erro quadrático médio*, MSE of the estimator at x); Viés = bias.
- gl (*graus de liberdade*) = degrees of freedom of a smoother; d = polynomial degree (a degree-d polynomial has $d + 1$ df).

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1 · U — supervised learning, and the two things one wants from it

The set-up (slides 2–5). One observes predictors X and a response Y — in ISLP's *Advertising* data, TV, radio and newspaper budgets and the sales of 200 markets. **Regression** if Y is quantitative, **classification** if it is a label (L02-04-logistic-regression, L02-05-discriminant-analysis). With no Y at all (clustering, PCA) the problem is *unsupervised* and out of this course's scope.

Two purposes, usually in tension (slides 10–12; ISLP §2.1.1, pp. 17–20):

	Prediction	**Inference**
Treat $\hat{\psi}$ as	a black box	an object to read
Success means	small $E[(Y - \hat{\psi}(X))^2]$ on *new* data	knowing which predictors matter, and how
Typical question	"what sales will this market have?"	"is TV associated with sales? what does +25% newspaper do?"
Favours	flexible methods (trees, boosting, SVMs)	rigid, interpretable ones (least squares, subset selection, lasso)

> **Flexibility buys accuracy and costs interpretability** (slide 12, after ISLP Fig. 2.7). This note is about the prediction side; L02-02-linear-log-linear-regression is the inference side of the same linear learners.

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2 · C — the regression function minimises the expected prediction error

Problem (slide 6). Find $\varphi : \mathbb{R}^p \to \mathbb{R}$ minimising the **expected prediction error** (also *expected test error*, or just *test error*)

$$E\big[(Y - \varphi(X))^2\big] = \int (y - \varphi(x))^2 \, dF_{X,Y}(x, y). \tag{1}$$

Step 1 — condition on X (slide 7). By iterated expectations,

$$E\big[(Y - \varphi(X))^2\big] = E\Big[E\big[(Y - \varphi(X))^2 \mid X\big]\Big] = \int E\big[(Y - \varphi(x))^2 \mid X = x\big] \, dF_X(x). \tag{2}$$

Every integrand in (2) is non-negative and involves φ only through the single number $\varphi(x)$, so **(2) is minimised by minimising the integrand separately at each x.**

Step 2 — one variable at a time (slide 8). Fix x and write $t = \varphi(x)$. Expanding the square inside the conditional expectation:

$$g(t) := E\big[(Y - t)^2 \mid X = x\big] = E[Y^2 \mid X = x] - 2t\, E[Y \mid X = x] + t^2. \tag{3}$$

Step 3 — first- and second-order conditions. Differentiating (3) in t,

$$g'(t) = -2\, E[Y \mid X = x] + 2t, \tag{4}$$

$$g'(t) = 0 \iff t = E[Y \mid X = x], \qquad g''(t) = 2 > 0. \tag{5}$$

So the minimiser is unique at every x, and putting the pointwise solutions back into (2):

$$\boxed{\arg\min_{\varphi} E\big[(Y - \varphi(X))^2\big] = \psi(x) = E[Y \mid X = x]} \tag{6}$$

The same result as a decomposition (AME Teorema 1, p. 10). For any fixed g, add and subtract $\psi(X)$:

$$E\big[(Y - g(X))^2\big] = E\big[(Y - \psi(X) + \psi(X) - g(X))^2\big]$$

$$= E\big[(\psi(X) - g(X))^2\big] + E\big[(Y - \psi(X))^2\big] + 2E\big[(\psi(X) - g(X))(Y - \psi(X))\big]. \tag{7}$$

The cross term vanishes by conditioning on X, because $\psi(X) - g(X)$ is then a constant:

$$E\big[(\psi(X) - g(X))(Y - \psi(X))\big] = E\Big[(\psi(X) - g(X))\,(E[Y \mid X] - \psi(X))\Big] = E\big[(\psi(X) - g(X)) \cdot 0\big] = 0, \tag{8}$$

and the middle term is $E\big[E[(Y - \psi(X))^2 \mid X]\big] = E\big[V[Y \mid X]\big]$. Substituting (8) into (7):

$$\underbrace{E\big[(Y-g(X))^2\big]}_{R_{\mathrm{pred}}(g)} = \underbrace{E\big[(\psi(X)-g(X))^2\big]}_{R_{\mathrm{reg}}(g)} + E\big[V[Y \mid X]\big]. \tag{9}$$

Reading (9). The second term does not depend on g. So **predicting Y well and estimating the regression function well are the same problem** under squared loss — the reason the whole course of regression methods is also a course in prediction. With absolute loss the answer would be the conditional median instead (slide 6, remark 2; AME Observação 1.2).

The catch (slide 9). ψ is a functional of the unknown $F_{X,Y}$. Learning means building $\hat{\psi} = \mathcal{L}[\mathcal{D}]$ from the training data; each method (OLS, a spline, a tree) is one functional $\mathcal{L}$.

---

3 · C — the additive model: reducible and irreducible error

The model (slides 13–14; ISLP eq. 2.1, p. 15):

$$Y = \psi(X) + \varepsilon, \qquad Y_i = \psi(X_i) + \varepsilon_i, \quad i = 1, \dots, n, \tag{10}$$

with $X, X_1, \dots, X_n$ i.i.d. F_X; $\varepsilon, \varepsilon_1, \dots, \varepsilon_n$ i.i.d., mean 0, variance σ^2; and X**, the X_i's and the ε's all mutually independent.** Then $E[Y \mid X = x] = \psi(x) + E[\varepsilon] = \psi(x)$, so ψ in (10) *is* the regression function of §2. The prediction for the new point is $\hat{Y} = \hat{\psi}(X)$.

Conditioning on the new X (slide 15), exactly as in (2):

$$E\big[(Y-\hat{\psi}(X))^2\big] = \int E\big[(Y-\hat{\psi}(x))^2 \mid X = x\big]\, dF_X(x). \tag{11}$$

The expectation now runs over **both** the new ε and the training sample $\mathcal{D}$ inside $\hat{\psi}$ (AME Observação 1.3, p. 11: the *expected* risk, as opposed to the conditional risk of one fixed $\hat{\psi}$).

Proof of the split (slide 16). Substitute (10) for Y at $X = x$:

$$E\big[(Y-\hat{\psi}(x))^2 \mid X = x\big] = E\big[(\psi(x) + \varepsilon - \hat{\psi}(x))^2\big]. \tag{12}$$

Write the bracket as $-(\hat{\psi}(x) - \psi(x)) + \varepsilon$ and expand:

$$= E\big[(\hat{\psi}(x)-\psi(x))^2\big] - 2\,E\big[(\hat{\psi}(x)-\psi(x))\,\varepsilon\big] + E[\varepsilon^2]. \tag{13}$$

$\hat{\psi}$ is built from $\mathcal{D}$ alone and the new ε is independent of $\mathcal{D}$, so the expectation of the product factorises:

$$E\big[(\hat{\psi}(x)-\psi(x))\,\varepsilon\big] = \big(E[\hat{\psi}(x)] - \psi(x)\big)\cdot E[\varepsilon] = \big(E[\hat{\psi}(x)] - \psi(x)\big)\cdot 0 = 0. \tag{14}$$

(The slide writes the cross term with a $+2$; the expansion gives -2. It is zero either way.) With $E[\varepsilon^2] = \mathrm{Var}[\varepsilon] + 0^2 = \sigma^2$, substituting (14) into (13):

$$\boxed{E[(Y-\hat{\psi}(x))^2 \mid X=x] = \underbrace{\mathrm{EQM}[\hat{\psi}(x)]}_{\text{reducible}} + \underbrace{\sigma^2}_{\text{irreducible}}} \tag{15}$$

and integrating (15) through (11):

$$E[(Y-\hat{\psi}(X))^2] = \int \mathrm{EQM}[\hat{\psi}(x)]\, dF_X(x) + \sigma^2. \tag{16}$$

The choice of method moves EQM**; nothing moves** σ^2 (slide 16). ISLP eq. (2.3), p. 18, is (15) with $\hat{\psi}$ and X held fixed, so its reducible part is the plain square $[\psi(X)-\hat{\psi}(X)]^2$. σ^2 **is a floor on the test error that is almost always unknown in practice** (ISLP p. 18) — it is the variation in Y due to everything not in X.

---

4 · C — the reducible part splits into squared bias plus variance

Definitions (slide 17), at a fixed x, with expectations over training samples $\mathcal{D}$:

$$\mathrm{Viés}[\hat{\psi}(x)] = E[\hat{\psi}(x)] - \psi(x), \qquad \mathrm{Var}[\hat{\psi}(x)] = E\big[(\hat{\psi}(x) - E[\hat{\psi}(x)])^2\big]. \tag{17}$$

Proof (slide 18) — the same add-and-subtract as Class 1's MSE = Var + bias^2 (L01 §5), now pointwise in x. Insert $E[\hat{\psi}(x)]$:

$$\mathrm{EQM}[\hat{\psi}(x)] = E\Big[\big((\hat{\psi}(x) - E[\hat{\psi}(x)]) - (\psi(x) - E[\hat{\psi}(x)])\big)^2\Big] \tag{18}$$

$$= E\big[(\hat{\psi}(x) - E\hat{\psi}(x))^2\big] - 2\,E\big[\hat{\psi}(x) - E\hat{\psi}(x)\big]\,(\psi(x) - E\hat{\psi}(x)) + (\psi(x) - E\hat{\psi}(x))^2. \tag{19}$$

The middle factor is a constant times $E[\hat{\psi}(x)] - E[\hat{\psi}(x)] = 0$, so

$$\mathrm{EQM}[\hat{\psi}(x)] = \mathrm{Var}[\hat{\psi}(x)] + \mathrm{Viés}^2[\hat{\psi}(x)]. \tag{20}$$

Substituting (20) into (16) gives the full factorisation:

$$\boxed{E[(Y-\hat{\psi}(X))^2] = \int \mathrm{Viés}^2[\hat{\psi}(x)]\, dF_X(x) + \int \mathrm{Var}[\hat{\psi}(x)]\, dF_X(x) + \sigma^2} \tag{21}$$

— expected squared bias, expected variance, irreducible error. ISLP eq. (2.7), p. 32, is the pointwise version of (21) at x_0 (stated there without proof); AME §1.5.2, p. 22, writes it with $V[Y \mid X=x]$ in place of σ^2, which also covers heteroskedastic noise.

Where the randomness lives — the point `bias-variance-R.txt` asks the reader to check:

Term in (21)	Random over	What drives it
σ^2	the new ε	the world; untouchable

Term in (21)	Random over	What drives it
Viés$^2[\hat{\psi}(x)]$	nothing (a number per x)	how far the *average* learned curve is from ψ: a rigid learner cannot bend to ψ
$\mathrm{Var}[\hat{\psi}(x)]$	the training sample $\mathcal{D}$	how much $\hat{\psi}$ moves when the sample is redrawn: a flexible learner chases its sample

"Variance" here is the variance of the *learned function* across training samples, not of Y. High-variance method: change 10% of the sample and $\hat{\psi}$ changes a lot (slide 19). High bias: the method can only learn a restricted class of functions — a straight line fitted to a strongly non-linear ψ.

4.1 · Worked — a high-bias, low-variance learner, quantified (`bias-variance-R.txt`)

The script draws $n = 1000$ points from $y = 1 - x + 2x^2 + \varepsilon$, $x \sim U(0,1)$, $\sigma = 0.1$, and refits a straight line to 90% subsamples: *"Can you even see the fitted regression line changing?"* The two terms of (21) can be computed exactly.

Bias. As n grows, the OLS line converges to the population projection of ψ on $\{1, x\}$. For $x \sim U(0,1)$, the projection of x^2 is $x - \frac{1}{6}$ (residual $x^2 - x + \frac{1}{6}$, orthogonal to 1 and x). Substituting into ψ:

$$\psi(x) = 1 - x + 2x^2 = 1 - x + 2(x - \tfrac{1}{6}) + 2(x^2 - x + \tfrac{1}{6}) = (\tfrac{2}{3} + x) + 2(x^2 - x + \tfrac{1}{6}). \tag{22}$$

So the bias of the line is $-2(x^2 - x + \frac{1}{6})$, and with $\int_0^1 (x^2 - x + \frac{1}{6})^2 dx = \frac{1}{180}$:

$$\int_0^1 \text{Viés}^2 \, dx = 4 \cdot \tfrac{1}{180} = \tfrac{1}{45} = 0.0222. \tag{23}$$

Variance. OLS with 2 coefficients on $n' = 900$ points averages $\sigma^2 \cdot 2/n'$ over the design:

$$\overline{\mathrm{Var}} \approx \frac{0.1^2 \times 2}{900} = 0.0000222. \tag{24}$$

Squared bias (0.0222) is 1000 times the variance (0.0000222) and more than twice the noise ($\sigma^2 = 0.01$). (Holding the design fixed, as (24) does. If every training set draws fresh x values the variance is about 8.6×10^{-5}, so bias² is "only" about 260 times larger; L02-09-python §1 computes both.) That is "high bias, low variance": the line is stable to the fourth decimal across subsamples *and* systematically wrong. The script's tall regression tree is the opposite corner (G layer: boosting attacks bias with many short trees, random forests attack variance by averaging many tall ones — named in the script, not developed in this course). Both numbers are checked in L02-09-python.

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5 · C — Worked: the tanh simulation (slides 20–27)

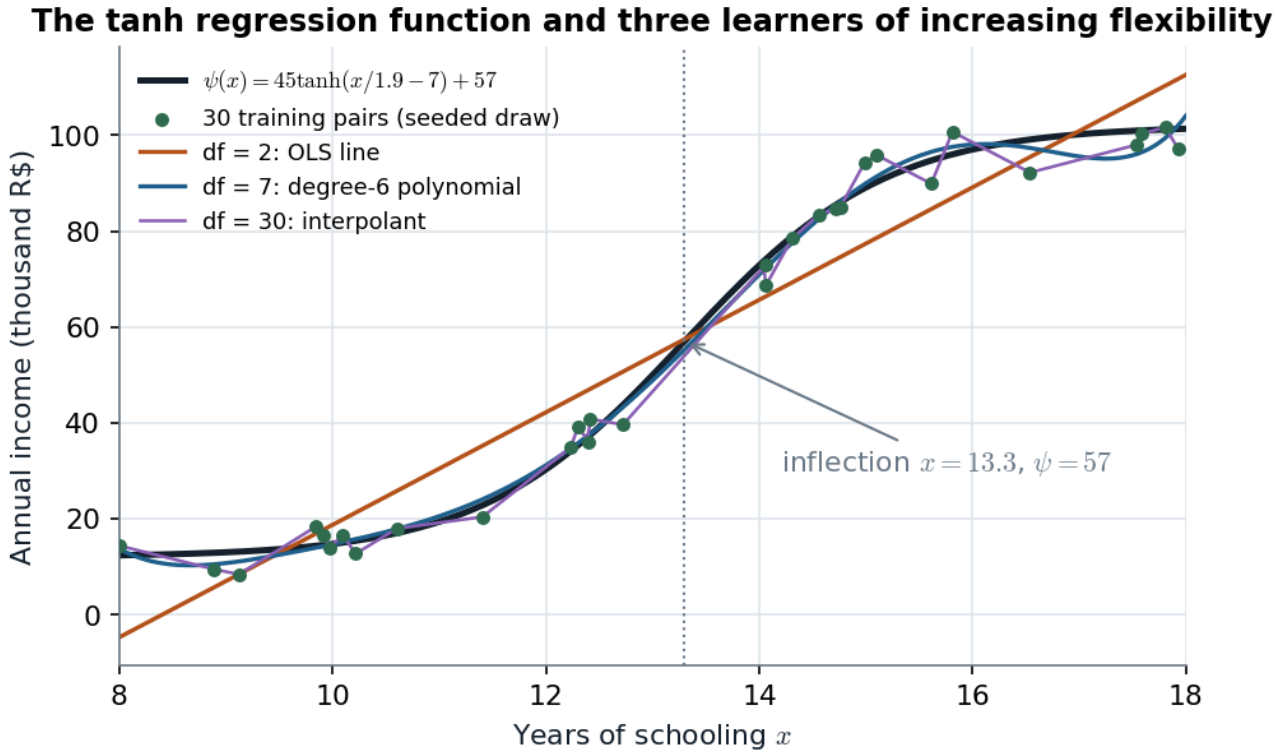
The world, fully known (slide 21). X = years of schooling, Y = annual income in thousand R$:

$$\psi(x) = 45\tanh\Big(\frac{x}{1.9} - 7\Big) + 57, \qquad X \sim U(8, 18), \qquad \varepsilon \sim N(0, 4^2), \qquad n = 30. \tag{25}$$

Reading (25) with numbers. The argument of tanh is zero at $x = 7 \times 1.9 = 13.3$, where $\psi = 45 \cdot 0 + 57 = 57$: the inflection point. At the ends,

$$\psi(8) = 45\tanh(-2.7895)+57 = 45(-0.99248)+57 = 12.339, \qquad \psi(18) = 45\tanh(2.4737)+57 = 45(0.98590)+57 = 1 \tag{26}$$

and the slope at the inflection is $\psi'(13.3) = \frac{45}{1.9}\operatorname{sech}^2(0) = 23.68$ thousand R$ per extra year — an S-curve from about 12 to about 101 that no straight line can follow. The irreducible error is $\sigma^2 = 4^2 = 16$, the dotted line of slide 26.



ψ of (25), 30 pairs from a seeded draw (not the slide's sample), and three learners: df = 2 (the OLS line, which is what the slide's smoothing spline gives at gl = 2), df = 7 (a degree-6 polynomial standing in for gl = 7), df = 30 (the interpolant, as gl = 30 on slide 24). Fundamentos slides 21–24.

The line is rigid: it under-predicts at 8 and 18 and over-predicts just below 13 years, and redrawing the sample would barely move it — bias without variance. The interpolant has zero training error (it passes through every point) but zigzags with the noise — variance without bias. The degree-6 curve tracks ψ to within a few thousand reais everywhere: the trade-off in one picture.

The slide's Monte Carlo (slide 25). For each gl $= 2, \ldots, n$: draw M test points $x^{(i)} \sim F_X$; for each, draw N training samples of size n, fit $\hat{\psi}_j^{(gl)}$, and estimate the two terms of (21) by

$$\widehat{\text{Viés}^2}(x^{(i)}) = \Big(\tfrac{1}{N}\textstyle\sum_{j=1}^{N} \hat{\psi}_j^{(gl)}(x^{(i)}) - \psi(x^{(i)})\Big)^2, \qquad \widehat{\text{Var}}(x^{(i)}) = \text{variance of the } N \text{ values } \hat{\psi}_j^{(gl)}(x^{(i)}), \tag{27}$$

then average each over the M points. Slide 26 shows the result: bias^2 falls to near zero by gl ≈ 7, variance climbs steadily, and the prediction error bottoms out near 20, above the dotted line at $\sigma^2 = 16$ — **"aproximadamente 7 graus de liberdade"** (slide 27).

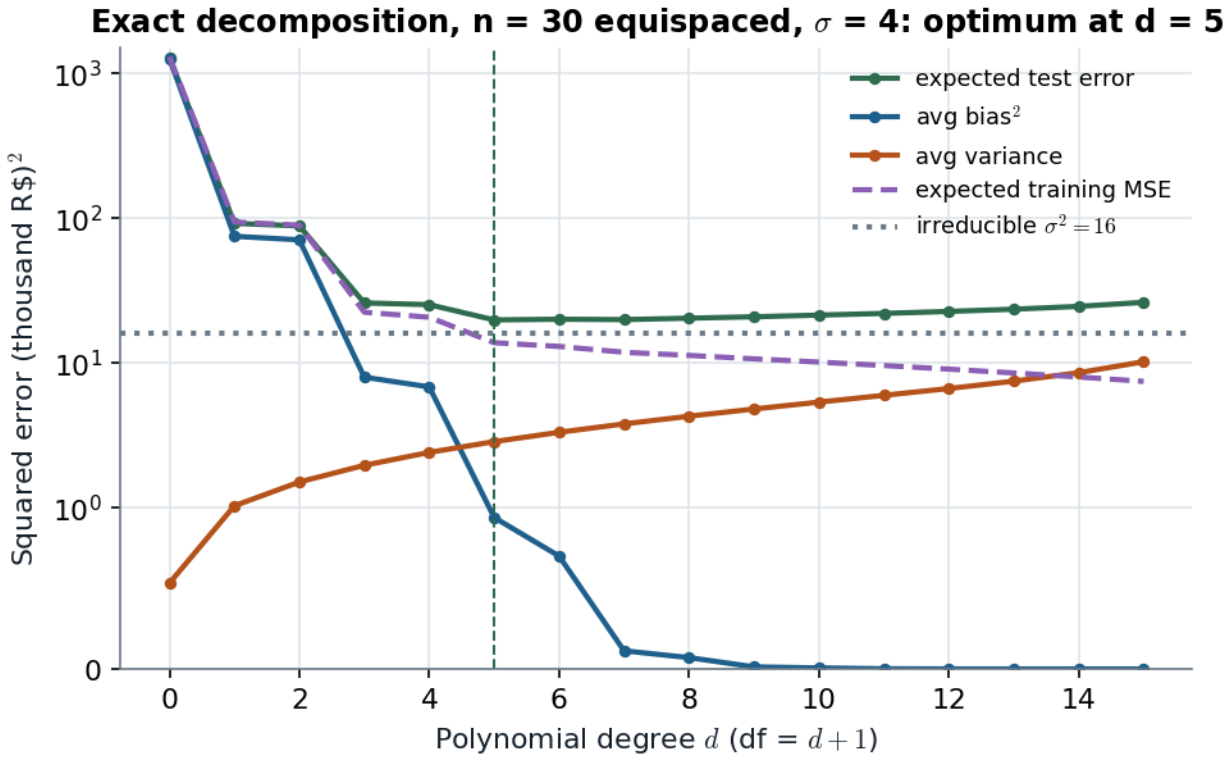
The same experiment, exactly. Swap the smoothing spline for an OLS polynomial of degree d on a fixed equispaced design of 30 points in $[8, 18]$. Then $\hat{\psi}(x_0) = b(x_0)'(B'B)^{-1}B'\mathbf{y}$ is linear in $\mathbf{y} = \psi + \varepsilon$ (B the $30 \times (d+1)$ basis matrix, $b(x_0)$ its row at x_0), so both terms of (17) are closed-form:

$$E[\hat{\psi}(x_0)] = b(x_0)'(B'B)^{-1}B'\psi, \qquad \mathrm{Var}[\hat{\psi}(x_0)] = \sigma^2\, b(x_0)'(B'B)^{-1}b(x_0), \tag{28}$$

averaged over a uniform grid on $[8, 18]$ in place of dF_X. The numbers (computed in `figures/make_figures.py`):

d (df)	avg bias^2	avg variance	σ^2	**expected test error**	expected training MSE
1 (2)	74.80	1.03	16	91.83	93.87
3 (4)	7.98	1.97	16	25.95	22.39
5 (6)	0.94	2.87	16	**19.81**	13.79
6 (7)	0.70	3.33	16	20.03	13.01
7 (8)	0.11	3.80	16	19.91	11.85
10 (11)	0.007	5.37	16	21.38	10.14
15 (16)	0.000	10.20	16	26.20	7.47

Check one row against (21): at $d = 5$, $0.94 + 2.87 + 16 = 19.81$. ✓



Exact (21) for an OLS polynomial on 30 equispaced points, σ = 4 (symmetric-log axis). Same world as Fundamentos slides 21 and 26; different learner and design, stated above.

The minimum is flat: degrees 5, 6 and 7 (6–8 df) are within 0.25 of each other, right where the slide's spline lands (about 7 df). Past it, bias² has nothing left to give and the variance curve carries the test error up; the training curve slides *below* $\sigma^2 = 16$ — the fit is now reproducing noise.

> **Interactive companion.** The Flexibility Dial (`companions/companion-bias-variance-polynomial.html`) — move the degree d, the training size n and the noise σ on this tanh model; watch bias², variance, training and test error, and where the optimal degree goes (larger n or smaller σ push it right).

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6 · C — why the training error cannot pick the flexibility

Training MSE (slide 28; ISLP eq. 2.5, p. 28; AME eq. 1.1, p. 14):

$$\mathrm{EQM}_{\mathrm{train}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\psi}(x_i)\right)^2. \tag{29}$$

Slides 29–30: as gl grows, (29) **falls monotonically** and hits **zero at gl = 30**, where the spline interpolates. Choosing gl by (29) always picks the most overfitted model. ISLP's test MSE (eq. 2.6, p. 29) is the average of $(y_0 - \hat{\psi}(x_0))^2$ over *previously unseen* pairs — the quantity (21) describes.

How optimistic is it? Exactly, for OLS (G layer, proved because it explains the AIC/C_p penalties of L01 §6). With hat matrix $H = B(B'B)^{-1}B'$, the residual vector is

$$\mathbf{e} = (I - H)\mathbf{y} = (I - H)\psi + (I - H)\varepsilon. \tag{30}$$

$I - H$ is symmetric and idempotent with trace $n - (d+1)$, and $E[\varepsilon] = \mathbf{0}$ kills the cross term, so

$$E\|\mathbf{e}\|^2 = \|(I - H)\psi\|^2 + E[\varepsilon'(I - H)\varepsilon] = \|(I - H)\psi\|^2 + \sigma^2(n - d - 1). \tag{31}$$

Dividing by n:

$$E[\mathrm{EQM}_{\mathrm{train}}] = \frac{\|(I - H)\psi\|^2}{n} + \sigma^2\Big(1 - \frac{d+1}{n}\Big). \tag{32}$$

For a *new* response at the same design points, the error is $\|(I - H)\psi\|^2/n + \sigma^2(d+1)/n + \sigma^2$ (bias² + variance + noise, averaged over the x_i). Subtracting (32):

$$E[\text{in-sample test error}] - E[\mathrm{EQM}_{\mathrm{train}}] = \frac{2\sigma^2(d+1)}{n}. \tag{33}$$

> **The training error understates the error on new responses by $2\sigma^2(d+1)/n$ — growing with every parameter.** Adding exactly that penalty back is Mallows' C_p / AIC; BIC uses $\log(n)$ in place of 2 (AME §1.5.1.1, p. 21). At $d = 5$, $n = 30$, $\sigma^2 = 16$: $2(16)(6)/30 = 6.4$. The other route — estimate the test error directly on held-out data — is cross-validation, L02-03-cross-validation-motorcycle.

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7 · G — named, not developed here

- **Smoothing splines** (the slide's learner, `smooth.spline()` with gl): a black box on purpose (slide 23). Polynomials stand in for them in this unit.
- **Trees, boosting, random forests** (`bias-variance-R.txt` comments): the bias–variance trade-off "in orthogonal directions". Outside this course's syllabus.
- **The same decomposition for classification** (slides 36–50: 0–1 loss, the Bayes classifier and its optimality proof, k-NN with k chosen by 10-fold CV, Bayes error 0.0183 by Monte Carlo): the classification side is L02-05-discriminant-analysis.
- **Ridge and lasso** — buying lower variance with deliberate bias — are Classes 3+4.

Links

- Next: reading the coefficients of the linear learners → L02-02-linear-log-linear-regression; estimating (21) from data → L02-03-cross-validation-motorcycle.
- Class 1: MSE = variance + bias² for a parameter estimator → L01 §5; OLS, hat matrix, R^2, AIC/BIC and overfitting → L01 §6.
- Unit: L02-07-formulas (equations (6), (15), (21), (33)), L02-08-pitfalls, L02-09-python.
- Rules pack: rules/06 (bias–variance trade-off). Class plan: Class 2.
- Reading: ISLP §2.1.1 pp. 17–20, §2.2.1–2.2.2 pp. 28–34; AME §1.4 pp. 9–11, §1.5.2 pp. 22–23 (in Portuguese).
- **Practice:** ISLP Ch. 2 exercises 1 and 3 (p. 63).

W §2.4, §6.2 · Reading a linear regression: levels, logs, elasticities and interactions

> **Claim.** "Linear" regression is linear in the **parameters**, not in the variables, so logs, squares and products of the regressors are all fitted by the same OLS. What changes is the **reading** of a coefficient: a unit change (level-level), an approximate percentage (log-level: 100β), an elasticity (log-log: β), or a slope that depends on another variable (interactions, quadratics). **The 100β reading is a first-order approximation; the exact change is $100(e^{\beta\Delta x}-1)$, and the gap is already 5 points at $\beta = 0.306$.** Prev: L02-01-prediction-error-bias-variance · Next: L02-03-cross-validation-motorcycle · Map: L02-00-map

Notation.

- y response, $x_1, \dots, x_k$ regressors, u error with $E[u \mid x] = 0$; $\beta_0, \dots, \beta_k$ population coefficients, $\hat{\beta}_j$ OLS estimates.
- Δ a discrete change; $\%\Delta y = 100\,\Delta y/y$ a percentage change; "ceteris paribus" = all other regressors and u held fixed ($\Delta u = 0$).
- log is the natural log throughout (W's convention).
- s_y, s_{x_j} sample standard deviations; a superscript * marks a standardised variable $(v - \bar{v})/s_v$.
- Dummies: $D = 1$ for the named category, 0 for the base category.

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1 · U — "linear" means linear in β

The model $y = \beta_0 + \beta_1 x + u$ is *linear* because it is linear in β_0, β_1; nothing stops y or x from being log(wage), $\sqrt{\text{inc}}$, x^2 or $x_1 x_2$ (W §2-4c, p. 40). The mechanics of OLS — $\hat{\beta} = (X'X)^{-1}X'y$ from L01 §6 — do not care how the columns of X were built.

> **The mechanics are free; the interpretation is not.** W p. 40: *"it is much more important to become proficient at interpreting coefficients than to become efficient at computing formulas."* The same fact is what makes polynomial regression a linear model (L02-03-cross-validation-motorcycle §2).

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2 · C — level-level: a coefficient is a slope in the variables' own units

For $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$, holding the other regressors and u fixed,

$$\Delta y = \beta_j \, \Delta x_j. \tag{1}$$

Units move the coefficient, never the fit (W §2-4a, pp. 36–37). If x_j is replaced by $c\,x_j$, (1) stays true only if the slope becomes β_j/c; if y is replaced by $c\,y$, every coefficient is multiplied by c. R^2 is unchanged. W's own case: the CEO-salary slope on *roe* is 18.501 (eq. 2.39, salary in thousand dollars per percentage point) and on $roedec = roe/100$ it is 1,850.1 (eq. 2.41, p. 37) — dividing x by 100 multiplied the slope by 100.

2.1 · Worked — the Credit best model (multiplelinearregression.pdf slide 44)

$$\widehat{\text{Balance}} = -499.7 - 7.84\,\text{Income} + 0.267\,\text{Limit} + 23.18\,\text{Cards} + 429.6\,\text{Student}, \tag{2}$$

$n = 400$, residual standard error 99.56 on 395 df, $R^2 = 0.9536$, adjusted $R^2 = 0.9531$. Balance is the average credit-card debt in dollars; Income is in **thousands** of dollars (ISLP §3.3.1, p. 91).

Coefficient	Reading by (1), others held fixed
Income −7.839	\$1,000 more income goes with \$7.84 *less* balance
Limit 0.2666	each extra dollar of credit limit goes with \$0.27 more balance
Cards 23.18	one more card: \$23.18 more balance
Student 429.6 (dummy)	a student carries \$429.6 more balance than a non-student with the same income, limit and cards

Check the printout's own arithmetic. The t-statistic is the estimate over its standard error:

$$t_{\text{Student}} = \frac{429.6}{16.61} = 25.86 \ \checkmark, \qquad t_{\text{Cards}} = \frac{23.18}{3.639} = 6.37 \ \checkmark \tag{3}$$

and adjusted $R^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$ with $k = 4$:

$$\bar{R}^2 = 1 - (1 - 0.9536)\,\frac{399}{395} = 1 - 0.0464 \times 1.010127 = 1 - 0.04687 = 0.9531 \ \checkmark \tag{4}$$

2.2 · Worked — rescaling in the houseprice data (slides 17 and 29)

Slide 17 converts the data before fitting: square feet to m² (`SqFt*0.092903`) and price from US dollars to **thousand reais** (`Price*3.2/1000`). The six-variable model of slide 29 has $\widehat{\beta}_{\text{size}} = 1.886008$: **R\$1,886 per extra m²**, other regressors fixed. Undoing the conversion with the rule under (1): since $\text{m}^2 = 0.092903\,\text{sqft}$, the slope per square foot is

$$\widehat{\beta}_{\text{sqft}} = 0.092903 \times 1.886008 = 0.17522 \text{ thousand R\$} = \text{R\$}175.2 \text{ per sqft}, \tag{5}$$

and at 3.2 reais per dollar that is $175.22/3.2 = \$54.75$ per square foot.

The CAPM regressions (`capm-regression-R.txt`) are level-level too: monthly *excess* returns of the food, durables and construction industries on the market's excess return (Ecdat `Capm`, 516 months, 1960–2002). There β reads "percentage points of industry excess return per percentage point of market excess return", and the intercept is the pricing error α. The script prints no output, so there are no numbers to work here; its conjugate-prior part belongs to Classes 3+4.

---

3 · C — interactions: the slope of one variable depends on another

With an interaction (W §6-2c, pp. 177–178, eq. 6.17),

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u \quad \Longrightarrow \quad \frac{\Delta y}{\Delta x_2} = \beta_2 + \beta_3 x_1. \tag{6}$$

β_2 alone is the effect of x_2 at $x_1 = 0$ — often meaningless (W's example: the effect of a bedroom on a house of zero square feet). Report (6) at interesting values of x_1, or centre x_1 at its mean before forming the product so that β_2 becomes the effect at the mean (W p. 178).

When x_1 is a dummy, (6) gives two slopes. `salaries-for-professors.txt` fits the full model `salary ~ sex.dum*yrs.service` (salary in thousand dollars, `sex.dum` = 1 for men):

$$E[\text{salary} \mid \text{male}, \text{yrs}] = \beta_0 + \beta_1 \,\text{male} + \beta_2 \,\text{yrs} + \beta_3 \,\text{male} \cdot \text{yrs}, \tag{7}$$

$$\frac{\partial E[\text{salary}]}{\partial \,\text{yrs}} = \begin{cases} \beta_2 & \text{women} \\ \beta_2 + \beta_3 & \text{men} \end{cases}, \qquad \text{gap(yrs)} = E[\cdot \mid \text{male}] - E[\cdot \mid \text{female}] = \beta_1 + \beta_3 \,\text{yrs}. \tag{8}$$

So β_1 is the gender gap **at zero years of service**, and β_3 is how much the gap widens per year. The reduced model `salary ~ sex.dum + yrs.service` forces $\beta_3 = 0$: parallel lines, one constant gap. The script prints no estimates; it compares the two models by a Bayes factor (Classes 3+4).

3.1 · Worked — the seven-variable houseprice model (slide 29)

`coef(regs,7)` includes `sizeN2` $= -0.3614412$ and `sizebrick` $= 0.2907379$ with $\hat{\beta}_{\text{size}} = 1.9374668$. By (6), the price of one more m² (thousand R$) is $1.9375 - 0.3614\,N2 + 0.2907\,\text{brick}$:

	not brick	brick
not neighbourhood 2	1.9375	$1.9375 + 0.2907 = 2.2282$
neighbourhood 2	$1.9375 - 0.3614 = 1.5760$	$1.9375 - 0.3614 + 0.2907 = 1.8668$

One coefficient per row of the printout, but four different "effects of size". A square metre is worth R$2,228 in a brick house outside N2 and R$1,576 in a non-brick house in N2.

---

4 · C — log-level: 100β is a semi-elasticity, and an approximation

Model (W eq. 2.42, p. 37):

$$\log y = \beta_0 + \beta_1 x + u. \tag{9}$$

Step 1. Two values of x, same u; subtract the two versions of (9):

$$\log y_1 - \log y_0 = \beta_1 (x_1 - x_0) = \beta_1 \Delta x. \tag{10}$$

Step 2 — exact. Exponentiate (10): $y_1/y_0 = e^{\beta_1 \Delta x}$, so

$$\%\Delta y = 100\Big(\frac{y_1}{y_0} - 1\Big) = 100(e^{\beta_1 \Delta x} - 1). \tag{11}$$

Step 3 — approximate. For small z, $e^z = 1 + z + z^2/2 + \cdots \approx 1 + z$. Substituting $z = \beta_1 \Delta x$ into (11):

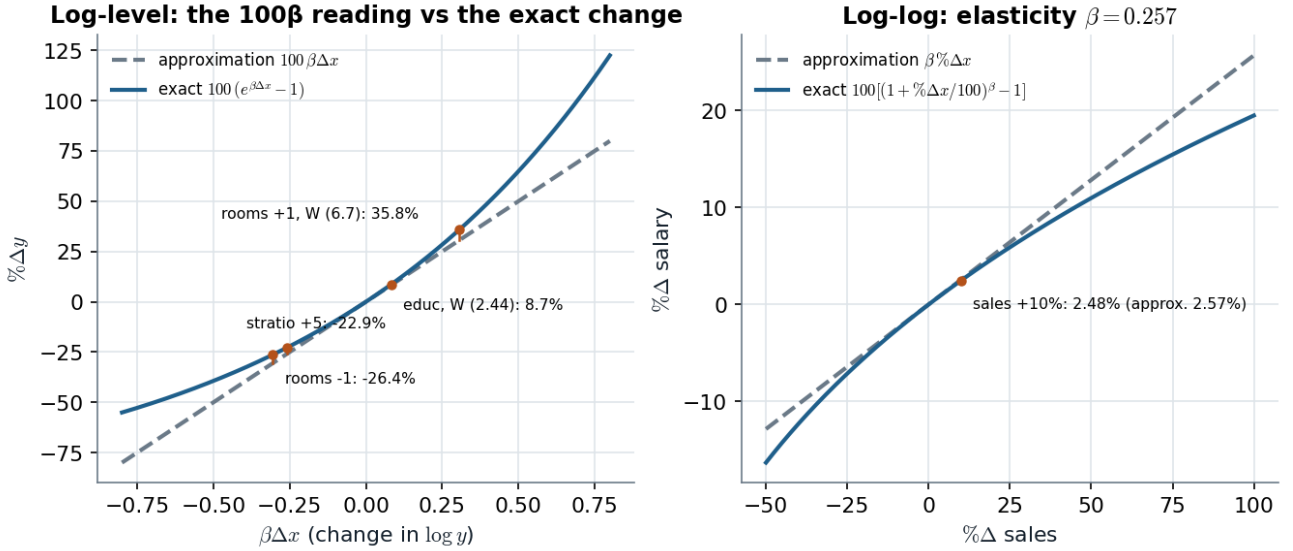
$$\%\Delta y \approx 100\,\beta_1 \Delta x, \tag{12}$$

W's eq. (2.43); $100\beta_1$ is the **semi-elasticity** of y with respect to x. Eq. (11) is W's (6.8), p. 171; with $\Delta x = 1$ it is (6.9). The error of (12) is about $100z^2/2$: negligible for $|z| \le 0.05$, five points at $z = 0.3$.

4.1 · Worked — W's own numbers, approximate vs exact

Source	$\hat{\beta}\,\Delta x$	(12): $100\hat{\beta}\Delta x$	(11): $100(e^{\hat{\beta}\Delta x}-1)$
Ex. 2.10, $\log(\text{wage}) = 0.584 + 0.083\,\text{educ}$ ($n = 526$, $R^2 = 0.186$), one more year	0.083	8.3%	$100(e^{0.083}-1) = 8.65\%$
(6.7) $\log(\text{price}) = 9.23 - 0.718\log(\text{nox}) + 0.306\,\text{rooms}$, one more room	0.306	30.6%	$100(e^{0.306}-1) = 35.8\%$
same, one *fewer* room ($\Delta x = -1$)	−0.306	−30.6%	$100(e^{-0.306}-1) = -26.4\%$
stratio coefficient −0.052, +1	−0.052	−5.2%	$100(e^{-0.052}-1) = -5.1\%$
same, +5	−0.260	−26%	$100(e^{-0.26}-1) = -22.9\%$

(W pp. 38, 171–172.) Note the room row: **the approximation** 30.6% **always lies between the exact decrease** (26.4%) **and the exact increase** (35.8%) — it is the symmetric, "arc" version (W p. 172).



Exact (11) vs approximation (12) at W's coefficients (eqs. 2.44, 6.7, pp. 38, 171–172), and the log-log analogue (16) at W's elasticity 0.257 (eq. 2.46, p. 39).

The vertical gaps on the left are the error of reading 100β: invisible at $\beta = 0.083$, five points at 0.306. Exact changes are asymmetric — up 35.8%, down 26.4% for the same $|\Delta\text{rooms}|$ — because a log-linear model is multiplicative in y.

Three cautions. (i) *Not* "log(wage) rises by 8.3%": the log rises by 0.083 (W p. 38). (ii) Rescaling y (dollars to thousands) changes only the intercept by $\log c$, never the slope (W p. 39). (iii) R^2 from a regression of $\log y$ cannot be compared with one of y — they explain the variation of different variables (W p. 173).

G — what (11) is a change of. With u held fixed, (11) is the change in $e^{\beta_0+\beta_1 x+u}$, i.e. in the median of y when u is symmetric. If u is independent of x, $E[y \mid x] = e^{\beta_0+\beta_1 x}E[e^u]$ and the ratio $E[y \mid x_1]/E[y \mid x_0] = e^{\beta_1 \Delta x}$ is the same, so (11) also holds for the mean. Plugging $\widehat{\beta}$ into (11) is consistent but not unbiased, since exp is non-linear (W p. 171).

---

5 · C — log-log and level-log

Log-log (W eq. 2.45, p. 39): $\log y = \beta_0 + \beta_1 \log x + u$. Differencing as in (10),

$$\Delta \log y = \beta_1 \, \Delta \log x \quad \Longrightarrow \quad \beta_1 = \frac{d \log y}{d \log x} = \frac{dy/y}{dx/x}, \tag{13}$$

the **elasticity** of y with respect to x. Using $\Delta \log v \approx \%\Delta v/100$ on both sides of (13):

$$\%\Delta y \approx \beta_1 \, \%\Delta x. \tag{14}$$

Exactly, exponentiating (13): $y_1/y_0 = (x_1/x_0)^{\beta_1}$, so

$$\%\Delta y = 100\Big[\Big(1 + \frac{\%\Delta x}{100}\Big)^{\beta_1} - 1\Big]. \tag{15}$$

Worked — W Example 2.11 (p. 39): $\widehat{\log(\text{salary})} = 4.822 + 0.257 \log(\text{sales})$, $n = 209$, $R^2 = 0.211$. A 1% rise in sales goes with a 0.257% rise in CEO salary. For a 10% rise, (14) gives $0.257 \times 10 = 2.57\%$ and (15) gives

$$100(1.10^{0.257} - 1) = 100(1.024797 - 1) = 2.48\%. \tag{16}$$

Level-log: $y = \beta_0 + \beta_1 \log x + u$ gives $\Delta y = \beta_1 \Delta \log x \approx (\beta_1/100)\,\%\Delta x$.

W Table 2.3 (p. 39), the summary to keep:

Model	Dependent	Independent	Interpretation of β_1
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$

When to log (W pp. 172–173): positive money amounts and large counts usually; years (education, experience, age) usually not; proportions either way — and then a "percentage" change (log) must not be confused with a "percentage point" change (level): unemployment from 8 to 9 is one point, 12.5%, and $\log 9 - \log 8 = 0.118$.

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6 · C — Worked: the site's log-family-income regression (Mroz data)

`regressaomultipla-return-to-education.html` (Lopes, 5/30/2023) uses Mroz's 1975 PSID sample of $n = 753$ married women and fits

$$y^* = \sum_{j=1}^{18} \beta_j^* x_j^* + e, \qquad y^* = \frac{\log(\text{FAMINC}) - \overline{\log \text{FAMINC}}}{s_y}, \quad x_j^* = \frac{x_j - \bar{x}_j}{s_{x_j}}, \tag{17}$$

by `lm(y~X-1)` — no intercept, because every variable has mean zero.

Reading a standardised log-level coefficient. Undo the standardisation in (17):

$$\log(\text{FAMINC}) = \overline{\log \text{FAMINC}} + s_y \sum_j \beta_j^* \frac{x_j - \bar{x}_j}{s_{x_j}} + s_y e \quad \Longrightarrow \quad \frac{\partial \log(\text{FAMINC})}{\partial x_j} = \beta_j^* \frac{s_y}{s_{x_j}}. \tag{18}$$

So β_j^* reads "**one standard deviation more of x_j goes with β_j^* standard deviations more of log family income**", and the semi-elasticity per unit of x_j is $100\beta_j^* s_y / s_{x_j}$ by (12). The page does not print s_y or s_{x_j} (and the data are not in the repo), so the percentage reading cannot be completed from the page — the SD reading is the one available.

Regressor	$\hat{\beta}^*$	SE	t	Reading
MTR (wife's marginal tax rate)	−0.7119	0.0329	−21.64	by far the strongest association; but tax rates are computed from income, so this is a predictive association, not an effect
HW (husband's wage)	0.1476	0.0320	4.62	+1 SD of husband's wage, +0.148 SD of log family income
WHRS (wife's hours)	0.1323	0.0285	4.65	
WE (wife's education)	−0.0053	0.0238	−0.22	no association **holding the wife's wage, hours and the other 15 fixed**

The page's "return to education" is nil — because of what is held fixed. Education works partly *through* the wife's wage (WW) and hours, which are in the regression; conditioning on them removes that channel. Which controls are legitimate is a Classes 5+6 question (bad controls).

Check the printout (residual SE 0.4511 on 735 df, $R^2 = 0.8011$, adjusted 0.7962).

t for MTR: $-0.711872/0.032891 = -21.64$ ✓. The residual sum of squares and, because y^* has mean 0 and variance 1, the total sum of squares $\sum y_i^{*2} = n - 1 = 752$:

$$\text{RSS} = 735 \times 0.4511^2 = 735 \times 0.203491 = 149.57, \qquad R^2 = 1 - \frac{149.57}{752} = 1 - 0.19889 = 0.8011 \checkmark \tag{19}$$

R's no-intercept adjusted R^2 uses n and $n - p$ with $p = 18$:

$$\bar{R}^2 = 1 - (1 - 0.8011)\,\frac{753}{735} = 1 - 0.1989 \times 1.02449 = 1 - 0.20377 = 0.7962 \checkmark \tag{20}$$

The page's "MLE" standard errors use $\hat{\sigma}^2_{\text{MLE}} = \text{RSS}/n$ instead of $\text{RSS}/(n-p)$, so each is the OLS one times $\sqrt{735/753} = 0.98798$: for LFP, $0.029839 \times 0.98798 = 0.029480$, the printed 0.02947983 ✓ (the MLE of σ^2 is the biased one, L01 §6).

The rest of the page (prior predictive densities −2253.0, −2107.2, −2015.1 and log Bayes factors 145.8, 238.0, 92.1 selecting LFP, WHRS, K618, HW, MTR) is Bayesian model comparison: Classes 3+4.

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7 · C — quadratics: a slope that changes with x

$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ (W eq. 6.10, p. 173). The slope is not β_1:

$$\frac{\Delta \hat{y}}{\Delta x} \approx \hat{\beta}_1 + 2\hat{\beta}_2 x, \tag{21}$$

zero at the turning point

$$x^* = \Big|\frac{\hat{\beta}_1}{2\hat{\beta}_2}\Big|. \tag{22}$$

Worked (W pp. 174–176). Wage on experience, $\hat{\beta}_1 = 0.298$, $\hat{\beta}_2 = -0.0061$ (eq. 6.12): $x^* = 0.298/(2 \times 0.0061) = 0.298/0.0122 = 24.4$ years. Example 6.2, log(price) with $\hat{\beta}_{\text{rooms}} = -0.545$, $\hat{\beta}_{\text{rooms}^2} = 0.062$: $x^* = 0.545/0.124 = 4.4$ rooms, and by (21) and (12)

$$\%\Delta\text{price} \approx 100\big[-0.545 + 2(0.062)\,\text{rooms}\big]\Delta\text{rooms} = (-54.5 + 12.4\,\text{rooms})\,\Delta\text{rooms}, \tag{23}$$

$= -54.5 + 12.4(5) = 7.5\%$ from five to six rooms and $-54.5 + 12.4(6) = 19.9\%$ from six to seven.

> **A quadratic is a degree-2 polynomial regression.** Raising the degree buys flexibility and, by L02-01-prediction-error-bias-variance, variance; choosing the degree from data is L02-03-cross-validation-motorcycle.

Links

- Prev: the prediction view of the same models → L02-01-prediction-error-bias-variance; next: polynomial degree by cross-validation → L02-03-cross-validation-motorcycle.
- Class 1: OLS, R^2, adjusted R^2, MLE vs unbiased $\hat{\sigma}^2$ → L01 §6.
- Unit: Poisson coefficients read as rate ratios use the same e^{β} step as (11) → L02-06-gaussian-vs-poisson-regression; logit coefficients as odds ratios → L02-04-logistic-regression; L02-07-formulas, L02-08-pitfalls, L02-09-python.
- Class plan: Class 2.
- **Practice:** W Ch. 2 Problem 9 (p. 55, units in log models); W Ch. 6 Computer Exercise C2 (p. 199, the experience quadratic with log(wage)).

ISLP §5.1–5.2, §7.1 · Choosing the polynomial order by cross-validation, and putting a bootstrap band on the winner

> **Claim.** The test error of L02-01-prediction-error-bias-variance is not observable, but it can be **estimated by refitting on part of the data and predicting the rest**. A single hold-out split is noisy (on the site, 100 random 80/20 splits pick order 12 only 63 times); **10-fold CV picks it 99 times out of 100**; LOOCV costs a *single* fit for least squares, thanks to $CV_{(n)} = \frac{1}{n}\sum\left(\frac{y_i-\hat{y}_i}{1-h_{ii}}\right)^2$. All of adjusted R^2, BIC, LOOCV and 10-fold land on **order 12** for the motorcycle data, and a pairs bootstrap draws the uncertainty band around that fit. Prev: L02-02-linear-log-linear-regression · Next: L02-04-logistic-regression · Map: L02-00-map

Notation.

- (x_i, y_i), $i = 1, \dots, n$: the data; for mcycle, x = time after impact (ms), y = head acceleration (g), both standardised; $n = 132$.
- d: polynomial order (tuning parameter); the model has $d + 1$ coefficients.
- X: the $n \times (d+1)$ design $[1, x, \dots, x^d]$; $\widehat{\beta} = (X'X)^{-1}X'y$; $\hat{y}_i = x_i'\widehat{\beta}$; $e_i = y_i - \hat{y}_i$.
- $H = X(X'X)^{-1}X'$ the hat matrix, $h_{ii} = x_i'(X'X)^{-1}x_i$ the **leverage** of point i.
- $\widehat{\beta}_{(i)}$, $\hat{y}_{(i)} = x_i'\widehat{\beta}_{(i)}$: fit and prediction with point i left out.
- k: number of folds; $\mathcal{F}_1, \dots, \mathcal{F}_k$ the index sets of the folds, $n_j = |\mathcal{F}_j|$.
- B: number of bootstrap resamples; $\widehat{f}^{*b}$ the fit on resample b.

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1 · U — two jobs for an error estimate

Fundamentos slide 27 names them: **model selection** (pick the flexibility — here d — that best balances bias and variance) and **model assessment** (say how well the chosen model will predict). The training MSE does neither: it falls with every added term (L02-01-prediction-error-bias-variance §6). Two fixes exist: **penalise** the training error (adjusted R^2, C_p, AIC, BIC — L01 §6), or **estimate the test error directly** by predicting observations the fit has not seen (this note).

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2 · C — polynomial regression and the mcycle data

Polynomial regression is a linear model (ISLP eq. 7.1, p. 290):

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i, \tag{1}$$

OLS on the columns $x, x^2, \dots, x^d$ (L02-02-linear-log-linear-regression §1). ISLP warns that d above 3 or 4 gives "very strange shapes", "especially near the boundary of the X variable" (p. 290) — keep that in mind when order 12 wins below.

The data (slides 2–3). MASS's `mcycle`: head acceleration in a simulated motorcycle crash, used to test helmets (Silverman 1985). The code keeps the first 132 rows, maps $x \mapsto x/\max x$, then $x \mapsto -1 + 2x$, then standardises; y likewise. **The first two maps are wasted work:** standardising $ax + b$ ($a > 0$) gives back the standardised x, because the mean shifts by b and scales by a while the SD scales by a:

$$\frac{(ax_i + b) - (a\bar{x} + b)}{a\, s_x} = \frac{x_i - \bar{x}}{s_x}. \tag{2}$$

The plot (slide 3) is flat and tight before impact, then a deep dip, a rebound and a noisy tail: **the noise variance is far from constant**, which matters for §7.

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3 · C — in-sample criteria: adjusted R^2 and BIC (slides 4–5)

Slide 4 fits $d = 1, 2, 5, 8, 10, 11, 12, 20$; the panel labels read

d	1	2	5	8	10	11	12	20
label "R2"	0.0769	0.1323	0.5259	0.6941	0.769	0.7686	**0.7798**	0.7688

(the html code prints adjusted R^2 in that legend, and slide 5's adjusted-R^2 curve peaks at the same ≈ 0.78, at $d = 12$). The html computes, with $\hat{\sigma}^2 = \mathrm{RSS}/(n - d - 1)$,

$$\bar{R}^2_d = 1 - (1 - R^2_d)\frac{n-1}{n-d-1}, \qquad \mathrm{BIC}_d = \frac{1}{n}(\mathrm{RSS}_d + \log(n)\,(d+1)\,\hat{\sigma}^2_d), \tag{3}$$

the BIC in AME's penalised-risk form $\mathrm{EQM}(g) + \log(n)\, p\, \hat{\sigma}^2/n$ with $p = d + 1$ (AME §1.5.1.1, p. 21). **Both pick** $d = 12$ (slide 5, dotted lines). Going from 12 to 20 adds eight coefficients and *lowers* $\bar{R}^2$ from 0.7798 to 0.7688: the penalty at work.

A number to carry. y is standardised, so $s^2_y = 1$ and $\bar{R}^2 = 1 - \hat{\sigma}^2/s^2_y$ gives at $d = 12$

$$\hat{\sigma}^2_{12} = 1 - 0.7798 = 0.2202, \qquad \hat{\sigma}_{12} = \sqrt{0.2202} = 0.4693. \tag{4}$$

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4 · C — the validation-set (hold-out) approach

Estimator (ISLP §5.1.1, pp. 202–204; AME eq. 1.2, p. 14). Split at random into a training part $\mathcal{T}$ and a validation part $\mathcal{V}$; fit on $\mathcal{T}$ only; then

$$\hat{R}(d) = \frac{1}{|\mathcal{V}|}\sum_{i \in \mathcal{V}} \left(y_i - \hat{f}_{\mathcal{T}}(x_i)\right)^2. \tag{5}$$

Since $\mathcal{V}$ played no part in the fit, (5) is an honest estimate (consistent by the law of large numbers, AME p. 14). **Two drawbacks** (ISLP §5.1.1 and p. 208): it varies a lot with the split, and it fits on fewer points than n, so it tends to *overestimate* the error of the full-sample fit.

One split (slide 6, 80% / 20%): the root-MSE curve falls from about 0.92 at $d \leq 4$ to its minimum at $d = 12$.

One hundred splits (slide 7) — the variability made visible. Times each order wins:

d	1–5	6	7	8	9	**10**	11	**12**	13	14
wins (of 100)	0	1	0	2	2	**15**	7	**63**	10	0

Total $1+2+2+15+7+63+10=100$ ✓. Order 12 wins 63% of the time; orders 10–13 together $15+7+63+10=95\%$. The same experiment drives the best-subset size k in the houseprice data (multiplelinearregression.pdf slides 36–38: 1,000 random splits at 50%, 60%, 70%, 80% training; the winners pile up at $k=5$–7).

> **A single split answers "which d?" with a random variable.** Repeating the split and reporting the distribution of winners is honest; reporting one split is not.

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5 · C — leave-one-out CV, and the one-fit shortcut

Definition (Fundamentos slide 31; ISLP eq. 5.1, p. 204; AME p. 15). For each i, fit on the other $n-1$ points and predict y_i:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_{(i)}\right)^2. \tag{6}$$

No randomness, and each fit uses $n-1$ points, so little bias (ISLP p. 205). AME Observação 1.5 (pp. 15–16): LOOCV on $n+1$ points is *exactly* unbiased for the expected risk of the method fitted on n.

The shortcut (ISLP eq. 5.2, p. 205, stated there without proof). For least squares,

$$\boxed{CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1-h_{ii}}\right)^2} \tag{7}$$

— one fit instead of n. **Derivation.** Write $A = X'X$ and $u = x_i$, so $h_{ii} = u'A^{-1}u$.

Step 1 — drop row i. Removing one row from X and y:

$$X'_{(i)}X_{(i)} = A - uu', \qquad X'_{(i)}y_{(i)} = X'y - u\,y_i. \tag{8}$$

Step 2 — Sherman–Morrison (for $1-u'A^{-1}u \neq 0$):

$$(A-uu')^{-1} = A^{-1} + \frac{A^{-1}uu'A^{-1}}{1-h_{ii}}. \tag{9}$$

Step 3 — substitute (8) and (9) into $\hat{\beta}_{(i)} = (X'_{(i)}X_{(i)})^{-1}X'_{(i)}y_{(i)}$:

$$\hat{\beta}_{(i)} = \left(A^{-1} + \frac{A^{-1}uu'A^{-1}}{1-h_{ii}}\right)(X'y - u\,y_i). \tag{10}$$

Multiply out, using $A^{-1}X'y = \hat{\beta}$, $u'\hat{\beta} = \hat{y}_i$ and $u'A^{-1}u = h_{ii}$:

$$\hat{\beta}_{(i)} = \hat{\beta} - A^{-1}u\,y_i + \frac{A^{-1}u\,\hat{y}_i}{1-h_{ii}} - \frac{A^{-1}u\,h_{ii}\,y_i}{1-h_{ii}}. \tag{11}$$

Put the last three terms over $1-h_{ii}$:

$$\hat{\beta}_{(i)} = \hat{\beta} + \frac{A^{-1}u\,[-y_i(1-h_{ii}) + \hat{y}_i - h_{ii}y_i]}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}u\,(y_i - \hat{y}_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i\,e_i}{1-h_{ii}}. \tag{12}$$

Step 4 — the left-out residual. Premultiply (12) by x_i':

$$y_i - \hat{y}_{(i)} = y_i - \hat{y}_i + \frac{h_{ii}\,e_i}{1-h_{ii}} = e_i\Big(1 + \frac{h_{ii}}{1-h_{ii}}\Big) = \frac{e_i}{1-h_{ii}}. \tag{13}$$

Substituting (13) into (6) gives (7). ■

> **Reading (13).** The left-out residual is the ordinary residual inflated by $1/(1-h_{ii})$ — "exactly the right amount" (ISLP p. 205). Since $\sum_i h_{ii} = \operatorname{tr} H = d+1$, the average leverage is $(d+1)/n$: at $d=12$, $n=132$, it is $13/132 = 0.0985$. But leverages are largest at the ends of the x range, which is where a degree-12 polynomial is least trustworthy. (7) holds for least squares and some linear smoothers, **not** in general — logistic regression or LDA must be refitted n times (ISLP p. 205).

On the site (slide 8): the LOOCV curve's minimum is at $d = 12$, about 0.48 on the slide's "Root MSE" axis.

Variance of $CV_{(n)}$ (Fundamentos slide 35). With $\mathrm{EQM}_i = (y_i - \hat{y}_{(i)})^2$,

$$\operatorname{Var}[CV_{(n)}] = \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}[\mathrm{EQM}_i] + \frac{2}{n^2}\sum_{1\le i<j\le n}\operatorname{Cov}[\mathrm{EQM}_i, \mathrm{EQM}_j]. \tag{14}$$

The n fits share $n-2$ of their $n-1$ points, so the covariances are large and positive: LOOCV is nearly unbiased but **high-variance** (ISLP §5.1.4, pp. 208–209).

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6 · C — k-fold CV

Definition (Fundamentos slide 33; ISLP eq. 5.3, p. 206). Split at random into k folds of about equal size; fit without fold j, compute MSE_j on fold j; average:

$$CV_{(k)} = \frac{1}{k}\sum_{j=1}^{k}\mathrm{MSE}_j, \qquad \mathrm{MSE}_j = \frac{1}{n_j}\sum_{i\in\mathcal{F}_j}\big(y_i - \hat{f}_{-j}(x_i)\big)^2. \tag{15}$$

$k = n$ is LOOCV (slide 35). AME (p. 15) and the site's code use the observation-weighted version

$$\widetilde{CV}_{(k)} = \frac{1}{n}\sum_{j=1}^{k}\sum_{i\in\mathcal{F}_j}\big(y_i - \hat{f}_{-j}(x_i)\big)^2 = \sum_{j=1}^{k}\frac{n_j}{n}\,\mathrm{MSE}_j, \tag{16}$$

equal to (15) only when all $n_j = n/k$.

The site's folds, worked. `L = trunc(n/fold)` $= \lfloor 132/10 \rfloor = 13$; folds 1–9 are 13 consecutive (shuffled) positions each, $9 \times 13 = 117$, and fold 10 takes the rest, $132 - 117 = 15$. So $n_j/n = 13/132$ or $15/132$, not $1/10$ — (16) and (15) differ slightly.

Why $k = 5$ or 10 (ISLP §5.1.4; Fundamentos slide 35):

Method	fit size here ($n = 132$)	bias of the error estimate	variance	cost (fits)
80/20 hold-out	≈ 105 ($0.8 \times 132 = 105.6$)	highest	high (one split)	1
10-fold	≈ 119 ($0.9 \times 132 = 118.8$)	small	moderate	10
LOOCV	131	smallest	highest (correlated fits)	n (1 with (7))

On the site: one 10-fold run (slide 9) has its minimum at $d = 12$; repeating the random fold assignment 100 times (slide 10), $d = 12$ **wins 99 times and** $d = 13$ **once**. Compare 63 for the hold-out in §4: every observation is predicted once in every run, so the only randomness left is the fold assignment.

> **The axis label is not the quantity — but the argmin is safe.** In the html code the LOO loop accumulates the *sum* of squared errors and the 10-fold loop divides it by n (an MSE), both plotted under "Root MSE". On the slides the curves sit near 0.48 at $d = 12$; by (4) an MSE there cannot be below about $\hat{\sigma}^2 = 0.22$, while $\sqrt{0.22} = 0.47$ — so the slides show root MSE. SSE, MSE and root MSE are increasing transforms of one another: **the chosen d is the same**, only the vertical scale differs.

---

7 · C — bootstrap bands for the chosen fit

Pairs bootstrap (slide 11; html §8; ISLP §5.2, pp. 212–214). With $d^* = 12$ fixed, for $b = 1, \ldots, B = 1000$: draw n indices with replacement, refit (1) on those pairs, keep the fitted curve $\hat{f}^{*b}$. Slide 11 draws all 1,000 curves in grey under the red order-12 fit. A pointwise 95% **percentile band** keeps, at each x, the 2.5% and 97.5% quantiles of $\{\hat{f}^{*b}(x)\}_{b=1}^{B}$; the bootstrap standard error at x is ISLP's (5.8), p. 214:

$$\mathrm{SE}_B(\hat{f}(x)) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{f}^{*r}(x) - \frac{1}{B}\sum_{r'=1}^{B}\hat{f}^{*r'}(x)\right)^2}. \tag{17}$$

The formula alternative (ISLP eq. 7.2 and footnote 1, p. 291): with $\ell_0 = (1, x_0, \ldots, x_0^d)'$ and $\hat{C} = \hat{\sigma}^2(X'X)^{-1}$,

$$\widehat{\mathrm{Var}}[\hat{f}(x_0)] = \ell_0' \hat{C}\, \ell_0, \qquad \hat{f}(x_0) \pm 2\sqrt{\ell_0' \hat{C}\, \ell_0}. \tag{18}$$

(18) assumes **one** error variance for all x. Resampling pairs does not: each resample carries the quiet pre-impact points and the noisy tail in their own proportions. With mcycle's heteroskedasticity (§2) the two bands disagree, the bootstrap one being wider where the noise is large.

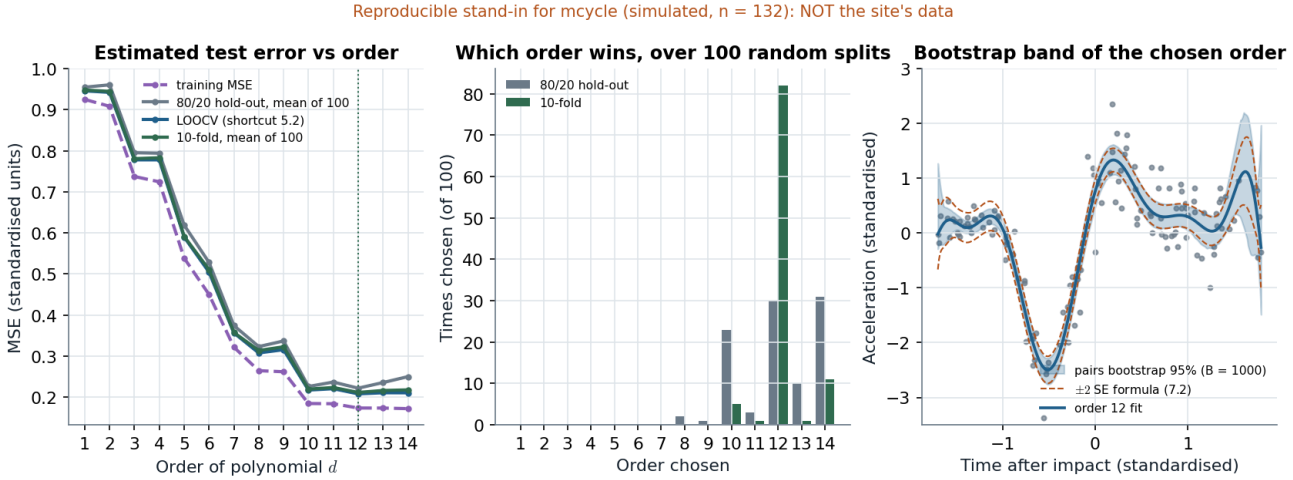
> **What the band does not cover:** the order was chosen on the same data and is held at 12 in every resample. Model-selection uncertainty is outside it (G: re-running the CV inside each resample would put it in).

8 · C — Worked: the whole pipeline on a reproducible stand-in

The `mcycle` data are not in the repo, so the figures below run every step on a **simulated stand-in** built to look like slide 3 — $n = 132$, a flat start, a dip to about -2 and a rebound to about $+1.3$ in standardised units, noise SD 0.15 before the dip and 0.45 after (seeded; definition in `figures/make_figures.py`). **Its numbers are not the site's.** That its CV also lands on 12 is a property of how the stand-in was drawn, not evidence about `mcycle`.

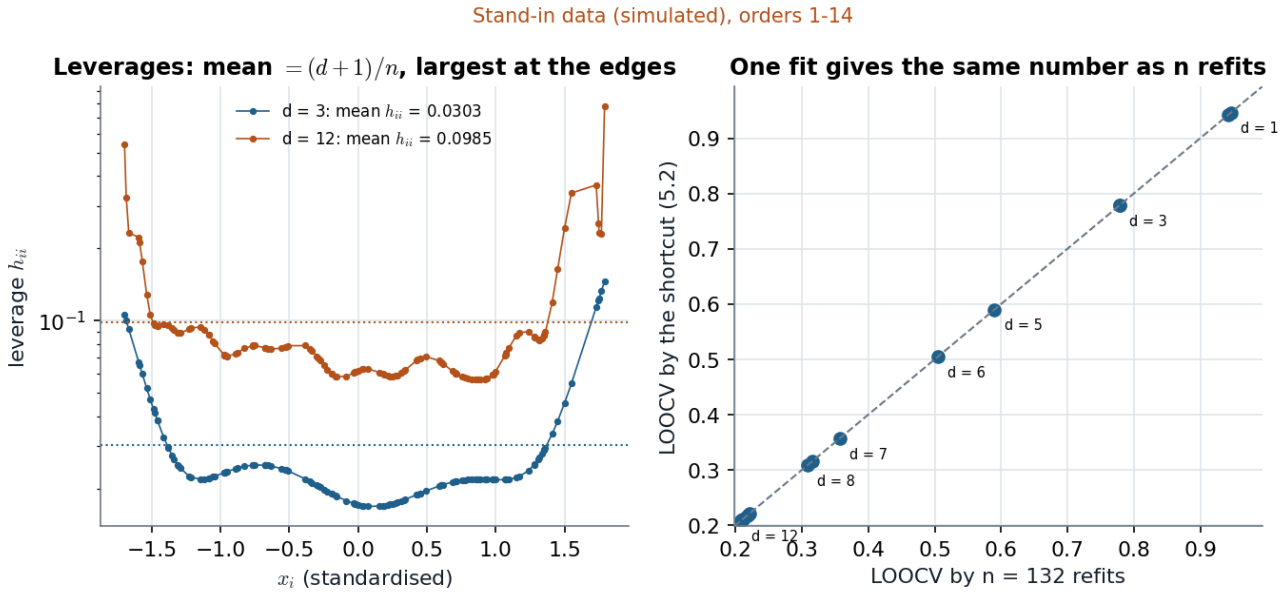
d	training MSE	80/20 hold-out (mean of 100)	LOOCV by (7)	10-fold (mean of 100)
1	0.9252	0.9550	0.9461	0.9481
3	0.7376	0.7954	0.7781	0.7810
5	0.5388	0.6192	0.5900	0.5917
7	0.3219	0.3746	0.3567	0.3577
10	0.1847	0.2262	0.2174	0.2197
12	0.1737	**0.2220**	**0.2082**	**0.2117**
13	0.1736	0.2357	0.2109	0.2164
14	0.1722	0.2501	0.2107	0.2179

Read across a row: training < LOOCV < 10-fold < hold-out, the ordering of §6's table (fit size $132 > 131 > \approx 119 > \approx 105$). Read down: training keeps falling to $d = 14$; the three honest estimates turn up after 12. Over 100 random splits, the hold-out winners scatter (order 14: 31, 12: 30, 10: 23, 13: 10, others 6) while 10-fold picks 12 82 times — the same contrast as the site's 63 vs 99.



Stand-in data (simulated, labelled on the figure): estimated test error by (5), (7), (16); winners over 100 splits; percentile band from $B = 1000$ pairs resamples vs the formula band (18). Mirrors motorcycle slides 6–11.

The right panel shows both lessons of §7 and §2 at once: in the noisy tail the bootstrap band is wider than the constant-variance formula band, and at both ends of the x range the band flares, the boundary misbehaviour ISLP warns about for high degrees.



Leverages h_{ii} (mean $(d+1)/n$: $4/132 = 0.0303$, $13/132 = 0.0985$) and the identity (7) checked against brute force (6), stand-in data. ISLP eq. 5.2, p. 205.

Every point sits on the diagonal (agreement to relative 10^{-8}, asserted in the script): the derivation (8)–(13) in numbers. The left panel shows where the inflation $1/(1-h_{ii})$ bites — at the edges, more so for $d = 12$.

> **Interactive companion.** The Flexibility Dial (`companions/companion-bias-variance-polynomial.html`) — the population version of the left panel: move the degree d, the training size n and the noise σ on the tanh model of L02-01 and watch the *true* test error that CV is trying to estimate, and where its minimum sits.

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9 · G — named, not developed here

- **One-standard-error rule**: pick the simplest model whose CV error is within one standard error of the minimum (ISLP Ch. 6). On the stand-in, orders 10 and 12 differ by less than 0.01 in CV error, so the rule could well prefer the simpler one.
- **Selection bias of the minimum**: the smallest of 14 CV estimates is optimistic for the chosen model; an untouched test set (AME's training / validation / test split, p. 16) or nested CV fixes it.
- **CV for classification** (ISLP eq. 5.4): the error indicator replaces the squared error → L02-04-logistic-regression, L02-05-discriminant-analysis.
- **Ridge and lasso**, whose penalty is chosen by CV (`cv.glmnet` in multiplelinearregression.pdf): Classes 3+4.

Links

- Prev: L02-02-linear-log-linear-regression; Next: L02-04-logistic-regression; the target being estimated: L02-01-prediction-error-bias-variance §4–§6.
- Class 1: R^2, adjusted R^2, AIC/BIC, overfitting → L01 §6; the bootstrap and where it fails ($x_{(n)}$) → L01 §5.

- Unit: L02-07-formulas ((7), (15), (17), (18)), L02-08-pitfalls, L02-09-python (CV by hand, the shortcut, the bootstrap band).
- Rules pack: rules/06 (training and test samples, LOOCV, k-fold). Class plan: Class 2.
- Reading: ISLP §5.1 pp. 202–209, §5.2 pp. 212–214, §7.1 pp. 290–291; AME §1.5.1 pp. 14–16 (in Portuguese).
- **Practice:** ISLP 5.3 (p. 225), 5.5 (p. 225), 5.8 (p. 227, LOOCV for polynomial degree); ISLP 7.6 (p. 327), 7.9 (p. 327, CV-chosen degree).

Logistic part, slides 4–38 · The logit model, its MLE by IRLS, and what the `default` and spam outputs actually say

> **Claim.** A binary y is Bernoulli with an individual probability π_i, and logistic regression makes the **log-odds** $\log\{\pi_i/(1-\pi_i)\}$ linear in x_i. The log-likelihood is concave, its score is $X'(y-\pi)$, its Hessian is $-X'WX$, and **one Newton–Raphson step is one weighted least-squares fit** on an adjusted response z (IRLS). Coefficients are read as **odds ratios** e^{β_j}, a probability cut-off c maps to a covariate cut-off $x^* = (\operatorname{logit} c - \beta_0)/\beta_1$, and a classifier is judged by its **confusion matrix**, not by one error rate. Six slips in the deck (slides 9, 10, 15, 18, 22 and 38) are corrected below, each where it arises, plus a reversed comment in the admission script (§10). Prev: L02-03-cross-validation-motorcycle · Next: L02-05-discriminant-analysis · Map: L02-00-map

Notation.

- $y_i \in \{0,1\}$: response of individual $i = 1, \dots, n$ (1 = "yes", default, spam, admitted).
- $x_i = (1, x_{i1}, \dots, x_{ip})'$: covariates with a leading 1; X the $n \times (p+1)$ matrix with rows x_i'.
- $\beta = (\beta_0, \dots, \beta_p)'$: coefficients; $\eta_i = x_i'\beta$ the **linear predictor**, $\eta_i \in \mathbb{R}$.
- $\pi_i = P(y_i = 1 \mid x_i) \in (0,1)$; the deck also writes it $g(x_i'\beta)$ and, on slide 9, p.
- $g(\mu) = e^\mu/(1+e^\mu)$: logistic function (W writes Λ, eq. 17.3); $g^{-1}(\pi) = \log\{\pi/(1-\pi)\} = \operatorname{logit}\pi$.
- $\ell(\beta)$: log-likelihood; $W = \operatorname{diag}\{\pi_i(1-\pi_i)\}$: the weight matrix; z: IRLS adjusted response.
- $c \in (0,1)$: probability cut-off of a classifier; x^*: the covariate value where $\hat\pi = c$.
- TN, FP, FN, TP: confusion-matrix counts (true negative, false positive, false negative, true positive).

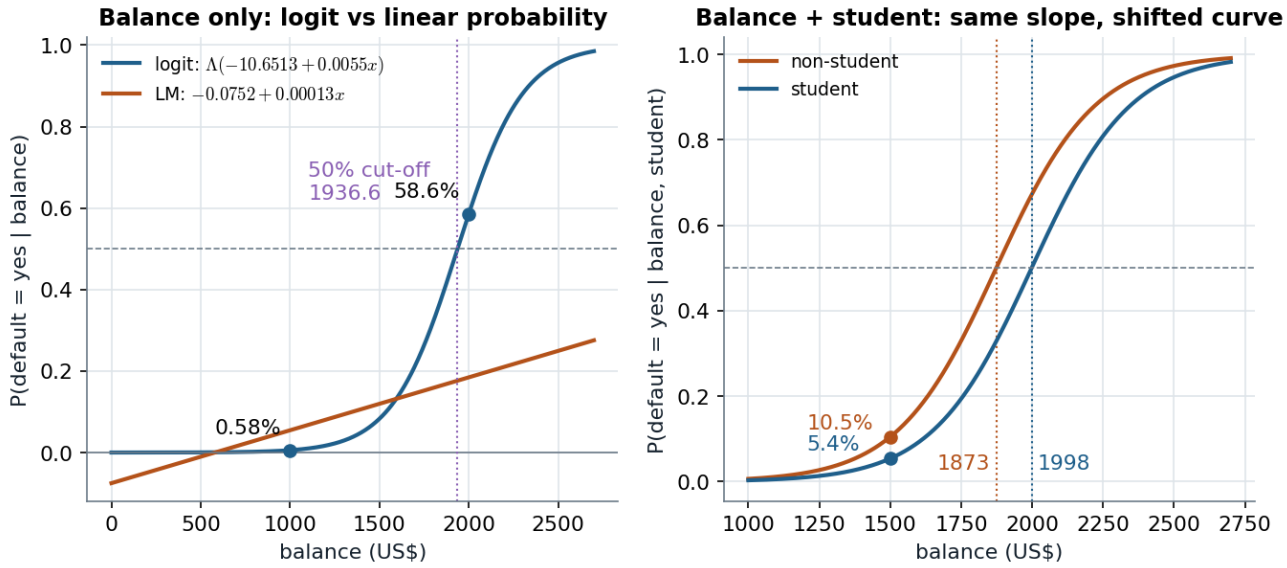
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1 · U — why a straight line fails on a 0/1 response

The `default` data (ISLR, $n = 10{,}000$; slide 4) has 9,667 non-defaulters and 333 defaulters (slide 12). Regressing the 0/1 indicator on `balance` by OLS — the **linear probability model** (I eq. 4.1, p. 139; W §17.1, p. 525) — gives (slide 13)

$$\hat\pi^{LM}(x) = -0.07519 + 0.0001299\, x.$$

It crosses zero at $x = 0.07519/0.0001299 = 578.8$, so **every customer with a balance under USD 579 gets a negative "probability"**, and at $x = 2000$ it says $-0.07519 + 0.2598 = 0.185$ where the logit says 0.586 (§5.1). W's two complaints (p. 525) are exactly these: fitted values outside $[0,1]$ and a **constant** partial effect. The fix is to model a transformation of π that is free to range over $\mathbb{R}$.



Left: the fitted logit (slides 13, 15) never leaves $[0, 1]$*; the LM line is negative below 578.8. Right: adding* `student` *shifts the curve, not its shape (slide 20). Curves drawn from the printed coefficients only — the data are not on disk.*

The logit is flat near 0 for balances below about 1,000, rises steepest where $\hat{\pi} = 1/2$, and saturates near 1; the straight line cannot bend, so it is too high at small balances, too low at large ones, and wrong in sign at the left end. On the right the two curves are horizontal translates of each other — the dummy moves the intercept, which is all a dummy can do in a logit without interactions.

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2 · C — the model: logistic function, odds, logit (slides 5–6)

Model (slide 5; I eq. 4.2, p. 139; W eq. 17.2–17.3, pp. 525–526).

$$\pi_i = P(y_i = 1 \mid x_i) = \frac{\exp\{x_i'\beta\}}{1 + \exp\{x_i'\beta\}} = g(x_i'\beta). \tag{1}$$

From (1) to the log-odds, one step at a time. Write $\eta_i = x_i'\beta$.

$$1 - \pi_i = 1 - \frac{e^{\eta_i}}{1 + e^{\eta_i}} = \frac{1 + e^{\eta_i} - e^{\eta_i}}{1 + e^{\eta_i}} = \frac{1}{1 + e^{\eta_i}}. \tag{2}$$

Dividing (1) by (2):

$$\frac{\pi_i}{1 - \pi_i} = \frac{e^{\eta_i}/(1 + e^{\eta_i})}{1/(1 + e^{\eta_i})} = e^{\eta_i} \qquad \text{(the odds; I eq. 4.3, p. 140).} \tag{3}$$

Taking logs of (3):

$$\log \frac{\pi_i}{1 - \pi_i} = x_i'\beta \qquad \text{(the log-odds or logit; I eq. 4.4, p. 140).} \tag{4}$$

The GLM reading (slide 6). $y_i \sim \text{Bernoulli}(\pi_i)$ with

$$E(y_i \mid x_i) = \pi_i = g(x_i'\beta), \qquad V(y_i \mid x_i) = \pi_i(1-\pi_i), \tag{5}$$

and $g^{-1} = \text{logit}$ is the **link**: it maps the mean to the scale on which the model is linear. Note 06 uses the same template with a log link (L02-06-gaussian-vs-poisson-regression §6).

2.1 · C — reading a coefficient: odds ratio and partial effect

Raise x_j by one unit, everything else fixed. From (3) the odds go from e^{η} to $e^{\eta+\beta_j}$, so

$$\frac{\text{odds}(x_j+1)}{\text{odds}(x_j)} = \frac{e^{\eta+\beta_j}}{e^{\eta}} = e^{\beta_j} \qquad \text{(odds ratio; I p. 140).} \tag{6}$$

The effect on the **probability** is not constant. Differentiate (1) with the quotient rule:

$$\frac{dg}{d\eta} = \frac{e^{\eta}(1+e^{\eta}) - e^{\eta}e^{\eta}}{(1+e^{\eta})^2} = \frac{e^{\eta}}{(1+e^{\eta})^2} = g(\eta)\,\{1-g(\eta)\}, \tag{7}$$

so by the chain rule

$$\frac{\partial \pi_i}{\partial x_{ij}} = \beta_j\,\pi_i(1-\pi_i) \le \frac{\beta_j}{4}, \tag{8}$$

with the maximum $\beta_j/4$ at $\pi_i = 1/2$ (W eq. 17.7, p. 527: partial effect $g(\beta_0 + x\beta)\beta_j$, and for the logit $g(0) = 0.25$).

Worked, `default` (slide 15: $\hat{\beta}_1 = 0.0055$ per USD of balance).

Quantity	Substitution	Value
odds ratio per USD 1	$e^{0.0055}$	1.0055
odds ratio per USD 100	$e^{0.0055\times 100} = e^{0.55}$	**1.733** (+73%)
$\partial\pi/\partial x$ at the 50% cut-off	$0.0055 \times 0.5 \times 0.5$	0.001375 per USD = 13.75 pp per USD 100
$\partial\pi/\partial x$ at $x = 1000$ ($\hat{\pi} = 0.005759$)	$0.0055 \times 0.005759 \times 0.994241$	0.0000315 per USD = 0.31 pp per USD 100

> **The odds ratio is the same everywhere; the probability effect is 44 times larger at the cut-off than at a balance of 1,000.** "A one-unit increase in balance raises the log-odds by 0.0055" (I Table 4.1 caption, p. 142) is exact; "raises the probability by 0.0055" is wrong.

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3 · C — likelihood, score and Hessian (slides 7–8)

Likelihood (slide 7; I eq. 4.5, p. 141; W eq. 17.10, p. 528). Independent Bernoulli draws:

$$p(y_{1:n} \mid x_{1:n}, \beta) = \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} = \prod_{i:y_i=1} \pi_i \prod_{i:y_i=0} (1-\pi_i). \tag{9}$$

This is the Bernoulli likelihood of L01 §1 with one change: θ is no longer common, each π_i is tied to x_i through (1).

Log-likelihood (W eq. 17.11).

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \right]. \tag{10}$$

Rewrite each term in η_i. Group the y_i terms:

$$y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) = y_i \log \frac{\pi_i}{1 - \pi_i} + \log(1 - \pi_i).$$

Substitute (4) for the log-odds and (2) for $1 - \pi_i$:

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \, (x_i'\beta) + \log \frac{1}{1 + e^{x_i'\beta}} \Big] = \sum_{i=1}^{n} \left[y_i \, x_i'\beta - \log(1 + e^{x_i'\beta}) \right]. \tag{11}$$

Score (slide 8). Differentiate (11); the derivative of $\log(1 + e^{\eta})$ is $e^{\eta}/(1 + e^{\eta}) = \pi$:

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^{n} \left[y_i \, x_i - \pi_i \, x_i \right] = \sum_{i=1}^{n} x_i \, (y_i - \pi_i) = X'(y - \pi). \tag{12}$$

Setting (12) to zero gives $p + 1$ equations, **nonlinear in** β because π_i is.

Hessian (slide 8). Differentiate (12) once more; only π_i depends on β, and by (7) $\partial \pi_i / \partial \beta' = \pi_i(1 - \pi_i) \, x_i'$:

$$\frac{\partial^2 \ell}{\partial \beta \, \partial \beta'} = -\sum_{i=1}^{n} x_i \, x_i' \, \pi_i(1 - \pi_i) = -X'WX, \qquad W = \operatorname{diag}\{\pi_i(1 - \pi_i)\}. \tag{13}$$

Two consequences.

1. Every $\pi_i(1 - \pi_i) > 0$, so $X'WX$ is positive definite when X has full column rank: ℓ is **strictly concave**, and a stationary point is the unique maximum. *(G: if some linear combination of x separates the 0s from the 1s perfectly, ℓ keeps rising as $\|\beta\| \to \infty$ and no finite MLE exists — the "unstable when classes are well-separated" bullet on slide 40.)*
2. The first row of (12) at $\hat{\beta}$ (the intercept's column of ones) reads $\sum_i (y_i - \hat{\pi}_i) = 0$: **the average fitted probability equals the sample share of 1s** — 0.0333 in `default` (I p. 140).

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4 · C — Newton–Raphson is iteratively reweighted least squares (slides 8–9)

Newton–Raphson (slide 8). Starting from β^{old},

$$\beta^{\text{new}} = \beta^{\text{old}} - \Big(\frac{\partial^2 \ell}{\partial \beta \, \partial \beta'} \Big)^{-1} \frac{\partial \ell}{\partial \beta}, \tag{14}$$

both derivatives evaluated at β^{old}.

Substitute (12) and (13) into (14):

$$\beta^{\text{new}} = \beta^{\text{old}} - \big(-X'WX\big)^{-1}X'(y-\pi) = \beta^{\text{old}} + (X'WX)^{-1}X'(y-\pi). \tag{15}$$

Factor $(X'WX)^{-1}$ out of both terms (write $\beta^{\text{old}} = (X'WX)^{-1}X'WX\beta^{\text{old}}$):

$$\beta^{\text{new}} = (X'WX)^{-1}\big[X'WX\beta^{\text{old}} + X'(y-\pi)\big].$$

Insert $WW^{-1} = I$ in front of $(y-\pi)$ (W is diagonal with positive entries):

$$\beta^{\text{new}} = (X'WX)^{-1}X'W\big[X\beta^{\text{old}} + W^{-1}(y-\pi)\big] = (X'WX)^{-1}X'Wz, \tag{16}$$

$$z = X\beta^{\text{old}} + W^{-1}(y-\pi), \qquad z_i = \eta_i^{\text{old}} + \frac{y_i - \pi_i}{\pi_i(1-\pi_i)}. \tag{17}$$

(16) is the **weighted least-squares** estimator of a regression of z on X with weights W, i.e.

$$\beta^{\text{new}} = \arg\min_{\beta}\,(z - X\beta)'W(z - X\beta), \tag{18}$$

whose normal equations $X'W(z-X\beta) = 0$ give (16). **Slide 9 prints the objective as $(z-X\beta)'(z-X\beta)$; the weight matrix is needed** — without it the minimiser is $(X'X)^{-1}X'z$, not (16). Recompute π and W at β^{new} and repeat: weights are "re-weighted" each pass, hence IRLS (R's `glm` reports "Fisher scoring iterations"; for the logit link Fisher scoring and Newton–Raphson coincide because (13) does not depend on y).

Standard errors come from the same matrix at convergence: $\widehat{\text{Var}}(\hat\beta) = (X'\hat WX)^{-1}$, the inverse observed information — the "Std. Error" column of every `glm` printout below, and the z-statistic is $\hat\beta_j/\text{SE}$ (I p. 141; W p. 529).

4.1 · Worked — IRLS by hand on the `student` table (slide 18)

The slide prints the full 2×2 table, so this fit **can be reproduced exactly** (it is the only `default` model whose data are fully on the page):

default \ student	No	Yes	Total
No	6850	2817	9667
Yes	206	127	333
Total	7056	2944	10000

Model: $\eta_i = \beta_0 + \beta_1 s_i$, $s_i = 1$ for a student.

Iteration 1 from $\beta = (0,0)$. Every $\pi_i = g(0) = 0.5$, every weight $0.5 \times 0.5 = 0.25$, and (17) gives

$$z_i = 0 + \frac{y_i - 0.5}{0.25} = 4y_i - 2.$$

With a constant $W = 0.25I$, (16) collapses to OLS of z on a dummy, i.e. **group means of z**:

$$\beta_0^{(1)} = 4 \cdot \frac{206}{7056} - 2 = 4(0.029195) - 2 = -1.88322,$$

$$\beta_0^{(1)} + \beta_1^{(1)} = 4 \cdot \frac{127}{2944} - 2 = 4(0.043139) - 2 = -1.82745 \;\Rightarrow\; \beta_1^{(1)} = 0.05577.$$

Subsequent iterations (computed in Python, `L02-09`):

iteration	β_0	β_1	-2ℓ
1	−1.88322	0.05577	4116.20
2	−2.78054	0.15372	3072.81
3	−3.31150	0.29836	2917.50
4	−3.48780	0.39184	2908.74
5	−3.50400	0.40477	2908.683
6	**−3.50413**	**0.40489**	**2908.683**

The limit in closed form. With one dummy, the two score equations (12) split by group: $\sum_{s_i=0}(y_i - \pi_0) = 0$ and $\sum_{s_i=1}(y_i - \pi_1) = 0$, so $\hat\pi_0 = 206/7056$ and $\hat\pi_1 = 127/2944$ — the group shares. Then by (4) and (6):

$$\hat\beta_0 = \log\frac{206}{6850} = -3.50413, \qquad \hat\beta_1 = \log\frac{127/2817}{206/6850} = \log 1.49913 = 0.40489.$$

Standard errors from $(X'\hat W X)^{-1}$ reduce to sums of reciprocal cell counts:

$$\mathrm{SE}(\hat\beta_0) = \sqrt{\tfrac{1}{206} + \tfrac{1}{6850}} = 0.07071, \qquad \mathrm{SE}(\hat\beta_1) = \sqrt{\tfrac{1}{206} + \tfrac{1}{6850} + \tfrac{1}{127} + \tfrac{1}{2817}} = 0.11502,$$

exactly the slide's 0.07071 and 0.11502; and $-2\ell(\hat\beta) = 2908.7$ is the slide's residual deviance (§7).

> **The deck's two probabilities on slide 18 are off by a factor of ten.** It prints $\hat P(y = 1 \mid \text{student}) = 0.43\%$ and 0.29%; the fitted values are $127/2944 = 0.0431 = \mathbf{4.31\%}$ and $206/7056 = 0.0292 = \mathbf{2.92\%}$, as I prints (p. 142). The odds ratio $e^{0.40489} = 1.499$: alone, being a student multiplies the odds of default by 1.5.

---

5 · C — the `default` outputs, read (slides 13, 15, 17, 18, 20, 22)

Model (slide)	$\hat\beta_0$	$\hat\beta$ balance	$\hat\beta$ student	Resid. dev. (df)	AIC
balance (13)	−10.65 (SE 0.3612)	0.005499 (SE 0.0002204)	—	1596.5 (9998)	1600.5
student (18)	−3.50413 (0.07071)	—	0.40489 (0.11502)	2908.7 (9998)	2912.7
balance + student (20)	−10.75 (0.3692)	0.005738 (0.0002318)	−0.7149 (0.1475)	1571.7 (9997)	1577.7

Null deviance 2920.6 on 9999 df in all three. AIC = deviance + 2(number of coefficients): $1596.5 + 4 = 1600.5$, $2908.7 + 4 = 2912.7$, $1571.7 + 6 = 1577.7$. z-values: $-10.65/0.3612 = -29.49$, $0.005499/0.0002204 = 24.95$, as printed.

5.1 · Balance only: plug-in probabilities (slide 15; I p. 142)

At $x = 1000$. Linear predictor, then (1):

$$\hat{\eta} = -10.6513 + 0.0055(1000) = -10.6513 + 5.5 = -5.1513,$$
$$\hat{\pi} = \frac{e^{-5.1513}}{1 + e^{-5.1513}} = \frac{0.005792}{1.005792} = 0.005759 = \mathbf{0.58}\%.$$

At $x = 2000$.

$$\hat{\eta} = -10.6513 + 0.0055(2000) = -10.6513 + 11 = 0.3487,$$
$$\hat{\pi} = \frac{e^{0.3487}}{1 + e^{0.3487}} = \frac{1.41722}{2.41722} = 0.5863 = \mathbf{58.6}\%.$$

(Slide 15's second line keeps the label "$x = 1000$"; the arithmetic is for 2000.)

5.2 · From a probability cut-off to a balance cut-off (slide 17)

Classify "yes" when $\hat{\pi}(x) \geq c$. Since g is increasing, $\hat{\pi}(x) \geq c \iff \hat{\eta}(x) \geq \operatorname{logit} c$; set equality and solve:

$$\hat{\beta}_0 + \hat{\beta}_1 x^* = \operatorname{logit} c \quad \Longrightarrow \quad x^* = \frac{\operatorname{logit} c - \hat{\beta}_0}{\hat{\beta}_1}. \tag{19}$$

At $c = 0.5$, $\operatorname{logit} 0.5 = \log 1 = 0$:

$$x^* = \frac{0 - (-10.6513)}{0.0055} = \frac{10.6513}{0.0055} = 1936.6.$$

The whole axis of slide 17, recomputed from (19):

c	$\operatorname{logit} c$	x^* with $\hat{\beta}_1 = 0.0055$	x^* with $\hat{\beta}_1 = 0.005499$	slide label
0.1	−2.1972	1537.1	1537.4	1537
0.2	−1.3863	1684.5	1684.9	1684
0.3	−0.8473	1782.5	1782.9	1782
0.4	−0.4055	1862.9	1863.2	1863
0.5	**0**	**1936.6**	**1937.0**	**1936**
0.6	0.4055	2010.3	2010.7	2010
0.7	0.8473	2090.7	2091.0	2091
0.8	1.3863	2188.7	2189.1	2189
0.9	2.1972	2336.1	2336.5	2336

The labels are the unrounded-slope column truncated to the dollar. **A classifier on one covariate is a threshold on that covariate**; moving c (note 05 §10) just slides x^*.

5.3 · Balance + student, and confounding (slides 19–22; I §4.3.4, pp. 142–144)

At balance 1500. For a student ($s = 1$):

$$\hat{\eta} = -10.75 + 0.005738(1500) - 0.7149(1) = -10.75 + 8.607 - 0.7149 = -2.8579,$$
$$\hat{\pi} = \frac{e^{-2.8579}}{1 + e^{-2.8579}} = \frac{0.057389}{1.057389} = 0.0543 = \mathbf{5.4}\%.$$

For a non-student ($s = 0$):

$$\hat{\eta} = -10.75 + 8.607 = -2.1430, \qquad \hat{\pi} = \frac{0.117302}{1.117302} = 0.1050 = \mathbf{10.5}\%.$$

50% cut-offs by group, from (19) with $\hat{\beta}_0 + \hat{\beta}_2 s$ in place of $\hat{\beta}_0$:

$$x^*_{\text{student}} = \frac{0 - (-10.75 - 0.7149)}{0.005738} = \frac{11.4649}{0.005738} = \mathbf{1998.1}, \qquad x^*_{\text{non-student}} = \frac{10.75}{0.005738} = \mathbf{1873.5}.$$

> **Slide 22 attaches 1873 to students and 1998 to non-students, and its legend colours the curves the same way; the printed coefficients say the reverse.** With $\hat{\beta}_2 = -0.7149 < 0$ a student needs a *higher* balance to reach 50%. Slide 25 (Bayes fit) draws the non-student curve above the student one, consistent with the coefficients.

Confounding. Alone, `student` has a **positive** coefficient (0.405: students default more, §4.1); given `balance` it is **negative** (−0.715; odds ratio $e^{-0.7149} = 0.489$, half the odds at the same balance). Both are right: students carry larger balances (slide 19's boxplots), and balance drives default. I's version adds `income` (Table 4.3, p. 143, income not significant, $p = 0.71$) and gets 0.058 vs 0.105 at balance 1,500 and income 40,000 (eqs. 4.8–4.9, p. 144) — the deck drops income and gets 0.054 vs 0.105.

Likelihood-ratio tests from the printed deviances (W eq. 17.12, p. 529: $LR = 2(\ell_{ur} - \ell_r)$, and for 0/1 data deviance $= -2\ell$, §7):

Comparison	LR = difference in deviance	df
balance vs null	$2920.6 - 1596.5 = 1324.1$	1
student vs null	$2920.6 - 2908.7 = 11.9$	1
balance + student vs balance	$1596.5 - 1571.7 = 24.8$	1

All far beyond $\chi^2_{1;0.95} = 3.84$. **McFadden's pseudo-R^2** (W p. 531), $1 - \ell_{ur}/\ell_0 = 1 - 1596.5/2920.6 = 0.453$ for balance alone, 0.462 with student, 0.004 for student alone.

---

6 · C — classifying a new individual: plug-in vs posterior predictive (slides 11, 23)

The deck **plugs in** the MLE (slide 11):

$$\hat{P}(y_{n+1} = 1 \mid x_{n+1}, y_{1:n}, x_{1:n}) = \frac{\exp\{x'_{n+1}\hat{\beta}_{MLE}\}}{1 + \exp\{x'_{n+1}\hat{\beta}_{MLE}\}}, \tag{20}$$

and classifies "yes" when (20) exceeds 1−(20), i.e. $c = 1/2$. The Bayesian answer (slide 23) **integrates** over the posterior instead:

$$P(y_{n+1} = 1 \mid x_{n+1}, y_{1:n}, x_{1:n}) = \int P(y_{n+1} = 1 \mid x_{n+1}, \beta)\, p(\beta \mid y_{1:n}, x_{1:n})\, d\beta, \tag{21}$$

and (20) is (21) with all posterior mass at $\hat{\beta}$. Slides 24–25 show the posterior of $P(y = 1 \mid 1500, \text{student})$ centred near 0.054 and 0.105 — **name only here; the Bayesian logit is Classes 3+4** (`Aula/class-03-04/materials/Bayesian-logistic-regression.html`).

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7 · C/G — deviance and residuals (slide 10)

Pearson residual $r_i = (y_i - \hat{\pi}_i)/\sqrt{\hat{\pi}_i(1-\hat{\pi}_i)}$; **deviance residual**

$$d_i = \text{sign}(y_i - \hat{\pi}_i)\sqrt{2\Big[y_i \log\frac{y_i}{\hat{\pi}_i} + (1-y_i)\log\frac{1-y_i}{1-\hat{\pi}_i}\Big]}. \tag{22}$$

For 0/1 data the deviance is $-2\ell(\hat{\beta})$. With $y_i \in \{0,1\}$, $y_i \log y_i = 0$ and $(1-y_i)\log(1-y_i) = 0$ (convention $0 \log 0 = 0$), so

$$d_i^2 = 2\big[-y_i \log\hat{\pi}_i - (1-y_i)\log(1-\hat{\pi}_i)\big] \quad \Rightarrow \quad D = \sum_i d_i^2 = -2\ell(\hat{\beta}) \text{ by (10)}. \tag{23}$$

Check on the null model: $\hat{\pi} = 0.0333$ for all, $-2[333 \log 0.0333 + 9667 \log 0.9667] = 2920.6$, the printed null deviance.

Two slips on slide 10: the statistics are $\chi^2 = \sum r_i^2$ and $D = \sum d_i^2$ (squares missing). *G caution:* with one observation per covariate pattern, the χ^2_{n-p} reference for D itself is unreliable; **differences** of deviances between nested models (§5.3) are the safe use.

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8 · G — regularized logistic regression (slides 26–27, 32–35), named only

The penalized objective (slide 26) is $-\ell(\beta)$ from (11) plus the elastic-net penalty:

$$\hat{\beta} = \arg\min_{\beta} \Big\{ -\sum_{i=1}^{n} (y_i x_i'\beta - \log(1 + e^{x_i'\beta})) + \lambda\big[(1-\alpha)\|\beta\|_2^2 + \alpha\|\beta\|_1\big] \Big\},$$

$\alpha = 1$ lasso, $\alpha = 0$ ridge, λ chosen by CV (`cv.glmnet`; AME §8.1.2, p. 143). Ridge/lasso/elastic net are **Classes 3+4** material; here they matter only as rows of the spam table.

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9 · C — the spam confusion matrix (slides 28, 36–38)

Data. 4,601 e-mails labelled spam ($y = 1$) or ham ($y = 0$), 40% spam, 57 word-frequency predictors (Efron & Hastie 2016, pp. 113–115, cited on slide 28); trained on 2,300, tested on 2,301 (slide 33).

Vocabulary (I Tables 4.6–4.7, p. 156; AME Tabela 7.1, p. 138), "positive" = spam:

	predicted ham ($\hat{y} = 0$)	predicted spam ($\hat{y} = 1$)
ham ($y = 0$)	TN	FP — *a ham sent to the spam folder*
spam ($y = 1$)	FN — *a spam let through*	TP

FN rate = $FN/(FN+TP)$ (1 − sensitivity); FP rate = $FP/(FP+TN)$ (1 − specificity, Type I error); **false discovery rate** = $FP/(FP+TP)$ (1 − precision).

Worked, the `glm` row (TN 1325, FP 80, FN 93, TP 803; total 2301; 896 spam in the test set):

Measure	Substitution	Value
error rate	$(80+93)/2301 = 173/2301$	**7.52%**
FN rate	$93/(93+803) = 93/896$	**10.38%**
FP rate	$80/(80+1325) = 80/1405$	**5.69%**
sensitivity	803/896	89.62%
specificity	1325/1405	94.31%
precision	$803/(803+80) = 803/883$	90.94%
FDR	80/883	**9.06%**

All five rows (error, FN and FP rates match the slide; the last two columns are recomputed):

Method	TN	FP	FN	TP	error %	FN %	FP %	FDR = $FP/(FP+TP)$ %	slide's "false.discovery" % = $FP/(FP+FN)$
glm	1325	80	93	803	7.52	10.38	5.69	9.06	8.20
ridge	1345	60	128	768	8.17	14.29	4.27	7.25	6.28
enet	1349	56	94	802	6.52	10.49	3.99	6.53	5.88
lasso	1347	58	95	801	6.65	10.60	4.13	6.75	6.08
bayes	1326	79	91	805	7.39	10.16	5.62	8.94	8.10

> **The slide defines false discoveries as "false positives over positives" but its column divides FP by FP + (all actual spam)**: $80/(80+896) = 8.20$ reproduces every row. Over *predicted* positives (the standard FDR, the slide's own words) the glm figure is 9.06%.

Reading. Elastic net has the lowest error (6.52%) and FP rate (3.99%); ridge lets the most spam through (14.29%). A mail filter cares more about FP (a lost real e-mail) than FN, so the 50% threshold behind these rows is not the right one for that loss — the cost-adjusted cut-off of L02-05-discriminant-analysis §10 is.

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10 · U — the graduate-school admission script (site code)

`graduate-school-admission.txt` (Albert & Hu 2020, Problem 8, pp. 480–481): 400 applicants, admission (0/1) on GRE and a dummy `GPA35` = 1{GPA ≥ 3.5}; it fits `lm` and `glm(..., family=binomial(link="logit"))` and then a Bayesian version by sampling–importance resampling. **The data are fetched from a URL and are not on disk**, and the script prints no coefficient table, so only the figures can be read:

- Page 3 of the PDF: the LM and GLM fitted values almost coincide over GRE 400–800 for both GPA groups and stay inside about 0.1–0.55. **When fitted probabilities stay in the middle of** $(0, 1)$ **the logistic curve is nearly linear** (slope $\approx \beta\pi(1-\pi)$ from (8) barely changes), so the LPM is a fair approximation there; the two separate only at the extreme (GRE 220: LM ≈ 0.06, GLM ≈ 0.11). In `default`, where most $\hat{\pi}$ are near 0, they disagree badly (§1).
- The horizontal line (sample admission rate) sits near 0.32, so about one third were admitted — the script's comment "2/3 admitted and 1/3 not admitted" has the shares reversed.
- Page 5 marks the MLE (blue dots) at roughly $\hat{\beta}_0 \approx -2.7$, $\hat{\beta}_{GRE} \approx 0.0027$, $\hat{\beta}_{GPA35} \approx 0.6$ — **figure readings, approximate**; the GPA odds ratio is then about $e^{0.6} \approx 1.8$.
- The prior/posterior pages are Bayesian logit (Classes 3+4).

Links

- Neighbours: L02-03-cross-validation-motorcycle (CV estimates the test error of a classifier as of a regression) · L02-05-discriminant-analysis (the generative route to the same posterior, cut-offs and costs) · L02-06-gaussian-vs-poisson-regression (same GLM/IRLS machinery, log link).
- Unit: L02-00-map · L02-01-prediction-error-bias-variance · L02-02-linear-log-linear-regression · L02-07-formulas · L02-08-pitfalls · L02-09-python (IRLS by hand with asserts).
- Class 1: L01 §1 (Bernoulli likelihood; iid vs logistic in `bernoulli-iid-logistic-R.txt`) · L01 §6 (OLS = MLE; IRLS is repeated weighted OLS).
- Later: Bayesian logit, ridge/lasso — Classes 3+4; probit and multinomial models — Class 7 (`Map/class-plan.md`).
- Rules pack: rules/06 (logistic regression, confusion matrix).
- **Practice** (statements in the books): I 4.1 (p. 193), I 4.9 (p. 195), I 4.12 (p. 196), I 4.13 (p. 196), I 4.14 (p. 197); W 17.C1 (p. 560).

Discriminant part, slides 40–65 · The Bayes rule, the Gaussian cut-offs, and what unequal costs do to them

> **Claim.** Discriminant analysis models $p(x \mid y = k)$ and turns it into $P(y = k \mid x)$ with Bayes' theorem; the **Bayes rule** allocates to the class with the largest $\pi_k p(x \mid y = k)$ and no rule has a smaller probability of error. With two classes everything runs through $h(x) = \log p_1(x) - \log p_2(x)$: allocate to 1 iff $h(x) > \log\{\pi_2 c_{12}/(\pi_1 c_{21})\}$. **Equal Gaussian variances make h linear** (one cut-off at the midpoint, LDA, Mahalanobis); **unequal variances make it quadratic** — in the deck's example class 1 wins *both* tails, with cut-offs 1.680 and 10.001 and a total error of 0.251 against 0.369 for the equal-variance case. Prev: L02-04-logistic-regression · Next: L02-06-gaussian-vs-poisson-regression · Map: L02-00-map

Notation.

- $y \in \{1, \dots, \kappa\}$: class label ($\kappa = 2$ unless stated); $x \in \mathbb{R}^p$: predictors.
- $\pi_k = P(y = k)$: prior probability, $\sum_k \pi_k = 1$ (here π_k is a **prior**, not the Bernoulli mean of note 04).
- $p_k(x) = p(x \mid y = k)$: class-conditional density (or pmf); μ_k, σ_k^2 (scalar) or μ_k, Σ / Σ_k (vector).
- R_k: allocation region of class k; $\phi_k(x) = 1\{x \in R_k\}$: allocation function.
- $h(x) = \log p_1(x) - \log p_2(x)$: discriminant function (two classes).
- $p_{ij} = \int \phi_i(x) p_j(x)\, dx$: probability of allocating to i an individual from j.
- $c_{ij} > 0$ $(i \neq j)$: cost of allocating to i when the truth is j; $c_{kk} = 0$.
- Φ: standard normal cdf; $\alpha = \Sigma^{-1}(\mu_1 - \mu_2)$, $\bar{\mu} = (\mu_1 + \mu_2)/2$.

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1 · U — two routes to the same posterior (slide 40)

Logistic regression (note 04) models $P(y = k \mid x)$ directly. **Discriminant analysis** models how x is distributed *within* each class and inverts with Bayes' theorem. When the classes are Gaussian the two give posteriors of the same form (§7.1), so why bother? Slide 40 (and I p. 146): the logit MLE is **unstable when the classes are well separated** (it may not exist — note 04 §3), **unstable when n is small**, and clumsy for $\kappa > 2$; DA has none of these problems, at the price of assuming a distribution for x.

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2 · C — discriminant rules and the Bayes rule (slides 41–42)

Discriminant rule (slide 41). A partition of $\mathbb{R}^p$ into disjoint regions $R_1, \dots, R_\kappa$ with $\bigcup_k R_k = \mathbb{R}^p$; allocate x to k iff $x \in R_k$. (The slide prints $\cap$; a partition needs the union.)

Bayes' theorem (slide 42; I eq. 4.15, p. 147).

$$P(y = k \mid x) = \frac{\pi_k\, p_k(x)}{\sum_{j=1}^{\kappa} \pi_j\, p_j(x)}. \tag{1}$$

Bayes discriminant rule. Allocate to

$$k^* = \arg\max_k P(y = k \mid x) = \arg\max_k \ \pi_k\, p_k(x), \tag{2}$$

the second form because the denominator of (1) is the same for every k. Equivalently $\phi_k(x) = 1$ iff $\pi_k p_k(x) = \max_j \pi_j p_j(x)$. **ML discriminant rule:** (2) with $\pi_1 = \cdots = \pi_\kappa = 1/\kappa$, i.e. $\arg\max_k p_k(x)$.

2.1 · C — why (2) minimises the probability of error (AME Teorema 9, p. 135)

Two classes. An error happens when $x \in R_2$ but $y = 1$, or $x \in R_1$ but $y = 2$:

$$P(\text{error}) = \pi_1 p_{21} + \pi_2 p_{12} = \pi_1 \int \phi_2(x) p_1(x)\, dx + \pi_2 \int \phi_1(x) p_2(x)\, dx.$$

Put both under one integral:

$$P(\text{error}) = \int [\phi_2(x)\, \pi_1 p_1(x) + \phi_1(x)\, \pi_2 p_2(x)] dx. \tag{3}$$

At each x exactly one of ϕ_1, ϕ_2 is 1, so the integrand is either $\pi_1 p_1(x)$ (if we allocate to 2) or $\pi_2 p_2(x)$ (if we allocate to 1). **The integral is smallest when, pointwise, we pay the smaller of the two** — allocate to 1 iff $\pi_1 p_1(x) > \pi_2 p_2(x)$, which is (2). AME writes the same rule as $P(Y = c_1 \mid x) \geq 1/2$ (p. 135); this is the 0–1-loss risk $R(g) = P(Y \neq g(X))$ of AME eq. 7.1 (p. 134).

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3 · C — two classes: the discriminant function (slide 49)

$$h(x) = \log p(x \mid y = 1) - \log p(x \mid y = 2). \tag{4}$$

ML rule: allocate to 1 if $h(x) > 0$, to 2 if $h(x) < 0$.

Bayes rule from (2). Allocate to 1 iff

$$\pi_1 p_1(x) > \pi_2 p_2(x) \iff \frac{p_1(x)}{p_2(x)} > \frac{\pi_2}{\pi_1} \iff h(x) > \log \frac{\pi_2}{\pi_1}, \tag{5}$$

dividing by $\pi_1 p_2(x) > 0$, then taking logs (log is increasing).

The posterior is a logistic function of h. Divide numerator and denominator of (1) (with $k = 1$) by $\pi_1 p_1(x)$:

$$P(y = 1 \mid x) = \frac{1}{1 + \dfrac{\pi_2 p_2(x)}{\pi_1 p_1(x)}} = \frac{1}{1 + \exp\{-h(x) - \log(\pi_1/\pi_2)\}} = g(h(x) + \log \tfrac{\pi_1}{\pi_2}), \tag{6}$$

with g the logistic function of note 04. Rule (5) is $P(y = 1 \mid x) > 1/2$ written on the h scale.

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4 · C — worked: the 0-1 predictor and the multinomial (slides 43–44)

Bernoulli (slide 43). $x \mid y=1 \sim \text{Bernoulli}(1/2)$, $x \mid y=2 \sim \text{Bernoulli}(3/4)$. By (4):

$$h(0)=\log\frac{1/2}{1/4}=\log 2=0.693>0 \;\Rightarrow\; \text{class 1}, \qquad h(1)=\log\frac{1/2}{3/4}=\log\tfrac{2}{3}=-0.405<0 \;\Rightarrow\; \text{class 2}.$$

ML misallocation probabilities: $p_{21}=P(x=1 \mid y=1)=1/2$, $p_{12}=P(x=0 \mid y=2)=1/4$; with equal priors $P(\text{error})=\frac{1}{2}(\frac{1}{2}+\frac{1}{4})=3/8$.

Priors move the rule (5). $x=0$ goes to 1 iff $\log 2>\log(\pi_2/\pi_1) \iff \pi_2/\pi_1<2 \iff \pi_1>1/3$; $x=1$ goes to 1 iff $\log\frac{2}{3}>\log(\pi_2/\pi_1) \iff \pi_1>3/5$. Hence

prior π_1	Bayes rule
$<1/3$	everything to class 2
1/3 to 3/5	the ML rule ($x=0\to 1$, $x=1\to 2$)
$>3/5$	everything to class 1

Multinomial (slide 44). $x \mid y=1 \sim \text{Mult}(n;\alpha_1,\dots,\alpha_\kappa)$, $x \mid y=2 \sim \text{Mult}(n;\beta_1,\dots,\beta_\kappa)$. The factor $n!/(x_1!\cdots x_\kappa!)$ is common and cancels in (4):

$$h(x)=\sum_{k=1}^{\kappa} x_k\log\alpha_k-\sum_{k=1}^{\kappa} x_k\log\beta_k=\sum_{k=1}^{\kappa} x_k\log\frac{\alpha_k}{\beta_k},$$

so the ML rule allocates to class 1 iff $\sum_k x_k\log(\alpha_k/\beta_k)>0$. **Slide 44 prints "$<0$" — a sign slip**: check with $\kappa=2$, $n=1$, $\alpha=(1/2,1/2)$, $\beta=(1/4,3/4)$, which is the Bernoulli example: $x=(1,0)$ gives $\log 2>0$ and must go to class 1.

---

5 · C — univariate Gaussians: linear or quadratic h (slides 45–47)

$x \mid y=k \sim N(\mu_k,\sigma_k^2)$, so $\log p_k(x)=-\log\sigma_k-\frac{1}{2}\log 2\pi-\dfrac{(x-\mu_k)^2}{2\sigma_k^2}$. Substitute into (4); the $\log 2\pi$ cancels:

$$h(x)=\log\frac{\sigma_2}{\sigma_1}-\frac{1}{2}\Big[\Big(\frac{x-\mu_1}{\sigma_1}\Big)^2-\Big(\frac{x-\mu_2}{\sigma_2}\Big)^2\Big]. \tag{7}$$

ML rule $h>0$. Move the log term and multiply by -2 (flip):

$$\Big(\frac{x-\mu_1}{\sigma_1}\Big)^2-\Big(\frac{x-\mu_2}{\sigma_2}\Big)^2<2\log\frac{\sigma_2}{\sigma_1}.$$

Expand both squares and collect powers of x — this is slide 45's display:

$$x^2\Big(\frac{1}{\sigma_1^2}-\frac{1}{\sigma_2^2}\Big)-2x\Big(\frac{\mu_1}{\sigma_1^2}-\frac{\mu_2}{\sigma_2^2}\Big)+\Big(\frac{\mu_1^2}{\sigma_1^2}-\frac{\mu_2^2}{\sigma_2^2}\Big)<2\log\frac{\sigma_2}{\sigma_1}. \tag{8}$$

5.1 · Equal variances: the midpoint (slide 45, last lines; I eqs. 4.18–4.19, p. 148)

With $\sigma_1 = \sigma_2 = \sigma$ the x^2 term and the log vanish in (8):

$$-\frac{2x(\mu_1 - \mu_2)}{\sigma^2} + \frac{\mu_1^2 - \mu_2^2}{\sigma^2} < 0 \iff (\mu_1 - \mu_2)(\mu_1 + \mu_2 - 2x) < 0.$$

With $\mu_1 < \mu_2$ the first factor is negative, so the second must be positive:

$$x < \frac{\mu_1 + \mu_2}{2} \qquad \Big(\text{equivalently } |x - \mu_2| > |x - \mu_1|\Big), \tag{9}$$

and (7) becomes **linear**: $h(x) = \dfrac{\mu_1 - \mu_2}{\sigma^2}\Big(x - \dfrac{\mu_1 + \mu_2}{2}\Big)$.

5.2 · Worked — the deck's example, $\mu_1 = 1,\ \mu_2 = 5,\ \sigma_1 = 6,\ \sigma_2 = 2.5$ (slide 46)

Write (7) as $h(x) = ax^2 + bx + c$:

coefficient	expression	value
a	$-\frac{1}{2}(1/36 - 1/6.25) =$ $-\frac{1}{2}(0.027778 - 0.16)$	0.066111
b	$\mu_1/\sigma_1^2 - \mu_2/\sigma_2^2 =$ $1/36 - 5/6.25 = 0.027778 - 0.8$	-0.772222
c	$-\frac{1}{2}(1/36 - 25/6.25) +$ $\log(2.5/6) = 1.986111 - 0.875469$	1.110642

Roots of $h(x) = 0$:

$$x = \frac{0.772222 \pm \sqrt{0.772222^2 - 4(0.066111)(1.110642)}}{2(0.066111)} = \frac{0.772222 \pm \sqrt{0.596327 - 0.293703}}{0.132222} = \frac{0.772222 \pm 0.550113}{0.132222},$$

$$x_L = \frac{0.222109}{0.132222} = \mathbf{1.680}, \qquad x_U = \frac{1.322335}{0.132222} = \mathbf{10.001}.$$

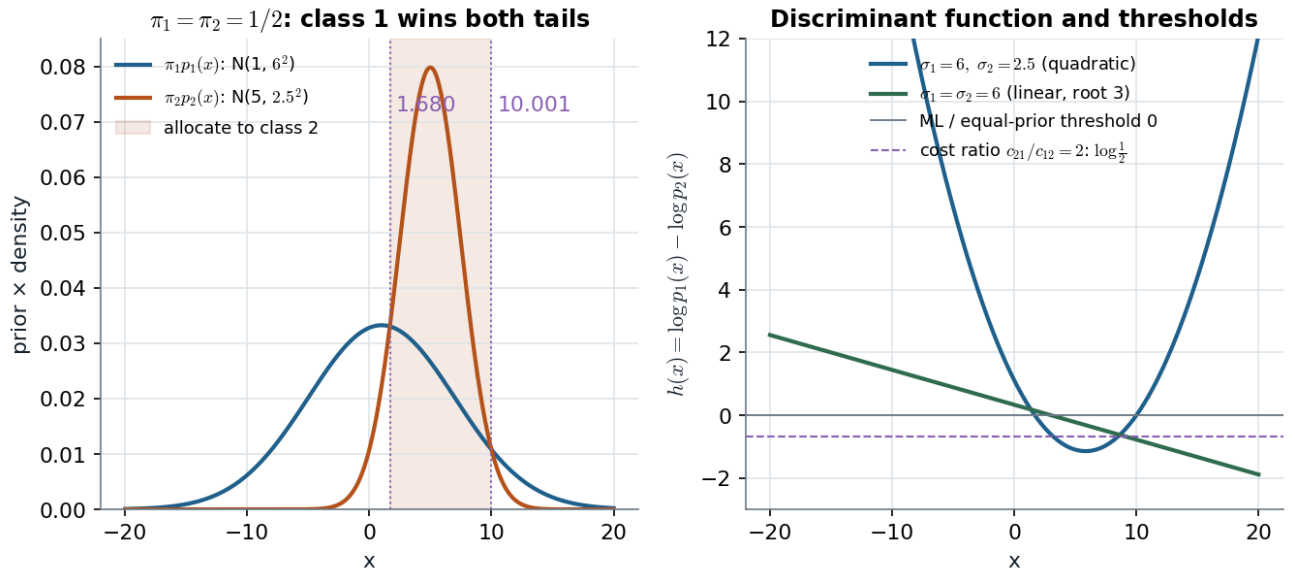
Because $a > 0$, h is a U: $h < 0$ (class 2) **between** the roots, $h > 0$ (class 1) **outside**:

$$R_1 = (-\infty, 1.680) \cup (10.001, \infty), \qquad R_2 = (1.680,\ 10.001).$$

Checks against slide 46's right panel: $h(-20) = 43.0$, $h(20) = 12.1$, minimum $h = -1.144$ at $x = -b/(2a) = 5.840$.

> **A large x, say 12 — seven units above $\mu_2 = 5$ — is allocated to class 1, whose mean is 1.** Class 1's larger σ gives it fatter tails, so far from both means it is the likelier source. That is what "quadratic" buys and what a single midpoint cut-off cannot do.

Equal variances, $\sigma_1 = \sigma_2 = 6$ (slide 47). By (9) the cut-off is $(1+5)/2 = 3$ and $h(x) = \frac{1-5}{36}(x-3) = -\frac{x-3}{9}$: $h(-20) = 23/9 = 2.556$, $h(20) = -17/9 = -1.889$, the slide's line.



Left: with equal priors class 2 owns only the band where its tall, narrow density is on top. Right: (7) for both cases; the dashed line is the cost-adjusted threshold of §10. Source: slides 45–47, eqs. (7)–(9).

On the left the two curves cross twice, and the shaded band is exactly R_2; outside it the wide class-1 density is higher. On the right the parabola dips below zero only between 1.680 and 10.001, and below the dashed line $\log\frac{1}{2}$ only on a narrower band — lowering the threshold shrinks R_2. The green line crosses zero once, at 3.

> **Interactive companion.** The Two-Gaussian Classifier (`companions/companion-gaussian-discriminant.html`) — move the class means, SDs, prior and cost ratio; watch the allocation regions, the posterior threshold and the total misclassification probability. Set $\sigma_1 = \sigma_2$ to see the two cut-offs collapse to the midpoint.

---

6 · C — probabilities of misallocation (slide 50) via Φ

$p_{ij} = \int \phi_i(x) p_j(x) dx$ is the probability that the region of class i catches an individual from j. For Gaussians, standardize and use Φ.

Equal variances ($\sigma = 6$, cut-off 3).

$$p_{21} = P(x > 3 \mid y = 1) = 1 - \Phi\Big(\frac{3-1}{6}\Big) = 1 - \Phi(0.3333) = 1 - 0.6306 = 0.3694,$$

$$p_{12} = P(x < 3 \mid y = 2) = \Phi\Big(\frac{3-5}{6}\Big) = \Phi(-0.3333) = 0.3694.$$

Total with equal priors: $\frac{1}{2}(0.3694 + 0.3694) = \mathbf{0.3694}$.

Unequal variances (cut-offs 1.680, 10.001).

$$p_{21} = P(1.680 < x < 10.001 \mid y = 1) = \Phi\Big(\frac{10.001-1}{6}\Big) - \Phi\Big(\frac{1.680-1}{6}\Big) = \Phi(1.5001) - \Phi(0.1133) = 0.93321 - 0.54510 =$$

$$p_{12} = \Phi\Big(\frac{1.680-5}{2.5}\Big) + 1 - \Phi\Big(\frac{10.001-5}{2.5}\Big) = \Phi(-1.3281) + 1 - \Phi(2.0003) = 0.09208 + 0.02273 = 0.1148.$$

Total with equal priors: $\frac{1}{2}(0.3881 + 0.1148) = \mathbf{0.2515}$.

> **The ML rule is not symmetric in its errors:** it misallocates 39% of class 1 but only 11% of the tight class 2. Equal *total* treatment of errors (0–1 loss) is a choice; §10 changes it.

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7 · C — p predictors, common covariance: LDA and the Mahalanobis distance (slide 48)

$x \mid y = k \sim N_p(\mu_k, \Sigma)$, Σ positive definite, so

$$\log p_k(x) = -\tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log|\Sigma| - \tfrac{1}{2}(x-\mu_k)'\Sigma^{-1}(x-\mu_k).$$

The first two terms do not depend on k, so the **ML rule** is

$$k^* = \arg\min_k (x-\mu_k)'\Sigma^{-1}(x-\mu_k), \tag{10}$$

the class with the smallest **squared Mahalanobis distance**.

Two classes: h is linear. From (4),

$$h(x) = -\tfrac{1}{2}\big[(x-\mu_1)'\Sigma^{-1}(x-\mu_1) - (x-\mu_2)'\Sigma^{-1}(x-\mu_2)\big].$$

Expand each quadratic form, using $\mu_k'\Sigma^{-1}x = x'\Sigma^{-1}\mu_k$ (Σ^{-1} symmetric):

$$(x-\mu_k)'\Sigma^{-1}(x-\mu_k) = x'\Sigma^{-1}x - 2\mu_k'\Sigma^{-1}x + \mu_k'\Sigma^{-1}\mu_k.$$

Substitute both: $x'\Sigma^{-1}x$ cancels,

$$h(x) = (\mu_1-\mu_2)'\Sigma^{-1}x - \tfrac{1}{2}(\mu_1'\Sigma^{-1}\mu_1 - \mu_2'\Sigma^{-1}\mu_2).$$

The bracket factors as $(\mu_1-\mu_2)'\Sigma^{-1}(\mu_1+\mu_2)$ (the cross terms $\mu_1'\Sigma^{-1}\mu_2 = \mu_2'\Sigma^{-1}\mu_1$ cancel), so

$$h(x) = (\mu_1-\mu_2)'\Sigma^{-1}\Big(x - \frac{\mu_1+\mu_2}{2}\Big) = \alpha'(x-\bar\mu), \qquad \alpha = \Sigma^{-1}(\mu_1-\mu_2),\ \bar\mu = \tfrac{\mu_1+\mu_2}{2}. \tag{11}$$

ML: allocate to 1 iff $\alpha'(x-\bar\mu) > 0$ (slide 48); Bayes: iff $\alpha'(x-\bar\mu) > \log(\pi_2/\pi_1)$. The boundary is a hyperplane. I writes the same rule as the largest $\delta_k(x) = x'\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k'\Sigma^{-1}\mu_k + \log\pi_k$ (eq. 4.24, p. 151); $p = 1$ is eq. 4.18 (p. 148). In practice μ_k, Σ, π_k are replaced by class means, the pooled covariance and n_k/n (I eqs. 4.20–4.21, p. 149; AME §8.1.4.1, p. 148) — a **plug-in** Bayes rule.

7.1 · C — LDA's posterior is a logistic regression

Substitute (11) into (6):

$$P(y=1 \mid x) = g(\alpha'(x-\bar{\mu}) + \log\tfrac{\pi_1}{\pi_2}) = g(\gamma_0 + \alpha' x), \qquad \gamma_0 = \log\tfrac{\pi_1}{\pi_2} - \alpha'\bar{\mu}.$$

Log-odds linear in x — the logit model (I eq. 4.32, p. 162; "identical to the linear form of LDA", p. 163). **Same functional form, different estimator:** LDA plugs in Gaussian MLEs of μ_k, Σ; logistic regression maximizes the conditional likelihood of y given x and does not need x to be Gaussian.

7.2 · Worked — iris on the two sepal measurements (slides 53–56)

Slide 54 prints the class means and ML covariances ($n_k = 50$ each):

	setosa ($k=1$)	versicolor ($k=2$)	virginica ($k=3$)
mean (length, width)	(5.006, 3.428)	(5.936, 2.770)	(6.588, 2.974)
var length, cov, var width	0.122, 0.097, 0.141	0.261, 0.083, 0.096	0.396, 0.092, 0.102

Pooled covariance (equal n_k: the average of the three):

$$\hat{\Sigma} = \begin{pmatrix} 0.25967 & 0.09067 \\ 0.09067 & 0.11300 \end{pmatrix}, \quad |\hat{\Sigma}| = 0.25967(0.113) - 0.09067^2 = 0.021122, \quad \hat{\Sigma}^{-1} = \begin{pmatrix} 5.3499 & -4.2925 \\ -4.2925 & 12.2937 \end{pmatrix}.$$

Flower 51 (versicolor, $x = (7.0, 3.2)$), distance to virginica: $d = x - \mu_3 = (0.412, 0.226)$,

$$d'\hat{\Sigma}^{-1}d = 5.3499(0.412)^2 + 2(-4.2925)(0.412)(0.226) + 12.2937(0.226)^2 = 0.9081 - 0.7994 + 0.6279 = 0.7367.$$

All three printed flowers (slide 54), squared distances from (10):

flower (true species)	x	to setosa	to versicolor	to virginica	ML allocation
1 (setosa)	(5.1, 3.5)	**0.053**	15.530	21.966	setosa ✓
51 (versicolor)	(7.0, 3.2)	25.814	4.402	**0.737**	virginica ✗
101 (virginica)	(6.3, 3.3)	10.581	**2.506**	2.556	versicolor ✗ (near tie)

Setosa vs versicolor in the form (11): $\hat{\alpha} = \hat{\Sigma}^{-1}(\mu_1 - \mu_2) = \hat{\Sigma}^{-1}(-0.930, 0.658)' = (-7.800, 12.081)'$, $\bar{\mu} = (5.471, 3.099)$; flower 1: $\hat{\alpha}'(x - \bar{\mu}) = -7.800(-0.371) + 12.081(0.401) = 2.894 + 4.844 = 7.738 > 0$ $\rightarrow$ setosa.

> **Sepals separate setosa and confuse the other two** — two of the three printed flowers land in the wrong class, in line with the partition plots' "app. error rate: 0.2" for both LDA and QDA (slides 56–57). (R's `lda` divides the pooled covariance by $n - \kappa$; rescaling Σ scales every distance by the same factor, so the equal-prior allocation is unchanged.)

8 · C — QDA: class-specific covariances (slide 48, last bullet; I §4.4.3, pp. 156–158)

With $x \mid y = k \sim N_p(\mu_k, \Sigma_k)$ the $\log|\Sigma_k|$ and $x'\Sigma_k^{-1}x$ terms no longer cancel:

$$h(x) = \frac{1}{2}\log\frac{|\Sigma_2|}{|\Sigma_1|} - \frac{1}{2}\big[(x-\mu_1)'\Sigma_1^{-1}(x-\mu_1) - (x-\mu_2)'\Sigma_2^{-1}(x-\mu_2)\big], \tag{12}$$

quadratic in x through $-\frac{1}{2}x'(\Sigma_1^{-1} - \Sigma_2^{-1})x$ (I eq. 4.28, p. 156). §5.2 is the $p = 1$ case. **Bias–variance:** LDA estimates κp means and one covariance, $p(p+1)/2$ parameters; QDA estimates $\kappa p(p+1)/2$ covariance parameters (I p. 157) — more flexible, noisier with small n.

Worked — the admission data (slides 58–65; Ledolter). 85 applicants, three classes; the priors printed by `lda` are the class shares $31/85 = 0.3647$, $26/85 = 0.3059$, $28/85 = 0.3294$. For an applicant with GPA 3.21 and GMAT 497:

	admit	border	notadmit	allocation
LDA posterior (slide 61)	0.5180	0.4816	0.0004	admit (near tie)
QDA posterior (slide 62)	0.9227	0.0769	0.0005	admit

Partition plots: apparent error 0.082 (LDA, slide 64) vs 0.035 (QDA, slide 65). **Both are training (resubstitution) errors**, optimistic by construction; the honest comparison is a CV estimate (L02-03-cross-validation-motorcycle). The "coefficients of linear discriminants" LD1/LD2 on slide 61 are Fisher's canonical directions, not α of (11) — named only. *(Data loaded from a URL, not on disk: these numbers are quoted, not recomputed.)*

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9 · C — admissibility (slides 50, 52)

Rule d is **as good as** d' if $p_{kk} \geq p'_{kk}$ for every k, **better** if at least one inequality is strict, **admissible** if no rule is better.

- **Theorem** (slides 50, 52): all Bayes rules, the ML rule included, are admissible.
- **Theorem:** under priors π_k, no rule has a larger posterior probability of correct allocation (smaller Bayes risk) than the Bayes rule for those priors — §2.1 is its proof for two classes.

Worked on the Bernoulli example (§4). The four possible rules:

rule ($x = 0 \to$, $x = 1 \to$)	p_{11}	p_{22}	status
(1, 2) — ML	1/2	3/4	admissible (Bayes for $1/3 < \pi_1 < 3/5$)
(2, 2)	0	1	admissible (Bayes for $\pi_1 < 1/3$)
(1, 1)	1	0	admissible (Bayes for $\pi_1 > 3/5$)

rule ($x=0\to$, $x=1\to$)	p_{11}	p_{22}	status
(2, 1) — reversed	1/2	1/4	**inadmissible**: ML has the same p_{11} and a larger p_{22}

Admissible is a weak property — "all to class 2" is admissible. It only rules out rules that are beaten everywhere; choosing among admissible rules needs priors and costs.

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10 · C — decision theory and unequal costs (slides 51–52)

Loss $K(i,j)=c_{ij}$ for allocating to i when the truth is j, $K(j,j)=0$. **Risk** of rule d for class k and **Bayes risk** (slide 51):

$$R(d,k)=\sum_{j=1}^{\kappa} c_{jk}\,p_{jk}, \qquad r(d,\pi)=\sum_{k=1}^{\kappa}\pi_k R(d,k). \tag{13}$$

Two classes. Only c_{21} (truth 1 sent to 2) and c_{12} (truth 2 sent to 1) survive:

$$r(d,\pi)=\pi_1 c_{21}p_{21}+\pi_2 c_{12}p_{12}=\int\left[\phi_2(x)\,\pi_1 c_{21}p_1(x)+\phi_1(x)\,\pi_2 c_{12}p_2(x)\right]dx,$$

the same structure as (3) with each term weighted by its cost. Minimise pointwise: allocate to 1 iff

$$\pi_1 c_{21}p_1(x)>\pi_2 c_{12}p_2(x) \iff h(x)>\log\frac{\pi_2\,c_{12}}{\pi_1\,c_{21}}. \tag{14}$$

On the posterior scale. Divide $\pi_1 c_{21}p_1>\pi_2 c_{12}p_2$ by $\sum_j \pi_j p_j(x)$ and use (1):

$$c_{21}P(y=1\mid x)>c_{12}\big[1-P(y=1\mid x)\big] \iff P(y=1\mid x)>\frac{c_{12}}{c_{12}+c_{21}}. \tag{15}$$

This is AME's cut-off $l_1/(l_0+l_1)$ (p. 170), with $l_1=c_{12}$ the cost of a false "1" and $l_0=c_{21}$ the cost of a missed "1". Equal costs give back 1/2 and (5).

Worked 1 — the deck's Gaussians, missing class 1 twice as costly ($c_{21}/c_{12}=2$, equal priors; the ratio is illustrative, the densities are slide 46's). Threshold (14): $\log(1/2)=-0.693$. Solve $h(x)=-0.693$ with the coefficients of §5.2 ($c\to 1.110642+0.693147=1.803789$): roots **3.228** and **8.453** — R_2 shrinks from $(1.680, 10.001)$ to $(3.228, 8.453)$.

rule	p_{21}	p_{12}	P(error)	expected cost $\frac{1}{2}(2p_{21})+\frac{1}{2}p_{12}$
ML (threshold 0)	0.3881	0.1148	**0.2515**	$0.3881+0.0574=0.4455$
cost rule (threshold $\log\frac{1}{2}$)	0.2481	0.3228	0.2855	$0.2481+0.1614=$ **0.4095**

The cost rule makes more errors and costs less — it trades cheap errors for expensive ones. Push the ratio past $e^{1.144} = 3.14$ (the depth of h's minimum) and the threshold sits below $\min h$: R_2 is empty and everything is allocated to class 1.

Worked 2 — I's `default` threshold (eqs. 4.26–4.27, pp. 153–154). Lowering the cut-off on $P(\text{default} \mid x)$ from 0.5 to 0.2 is (15) with $c_{12}/(c_{12}+c_{21}) = 0.2$, i.e. $0.8c_{12} = 0.2c_{21}$, **a missed defaulter costing 4 times a false alarm**. I's LDA tables:

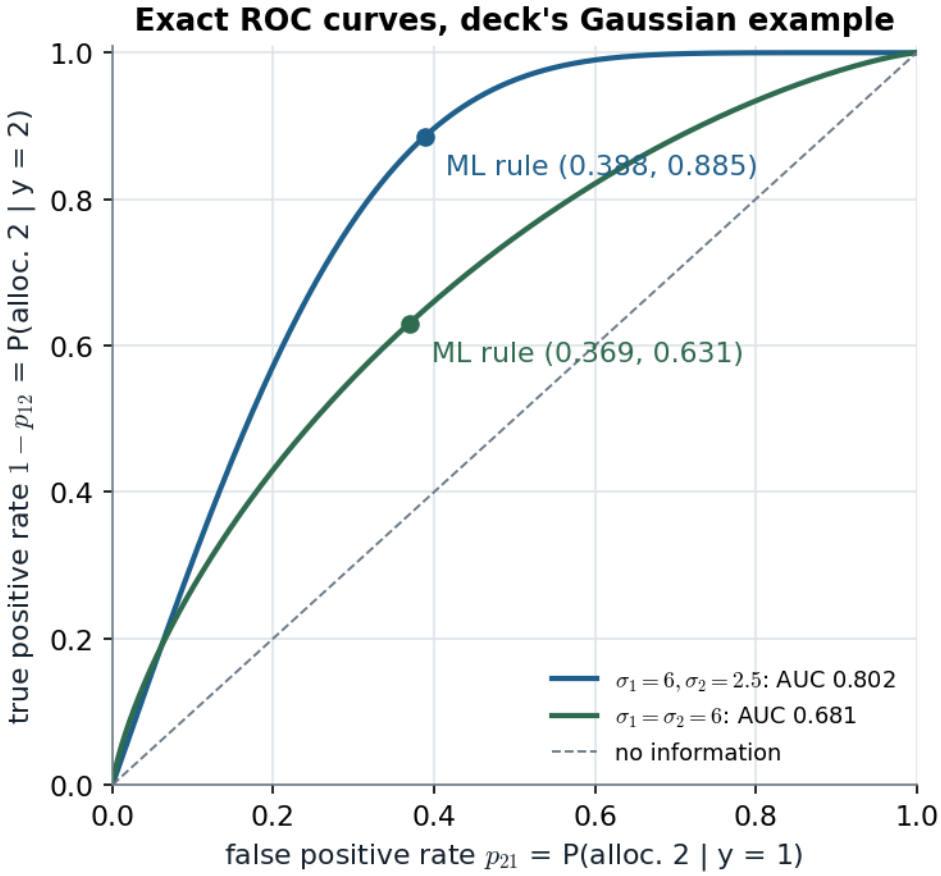
cut-off	missed defaulters	false alarms	error rate	cost at $c_{21} = 4, c_{12} = 1$
0.5 (Table 4.4, p. 152)	252 of 333 (75.7%)	23	2.75%	$4(252)+23 = 1031$
0.2 (Table 4.5, p. 154)	138 of 333 (41.4%)	235	3.73%	$4(138) + 235 =$ **787**

Slide 52's polio/flu sentence is the same idea: misdiagnosing polio as flu is the costly error, so the threshold for "polio" is lowered. For the spam filter of note 04 §9 the costly error is the false positive, so the spam cut-off should rise above 1/2.

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11 · C — the ROC curve (AME §9.1, pp. 169–170; I pp. 154–155)

Sweeping the threshold in (14) from $-\infty$ to ∞ traces every attainable pair (false-positive rate, true-positive rate); the **ROC curve** plots them, and the area under it (AUC) summarises all thresholds at once (I reports AUC 0.95 for LDA on `default`, p. 155). AME insists the curve be drawn on validation or test data (p. 170).



Class 2 is "positive": FPR $= p_{21}$*, TPR* $= 1 - p_{12}$*, each point a threshold* k *in "allocate to 2 iff* $h(x) < k$*". Exact, from (7) and* Φ*; no data.*

Both curves start at (0,0) (nothing allocated to 2) and end at (1,1) (everything). The unequal-variance curve lies above the equal-variance one almost everywhere: the different spreads are extra information the quadratic rule exploits. The dots are the ML rules of §6; moving along a curve is moving the cost ratio of §10.

G — the equal-variance AUC in closed form. With the score x itself, AUC $= P(x_{(2)} > x_{(1)})$ for independent draws from the two classes; $x_{(2)} - x_{(1)} \sim N(5 - 1,\ 36 + 36)$, so AUC $= \Phi(4/\sqrt{72}) = \Phi(0.4714) = 0.681$. The unequal-variance AUC (0.802) is computed numerically.

Links

- Neighbours: L02-04-logistic-regression (the discriminative route; plug-in cut-offs, confusion matrix) · L02-06-gaussian-vs-poisson-regression.
- Unit: L02-00-map · L02-01-prediction-error-bias-variance (bias–variance, here LDA vs QDA) · L02-02-linear-log-linear-regression · L02-03-cross-validation-motorcycle (estimate test error, not apparent error) · L02-07-formulas · L02-08-pitfalls · L02-09-python (LDA/QDA by hand with asserts).
- Class 1: L01 §1 (Bernoulli likelihoods behind §4) · L01 §6 (Gaussian log-density algebra).
- Companion: The Two-Gaussian Classifier (`companions/companion-gaussian-discriminant.html`).
- Later: priors and posterior probabilities as a full method — Classes 3+4 (`rules/07_bayes.md`).

- Rules pack: rules/06 (confusion matrix, ROC/AUC).
- **Practice** (statements in the books): I 4.12 (p. 196), I 4.13 (p. 196), I 4.14 (p. 197) — LDA/QDA on data; also I 4.2 and 4.3 (p. 193) — the LDA and QDA derivations of §5 and §8.

Gaussian-vs-Poisson-regression.html · Counts with a log link: the blog-visitor example, rate ratios and the mean–variance check

> **Claim.** A count is modelled as Poisson with $\log \lambda_i = x_i'\beta$, so a coefficient is a **rate ratio** e^{β_j}, fitted means are always positive, and the variance is tied to the mean. On the site's 28 days of blog visits, Gaussian and Poisson regressions on a weekend dummy give the **same fitted means** (109 and 83.33 visitors) because the model is saturated in two groups; they differ in the **variance they assume** — constant 140.4 vs equal to the mean — and therefore in their standard errors. The data are mildly overdispersed ($\hat{\sigma}^2 = 1.44$), which widens the Poisson SE by 20% and changes no conclusion. Prev: L02-05-discriminant-analysis · Next: L02-07-formulas · Map: L02-00-map

Notation.

- $y_i \in \{0, 1, 2, \dots\}$: visitors on day $i = 1, \dots, 28$ (June 2018).
- $w_i \in \{0, 1\}$: the page's `weekend` dummy, $w_i = 1$ for $t \leq 3$ (see §1).
- $\lambda_i = E(y_i \mid w_i) > 0$: Poisson mean; α, β: intercept and slope of $\log \lambda_i = \alpha + \beta w_i$ (the page's letters).
- β_0, β_1: OLS intercept and slope of the Gaussian model; s: its residual standard error.
- $n_0 = 16,\ S_0 = 1744$: number of weekdays and their total; $n_1 = 12,\ S_1 = 1000$: same for the weekend group.
- σ^2: dispersion factor in $\mathrm{Var}(y \mid x) = \sigma^2 E(y \mid x)$ (W eq. 17.35).

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1 · U — the data and the question

Visitors to a baseball-analytics blog, 28 consecutive days (the page's vector `y`):

```
95 81 85 100 111 130 113 | 92 65 78 96 118 120 104 | 91 91 79 106 91 114 110 | 98 61 84 96 126 119 90
```

The page labels the weekdays `fri, sat, sun, mon, tue, wed, thu` and sets `weekend = 1` for $t \leq 3$ — **Friday, Saturday and Sunday** (1 June 2018 was a Friday). Group summaries (page output, recomputed):

group	days	total	mean	sd	variance	variance / mean
weekday (Mon–Thu), $w = 0$	16	1744	**109.00**	12.15	147.73	1.36
"weekend" (Fri–Sun), $w = 1$	12	1000	**83.33**	11.42	130.42	1.57

The question: how much lower is traffic on weekend days, and how sure are we? A count is non-negative, integer, and typically noisier when its mean is higher — three things the Gaussian linear model ignores (I §4.6.1, pp. 167–169; W p. 543).

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2 · C — the Gaussian regression

$$y_i = \beta_0 + \beta_1 w_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma_\varepsilon^2) \text{ iid.} \tag{1}$$

With one dummy, OLS reproduces the group means. The normal equations $\sum_i (y_i - \beta_0 - \beta_1 w_i) = 0$ and $\sum_i w_i(y_i - \beta_0 - \beta_1 w_i) = 0$ give, subtracting the second from the first, $\sum_{w_i=0}(y_i - \beta_0) = 0$, so $\hat\beta_0 = S_0/n_0$; the second alone gives $\hat\beta_0 + \hat\beta_1 = S_1/n_1$:

$$\hat\beta_0 = \frac{1744}{16} = 109.000, \qquad \hat\beta_1 = \frac{1000}{12} - 109 = 83.333 - 109 = -25.667.$$

Standard errors (pooled residual variance, $28 - 2 = 26$ df). The page prints $s = 11.85$:

$$\mathrm{SE}(\hat\beta_1) = s\sqrt{\frac{1}{n_0} + \frac{1}{n_1}} = 11.8495\sqrt{\frac{1}{16} + \frac{1}{12}} = 11.8495\,(0.38188) = 4.525, \qquad \mathrm{SE}(\hat\beta_0) = \frac{s}{\sqrt{n_0}} = \frac{11.8495}{4} = 2.962.$$

$t = -25.667/4.525 = -5.67$ ($p = 5.8 \times 10^{-6}$), $R^2 = 0.553$ — all as printed. **Reading:** weekend days have 25.7 fewer visitors, in absolute terms, with one common spread $s^2 = 140.4$ for both groups.

---

3 · C — the Poisson regression

Model (I eqs. 4.35–4.37, pp. 169–170; W eqs. 17.31–17.32, p. 544):

$$y_i \mid w_i \sim \mathrm{Poisson}(\lambda_i), \qquad P(y_i = k) = \frac{e^{-\lambda_i}\lambda_i^k}{k!}, \qquad \log\lambda_i = \alpha + \beta w_i. \tag{2}$$

Log-likelihood (I eq. 4.38, p. 170; W eq. 17.33, p. 545). Take logs of the pmf and sum:

$$\ell(\alpha, \beta) = \sum_{i=1}^{28} \left[y_i \log\lambda_i - \lambda_i - \log y_i!\right] = \sum_{i=1}^{28} \left[y_i(\alpha + \beta w_i) - e^{\alpha+\beta w_i} - \log y_i!\right]. \tag{3}$$

Score. Differentiate (3); $\partial e^{\alpha+\beta w_i}/\partial\alpha = \lambda_i$ and $\partial e^{\alpha+\beta w_i}/\partial\beta = w_i\lambda_i$:

$$\frac{\partial\ell}{\partial\alpha} = \sum_i (y_i - \lambda_i) = 0, \qquad \frac{\partial\ell}{\partial\beta} = \sum_i w_i(y_i - \lambda_i) = 0. \tag{4}$$

In matrix form $X'(y - \lambda) = 0$ — **the same shape as the logit score** (note 04 eq. 12): both are canonical-link GLMs.

Solve (4) in closed form. The second equation involves only weekend days, where $\lambda_i = e^{\alpha+\beta}$:

$$S_1 - n_1 e^{\alpha+\beta} = 0 \;\Rightarrow\; e^{\hat\alpha+\hat\beta} = \frac{1000}{12} = 83.333.$$

Subtract it from the first; what remains are the weekdays, where $\lambda_i = e^\alpha$:

$$S_0 - n_0 e^{\alpha} = 0 \;\Rightarrow\; e^{\hat\alpha} = \frac{1744}{16} = 109.$$

Take logs:

$$\hat\alpha = \log 109 = \mathbf{4.69135}, \qquad \hat\beta = \log\frac{83.333}{109} = \log 0.764526 = \mathbf{-0.26850},$$

the page's estimates. With no covariate at all, (4) would give $\hat\lambda = \bar y$ — the Poisson MLE of L01 §2; here each group gets its own $\bar y$.

Hessian and standard errors. Differentiate (4) again:

$$-\frac{\partial^2 \ell}{\partial\theta\,\partial\theta'} = \sum_i \lambda_i \begin{pmatrix} 1 & w_i \\ w_i & w_i \end{pmatrix} = \begin{pmatrix} n_0 e^{\alpha} + n_1 e^{\alpha+\beta} & n_1 e^{\alpha+\beta} \\ n_1 e^{\alpha+\beta} & n_1 e^{\alpha+\beta} \end{pmatrix}.$$

At the MLE, $n_0 e^{\hat\alpha} = 1744$ and $n_1 e^{\hat\alpha+\hat\beta} = 1000$:

$$\hat I = \begin{pmatrix} 2744 & 1000 \\ 1000 & 1000 \end{pmatrix}, \qquad |\hat I| = 2744(1000) - 1000^2 = 1{,}744{,}000,$$

$$\hat I^{-1} = \frac{1}{1{,}744{,}000}\begin{pmatrix} 1000 & -1000 \\ -1000 & 2744 \end{pmatrix} \Rightarrow \mathrm{SE}(\hat\alpha) = \sqrt{\frac{1}{1744}} = 0.02395, \quad \mathrm{SE}(\hat\beta) = \sqrt{\frac{1}{1744} + \frac{1}{1000}} = 0.03967,$$

the printed 0.02395 and 0.03967: **each SE is built from the group totals alone**. $z = -0.26850/0.03967 = -6.77$. (The page's "Number of Fisher Scoring iterations: 4" is IRLS with $W = \mathrm{diag}(\lambda_i)$ and $z_i = \eta_i + (y_i - \lambda_i)/\lambda_i$ — note 04 §4 with the log link.)

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4 · C — reading the two slopes

Gaussian: additive, in visitors: $\hat\beta_1 = -25.667$ per day.

Poisson: multiplicative. From (2), $\lambda(w=1)/\lambda(w=0) = e^{\alpha+\beta}/e^{\alpha} = e^{\beta}$ — the **rate ratio** (I p. 171: "an increase in X_j by one unit is associated with a change in $E(Y)$ by a factor of $\exp(\beta_j)$"):

$$e^{\hat\beta} = e^{-0.26850} = 0.7645 \quad \Rightarrow \quad \text{weekend traffic is } 100(0.7645 - 1) = -23.5\% \text{ of weekday traffic.}$$

W's shortcut $100\hat\beta = -26.85\%$ (p. 544) is the log approximation; for a dummy the exact change is $100[\exp(\hat\beta) - 1] = -23.55\%$ (same page), and at $|\beta| \approx 0.27$ the two differ by 3.30 points.

Same means, different variance. The page's comparison table:

model	weekday mean	weekend mean
Gaussian: $\hat\beta_0$, $\hat\beta_0 + \hat\beta_1$	109.0000	83.333

model	weekday mean	weekend mean
Poisson: $e^{\hat{\alpha}}$, $e^{\hat{\alpha}+\hat{\beta}}$	109.0002	83.33345

(The fourth-decimal gaps come from exponentiating the five-decimal printed coefficients.) With **two groups and two parameters both models are saturated in the means**; what differs is the variance each assumes, hence $t = -5.67$ vs $z = -6.77$:

	Gaussian (1)	Poisson (2)
$\mathrm{Var}(y \mid w = 0)$	$s^2 = 140.4$	$\lambda = 109$
$\mathrm{Var}(y \mid w = 1)$	$s^2 = 140.4$	$\lambda = 83.3$
fitted values	can be negative in general	always > 0 (I p. 172)
slope reading	-25.7 visitors	$\times 0.7645$ (-23.5%)

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5 · C — the mean–variance check and overdispersion

The Poisson model says $\mathrm{Var}(y \mid x) = E(y \mid x)$ (W eq. 17.34, p. 545; I p. 172). The sample says 147.7 vs 109 and 130.4 vs 83.3 — ratios 1.36 and 1.57.

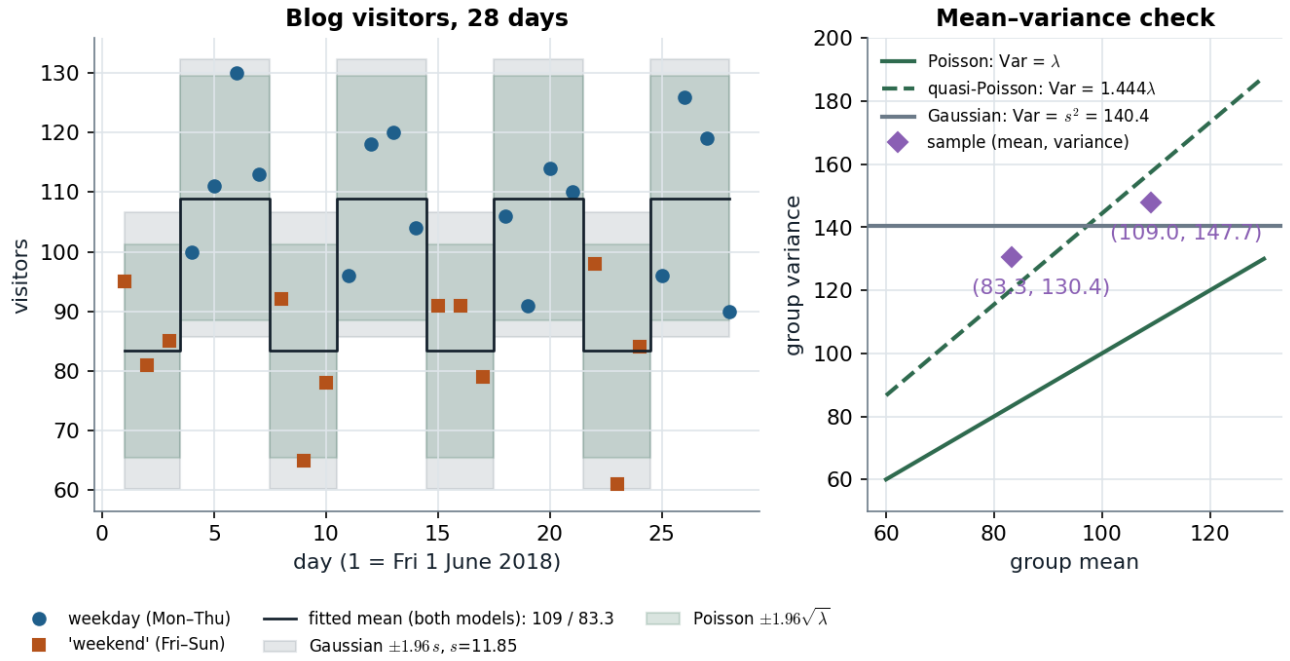
Dispersion estimate (W p. 546): $\hat{\sigma}^2 = \frac{1}{n-k-1}\sum_i \hat{u}_i^2/\hat{y}_i$ with $\hat{u}_i = y_i - \hat{\lambda}_i$, i.e. the Pearson χ^2 over its df:

$$\hat{\sigma}^2 = \frac{37.546}{26} = 1.444, \qquad \hat{\sigma} = 1.202.$$

Quasi-Poisson correction (W eq. 17.35 and p. 546): multiply the SEs by $\hat{\sigma}$:

$$\mathrm{SE}_Q(\hat{\beta}) = 1.202(0.03967) = 0.0477, \qquad z_Q = \frac{-0.26850}{0.0477} = -5.63.$$

Deviance view. Residual deviance 38.324 on 26 df (ratio 1.47); $P(\chi^2_{26} > 38.324) = 0.056$ — borderline evidence of lack of fit. The LR test of the weekend effect: $85.141 - 38.324 = 46.82$ on 1 df (W eq. 17.12; under (17.35) divide by $\hat{\sigma}^2$: $46.82/1.444 = 32.4$). AIC 221.76.



Left: both models fit the same step; the bands (±1.96 sd) differ. Right: the two groups' (mean, variance) against each model's variance function. Data and outputs: Gaussian-vs-Poisson-regression.html.

On the left the Gaussian band has the same width in both groups, while the Poisson band is narrower for the lower weekend mean ($\sqrt{83.3} = 9.1$ vs $\sqrt{109} = 10.4$); the weekday 130 and the weekend 65 and 61 sit outside the Poisson band (109 ± 20.5 and 83.3 ± 17.9) — three of 28 where about 1.4 are expected, which is the overdispersion. On the right both sample points lie **above** the Poisson line Var = mean and close to the quasi-Poisson line 1.444λ; the Gaussian constant 140.4 sits between them.

> **With means near 100 the Poisson and Gaussian models nearly agree** — a Poisson with a large mean is close to a normal. The log link earns its keep with **small counts**, where linear fits go negative (9.6% of I's Bikeshare fitted values, p. 168) and the variance visibly grows with the mean (I p. 172, recalling Fig. 4.14). **And Poisson SEs are only right if Var = mean**: always check, and inflate by $\hat{\sigma}$ when it fails (I footnote 5, p. 172: overdispersion "causes the Z-values to be inflated").

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6 · G — one template for linear, logistic and Poisson regression (I §4.6.3, pp. 172–173)

$$\eta(E(Y \mid X)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \qquad \text{(I eq. 4.42, p. 173)}$$

model	distribution	link $\eta(\mu)$	variance function	score	note
linear	Gaussian	μ	σ^2 (constant)	$X'(y - X\beta)/\sigma^2$	L01 §6
logistic	Bernoulli	$\log\{\mu/(1-\mu)\}$	$\mu(1-\mu)$	$X'(y - \pi)$	L02-04-logistic-regression
Poisson	Poisson	$\log \mu$	μ	$X'(y - \lambda)$	this note

Each link is the **canonical** one for its distribution, which is why the score is always "X'(data − fitted mean)" and why Newton–Raphson, Fisher scoring and IRLS coincide.

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7 · G — the page's Bayesian part, named only

The page continues with independent normal priors $\alpha \sim N(4.7, 0.1^2)$, $\beta \sim N(-0.3, 0.1^2)$, a grid posterior and sampling–importance resampling (SIR), ending with posterior densities of the weekday and weekend rates. **Classes 3+4** (`rules/07_bayes.md`, `rules/08_monte_carlo_mcmc.md`).

Links

- Neighbours: L02-05-discriminant-analysis · L02-07-formulas (next).
- Unit: L02-00-map · L02-01-prediction-error-bias-variance · L02-02-linear-log-linear-regression (log-level readings; here the log is on the *mean*, not on y) · L02-03-cross-validation-motorcycle · L02-04-logistic-regression (same score, same IRLS) · L02-08-pitfalls · L02-09-python (Poisson GLM via statsmodels with asserts).
- Class 1: L01 §2 (Poisson MLE $\hat{\lambda} = \bar{y}$, Fisher information n/λ — here $\mathrm{SE}(\hat{\alpha}) = 1/\sqrt{S_0}$ is the same object on the log scale) · L01 §3 (overdispersion test on counts).
- Later: GLMs again in Class 7 (`Map/class-plan.md`).
- Rules pack: rules/06.
- **Practice** (statements in the book): W 17.C4 (p. 561), W 17.C5 (p. 561).

Class 2 · Formulas — the exam specification mapped, every result, every number

Prev: L02-06-gaussian-vs-poisson-regression · Next: L02-08-pitfalls · Map: L02-00-map

Equation numbers are **the section notes' own**: "01 (21)" is equation (21) of L02-01-prediction-error-bias-variance. Unnumbered results are cited by section.

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0 · The exam specification, mapped

Three sources state what Class 2 covers: the site's class title, the site's class note on the motorcycle example (Aula/COURSE), and the class goals in Map/class-plan; the topic pack rules/06 adds the classification metrics. The exam has no code, so each item is "derive it, compute it by hand, read it".

#	Item (source)	Note, section	Be able to
1	*"Linear and log-linear regression modeling"* (site title)	L02-02-linear-log-linear-regression §2–§7	read a coefficient in level-level, log-level, log-log, level-log form; exact vs approximate % (11)/(12); interactions (6) and quadratics (21)–(22)
2	*"Training and testing samples"* (site title); training vs test error (plan)	L02-01-prediction-error-bias-variance §6; L02-03-cross-validation-motorcycle §1, §4	explain why training MSE (29) only falls; state its optimism (33)
3	*"Bias-variance trade-off"* (site title, plan)	L02-01-prediction-error-bias-variance §2–§5	prove (6), (15), (20), (21); say what each term is random over; read the tanh table
4	*"Validation"* (site title); validation set (plan)	L02-03-cross-validation-motorcycle §4	write (5); name its two drawbacks; read the 63-of-100 table
5	Replicated CV (plan; site class note)	L02-03-cross-validation-motorcycle §4, §6	explain why 100 hold-out splits disagree (63/100) and 100 ten-fold runs barely do (99/100)
6	LOOCV (plan; site class note)	L02-03-cross-validation-motorcycle §5	write (6); **derive the shortcut (7)** through (8)–(13); state when it fails; read (14)
7	10-fold CV (plan; site class note)	L02-03-cross-validation-motorcycle §6	write (15) and (16); explain the bias–variance ranking of hold-out, k-fold, LOOCV

#	Item (source)	Note, section	Be able to
8	Choose the polynomial order; "order 12 was the best" (site class note)	L02-03-cross-validation-motorcycle §2–§3, §6	read adjusted R^2 and BIC (3); explain that every criterion lands on 12 and what that does *not* prove
9	*"bootstrap algorithm to obtain approximate confidence bands"* (site class note)	L02-03-cross-validation-motorcycle §7	describe the pairs bootstrap, (17), the formula band (18), and why they differ under heteroskedasticity
10	Logistic regression (plan; syllabus topic 06)	L02-04-logistic-regression §2–§7	(1)–(8) readings; **derive score (12), Hessian (13) and IRLS (16)–(17)**; plug-in probabilities; cut-off (19); deviance and LR tests
11	Performance metrics for classification (syllabus; rules/06: confusion matrix, ROC, AUC)	L02-04-logistic-regression §9; L02-05-discriminant-analysis §6, §11	compute error, FPR, FNR, sensitivity, specificity, precision, FDR from a table; misallocation probabilities by Φ; read a ROC curve
12	Discriminant analysis (plan)	L02-05-discriminant-analysis §2–§10	Bayes and ML rules (2); $h(x)$ (4)–(8); the midpoint (9); the two Gaussian cut-offs; LDA (10)–(11); QDA (12); costs (14)–(15)
13	Gaussian vs Poisson regression for counts (plan)	L02-06-gaussian-vs-poisson-regression §2–§5	write (2)–(4), solve the dummy case in closed form, compute SEs from the information, read e^{β}, check Var = mean, apply the $\hat{\sigma}$ correction

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1 · Every definition and result

1.1 · Prediction error and bias–variance → L02-01-prediction-error-bias-variance

Eq.	Statement	Derived in
(1)	expected prediction error $E[(Y-\varphi(X))^2]$	§2

Eq.	Statement	Derived in
(2)–(5)	condition on X; minimise $g(t) = E[(Y-t)^2 \mid X = x]$ pointwise: $g'(t) = 0 \iff t = E[Y \mid X = x]$, $g'' = 2 > 0$	§2
(6)	$\arg\min_\varphi E[(Y - \varphi(X))^2] = \psi(x) = E[Y \mid X = x]$ — **the regression function**	§2
(7)–(9)	$R_{\text{pred}}(g) = R_{\text{reg}}(g) + E[V[Y \mid X]]$ (AME Teorema 1): predicting Y and estimating ψ are one problem	§2
(10)	additive model $Y = \psi(X) + \varepsilon$, $E\varepsilon = 0$, $\operatorname{Var}\varepsilon = \sigma^2$, all independent	§3
(15)	$E[(Y - \widehat{\psi}(x))^2 \mid X = x] = \text{EQM}[\widehat{\psi}(x)] + \sigma^2$ — reducible + irreducible	§3
(16)	integrated over F_X	§3
(17)	Viés $= E\widehat{\psi}(x) - \psi(x)$; Var $= E[(\widehat{\psi}(x) - E\widehat{\psi}(x))^2]$, over training samples	§4
(20)	$\text{EQM} = \text{Var} + \text{Viés}^2$	§4
(21)	$E[(Y - \widehat{\psi}(X))^2] = \int \text{Viés}^2 \, dF_X + \int \text{Var} \, dF_X + \sigma^2$	§4
(23)–(24)	straight line on $1 - x + 2x^2$, $x \sim U(0,1)$: $\int \text{Viés}^2 = 1/45$; $\overline{\text{Var}} = \sigma^2 \cdot 2/n'$ (design held fixed)	§4.1
(25)	the tanh world: $\psi(x) = 45\tanh(x/1.9 - 7) + 57$, $X \sim U(8,18)$, $\sigma = 4$, $n = 30$	§5
(27)	Monte Carlo estimators of bias² and variance (slide 25)	§5
(28)	linear smoother: $E\widehat{\psi}(x_0) = b(x_0)'(B'B)^{-1}B'\psi$, $\text{Var} = \sigma^2 b(x_0)'(B'B)^{-1}b(x_0)$	§5
(29)	training MSE $\frac{1}{n}\sum(y_i - \widehat{\psi}(x_i))^2$	§6
(32)	$E[\text{EQM}_{\text{train}}] = \|(I-H)\psi\|^2/n + \sigma^2(1 - \frac{d+1}{n})$ (G)	§6
(33)	optimism $= 2\sigma^2(d+1)/n$ — the C_p/AIC penalty (G)	§6

1.2 · Coefficient readings → L02-02-linear-log-linear-regression

Eq.	Statement	Derived in
(1)	level-level $\Delta y = \beta_j \Delta x_j$; rescaling x by c divides the slope by c	§2
(4)	$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$	§2.1
(6)	interaction: $\Delta y/\Delta x_2 = \beta_2 + \beta_3 x_1$ — β_2 alone is the slope at $x_1 = 0$	§3
(8)	dummy × continuous: two slopes β_2, $\beta_2 + \beta_3$; gap $\beta_1 + \beta_3$ yrs	§3
(9)–(10)	log-level: $\log y_1 - \log y_0 = \beta_1 \Delta x$	§4
(11)	**exact** $\%\Delta y = 100(e^{\beta_1 \Delta x} - 1)$ (W 6.8)	§4
(12)	approximate $\%\Delta y \approx 100\beta_1 \Delta x$ (W 2.43); error $\approx 100z^2/2$	§4
(13)–(15)	log-log: β_1 is the elasticity; $\%\Delta y \approx \beta_1 \%\Delta x$; exact $100[(1 + \%\Delta x/100)^{\beta_1} - 1]$	§5
W Table 2.3	level-log: $\Delta y = (\beta_1/100)\%\Delta x$	§5
(18)	standardised coefficient: $\partial \log y/\partial x_j = \beta_j^* s_y/s_{x_j}$	§6
(21)–(22)	quadratic slope $\hat{\beta}_1 + 2\hat{\beta}_2 x$; turning point $\lvert\hat{\beta}_1/(2\hat{\beta}_2)\rvert$	§7

1.3 · Validation, CV and the bootstrap → L02-03-cross-validation-motorcycle

Eq.	Statement	Derived in
(1)	polynomial regression is linear in β	§2
(2)	standardising $ax + b$ ($a > 0$) returns the standardised x	§2
(3)	$\bar{R}^2_d$; $\text{BIC}_d = \frac{1}{n}(\text{RSS}_d + \log(n)(d+1)\hat{\sigma}^2_d)$	§3
(5)	validation-set estimate $\frac{1}{\lvert\mathcal{V}\rvert}\sum_{\mathcal{V}}(y_i - \hat{f}_{\mathcal{T}}(x_i))^2$	§4
(6)	$CV_{(n)} = \frac{1}{n}\sum(y_i - \hat{y}_{(i)})^2$	§5
(7)	$CV_{(n)} = \frac{1}{n}\sum\left(\frac{y_i - \hat{y}_i}{1-h_{ii}}\right)^2$ **for least squares** (ISLP 5.2)	§5
(8)–(9)	drop row i: $A - uu'$; Sherman–Morrison	§5
(12)	$\hat{\beta}_{(i)} = \hat{\beta} - A^{-1}x_i e_i/(1 - h_{ii})$	§5
(13)	left-out residual $= e_i/(1 - h_{ii})$	§5
(14)	$\text{Var}[CV_{(n)}]$: n variances plus $n(n-1)$ large positive covariances	§5
(15)	$CV_{(k)} = \frac{1}{k}\sum_j \text{MSE}_j$ (ISLP 5.3)	§6

Eq.	Statement	Derived in
(16)	weighted $\widetilde{CV}_{(k)} = \sum_j \frac{n_j}{n}\text{MSE}_j$ (the site's code); equals (15) iff $n_j = n/k$	§6
(17)	bootstrap SE $\sqrt{\frac{1}{B-1}\sum_r (\hat{f}^{*r}(x) - \bar{f}^*(x))^2}$ (ISLP 5.8)	§7
(18)	formula band $\hat{f}(x_0) \pm 2\sqrt{\ell_0' \hat{C} \ell_0}$, $\hat{C} = \hat{\sigma}^2 (X'X)^{-1}$ (ISLP 7.2)	§7
—	$\sum_i h_{ii} = d + 1$: average leverage $(d+1)/n$	§5

1.4 · Logistic regression → L02-04-logistic-regression

Eq.	Statement	Derived in
(1)	$\pi_i = e^{x_i'\beta}/(1 + e^{x_i'\beta}) = g(x_i'\beta)$	§2
(2)–(4)	$1 - \pi_i = 1/(1 + e^{\eta_i})$; odds $= e^{\eta_i}$; **logit** $= x_i'\beta$	§2
(5)	GLM: $E(y_i \mid x_i) = \pi_i$, $V = \pi_i(1 - \pi_i)$, link = logit	§2
(6)	odds ratio e^{β_j} per unit of x_j, the same at every x	§2.1
(7)–(8)	$g' = g(1 - g)$; partial effect $\beta_j \pi_i (1 - \pi_i) \le \beta_j/4$	§2.1
(9)–(11)	likelihood; $\ell(\beta) = \sum [y_i x_i'\beta - \log(1 + e^{x_i'\beta})]$	§3
(12)	**score** $X'(y - \pi)$; the intercept row gives mean $\hat{\pi}$ = share of ones	§3
(13)	**Hessian** $-X'WX$, $W = \text{diag}\{\pi_i(1 - \pi_i)\}$: strictly concave	§3
(14)–(15)	Newton–Raphson $\beta^{\text{new}} = \beta^{\text{old}} + (X'WX)^{-1}X'(y - \pi)$	§4
(16)–(17)	**IRLS:** $\beta^{\mathbf{new}} = (X'WX)^{-1}X'Wz$**,** $z_i = \eta_i + (y_i - \pi_i)/\{\pi_i(1 - \pi_i)\}$	§4
(18)	the WLS objective $(z - X\beta)'W(z - X\beta)$	§4
—	$\widehat{\text{Var}}(\hat{\beta}) = (X'\hat{W}X)^{-1}$; $z = \hat{\beta}_j/\text{SE}$	§4
(19)	probability cut-off $c \leftrightarrow$ covariate cut-off $x^* = (\text{logit}\, c - \hat{\beta}_0)/\hat{\beta}_1$	§5.2
—	$LR =$ difference of deviances $\sim \chi^2_{\text{df}}$; McFadden $1 - \ell_{ur}/\ell_0$	§5.3

Eq.	Statement	Derived in
(20)–(21)	plug-in vs posterior predictive (21 is Classes 3+4)	§6
(22)–(23)	deviance residual; for 0/1 data $D = -2\ell(\widehat{\beta})$	§7
—	FNR $= FN/(FN+TP)$, FPR $= FP/(FP+TN)$, sensitivity, specificity, precision, **FDR** $= FP/(FP+TP)$	§9

1.5 · Discriminant analysis → L02-05-discriminant-analysis

Eq.	Statement	Derived in
(1)	$P(y=k \mid x) = \pi_k p_k(x) / \sum_j \pi_j p_j(x)$	§2
(2)	**Bayes rule** $\arg\max_k \pi_k p_k(x)$; ML rule: equal priors	§2
(3)	$P(\text{error}) = \int[\phi_2\pi_1 p_1 + \phi_1\pi_2 p_2]dx$ — minimised pointwise by (2) (AME Teorema 9)	§2.1
(4)–(5)	$h(x) = \log p_1 - \log p_2$; allocate to 1 iff $h(x) > \log(\pi_2/\pi_1)$	§3
(6)	$P(y=1 \mid x) = g(h(x) + \log\frac{\pi_1}{\pi_2})$	§3
(7)–(8)	univariate Gaussians: h quadratic, coefficients in μ_k, σ_k	§5
(9)	equal variances: allocate to 1 iff $x < (\mu_1+\mu_2)/2$ (when $\mu_1 < \mu_2$)	§5.1
—	p_{ij} by Φ over the regions	§6
(10)	common Σ: ML rule = smallest Mahalanobis distance	§7
(11)	$h(x) = \alpha'(x - \bar{\mu})$, $\alpha = \Sigma^{-1}(\mu_1 - \mu_2)$ — linear	§7
—	LDA posterior $g(\gamma_0 + \alpha' x)$: the logit form, a different estimator	§7.1
(12)	QDA: $\frac{1}{2}\log\frac{\lvert\Sigma_2\rvert}{\lvert\Sigma_1\rvert}$ plus two class-specific quadratic forms	§8
—	admissibility; all Bayes rules (ML included) are admissible	§9
(13)	risk $R(d,k) = \sum_j c_{jk} p_{jk}$; Bayes risk $\sum_k \pi_k R(d,k)$	§10
(14)	**allocate to 1 iff** $h(x) > \log\{\pi_2 c_{12}/(\pi_1 c_{21})\}$	§10
(15)	on the posterior scale: $P(y=1 \mid x) > c_{12}/(c_{12}+c_{21})$	§10

Eq.	Statement	Derived in
—	equal-variance AUC $= \Phi((\mu_2 - \mu_1)/\sqrt{2\sigma^2})$ (G)	§11

1.6 · Gaussian vs Poisson regression → L02-06-gaussian-vs-poisson-regression

Eq.	Statement	Derived in
(1)	Gaussian: $y_i = \beta_0 + \beta_1 w_i + \varepsilon_i$; with one dummy OLS = group means	§2
(2)	Poisson: $\log \lambda_i = \alpha + \beta w_i$	§3
(3)	$\ell = \sum[y_i(\alpha+\beta w_i) - e^{\alpha+\beta w_i} - \log y_i!]$	§3
(4)	score $X'(y-\lambda) = 0$ — the logit's shape; closed form $e^{\hat\alpha} = S_0/n_0$, $e^{\hat\alpha+\hat\beta} = S_1/n_1$	§3
—	information $\begin{pmatrix} S_0+S_1 & S_1 \\ S_1 & S_1 \end{pmatrix}$: $\mathrm{SE}(\hat\alpha) = 1/\sqrt{S_0}$, $\mathrm{SE}(\hat\beta) = \sqrt{1/S_0 + 1/S_1}$	§3
—	rate ratio e^β; exact % $100(e^\beta - 1)$	§4
—	$\mathrm{Var}(y \mid x) = E(y \mid x)$; $\hat\sigma^2 = \frac{1}{n-k-1}\sum \hat u_i^2/\hat y_i$; quasi-Poisson SE $\times\hat\sigma$ (W 17.35)	§5
I (4.42)	GLM template $\eta(E[Y \mid X]) = X'\beta$ (G)	§6

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2 · Every number and worked case

2.1 · Note 01

Case	Inputs	Result
tanh landmarks (26)	$x = 8$, 13.3, 18	$\psi = 12.339,\ 57,\ 101.365$; slope at 13.3: $45/1.9 = 23.68$
exact (21), polynomial OLS, 30 equispaced points, $\sigma = 4$	$d = 1, 3, 5, 6, 7, 10, 15$	test 91.83, 25.95, **19.81**, 20.03, 19.91, 21.38, 26.20; bias² 74.80, 7.98, 0.94, 0.70, 0.11, 0.007, 0.000; variance 1.03, 1.97, 2.87, 3.33, 3.80, 5.37, 10.20; training 93.87, 22.39, 13.79, 13.01, 11.85, 10.14, 7.47
the optimum		$d^* = 5$; $0.94 + 2.87 + 16 = 19.81$; degrees 5–7 within 0.25

Case	Inputs	Result
the slide's spline (slide 27)		about 7 df; prediction error bottoms near 20, floor 16
optimism (33)	$d = 5$, $n = 30$, $\sigma^2 = 16$	$2(16)(6)/30 = 6.4$
line on $1 - x + 2x^2$ (§4.1)	$n' = 900$, $\sigma = 0.1$	bias² $1/45 = 0.0222$; variance $0.01 \cdot 2/900 = 0.0000222$ — $1000\times$ smaller

2.2 · Note 02

Case	Result
W units (2.39, 2.41)	slope 18.501 on `roe` → 1,850.1 on `roedec`
Credit (2), slide 44	$t_{\text{Student}} = 429.6/16.61 = 25.86$; $t_{\text{Cards}} = 6.37$; $\bar{R}^2 = 0.9531$
houseprice (5), slides 17, 29	1.886008 per m² → $0.092903 \times 1.886008 = 0.17522$ thousand R$ per sqft = USD 54.75
houseprice interactions, slide 29	size effect 1.9375 / 2.2282 (brick) / 1.5760 (N2) / 1.8668 (both)
W Ex. 2.10, educ	8.3% approx., **8.65%** exact
W (6.7), rooms ±1	30.6% approx.; **+35.8% / −26.4%** exact
W (6.7), stratio +1 / +5	−5.2% / −26% approx.; −5.1% / −22.9% exact
W Ex. 2.11, sales +10%	2.57% approx., **2.48%** exact
unemployment 8 → 9	1 point, 12.5%, 0.118 log points
Mroz page (19)–(20)	RSS $= 735 \times 0.4511^2 = 149.57$; $R^2 = 0.8011$; $\bar{R}^2 = 0.7962$; $t_{\text{MTR}} = -21.64$; MLE SE factor $\sqrt{735/753} = 0.98798$
W (6.12), experience	turning point $0.298/0.0122 = 24.4$ years
W Ex. 6.2, rooms	turning point 4.4 rooms; +7.5% (5 → 6), +19.9% (6 → 7)

2.3 · Note 03

Case	Result
adjusted R^2 by order (slide 4)	0.0769 (1), 0.1323 (2), 0.5259 (5), 0.6941 (8), 0.769 (10), 0.7686 (11), **0.7798 (12)**, 0.7688 (20)
$\hat{\sigma}^2$ at $d = 12$ (4)	$1 - 0.7798 = 0.2202$; $\hat{\sigma} = 0.4693$
100 hold-out splits (slide 7)	wins: 6:1, 8:2, 9:2, 10:15, 11:7, **12:63**, 13:10; orders 10–13: 95
10-fold × 100 (slide 10)	**12: 99**, 13: 1
LOOCV minimum (slide 8)	$d = 12$, about 0.48 on the "Root MSE" axis
leverage	mean $(d+1)/n$: $13/132 = 0.0985$ ($d = 12$), $4/132 = 0.0303$ ($d = 3$)
the site's folds	$\lfloor 132/10 \rfloor = 13$: nine folds of 13, the last of 15
fit sizes	hold-out ≈ 105, 10-fold ≈ 119, LOOCV 131
bootstrap (slide 11)	$B = 1000$ pairs resamples of the order-12 fit

Case	Result
stand-in (simulated, **not** mcycle)	at $d = 12$: training 0.1737, hold-out 0.2220, LOOCV 0.2082, 10-fold 0.2117; 10-fold picks 12 in 82/100 runs, hold-out spreads (14: 31, 12: 30, 10: 23, 13: 10)

2.4 · Note 04

Case	Result
`default` counts	9,667 no / 333 yes; mean $\hat{\pi} = 0.0333$
LPM (slide 13)	$-0.07519 + 0.0001299x$: negative below 578.8; 0.185 at 2000
balance logit (slides 13, 15)	$-10.6513 + 0.0055x$ (SE 0.3612, 0.0002204; $z = -29.49$, 24.95); **0.58%** at 1000, **58.6%** at 2000
odds ratio, partial effects	$e^{0.55} = 1.733$ per USD 100; $\partial\pi/\partial x = 0.001375$ at the cut-off, 0.0000315 at 1000 (44×)
cut-offs (19), slide 17	$c = 0.1, \ldots, 0.9$: 1537, 1684, 1782, 1863, **1936.6**, 2010, 2091, 2189, 2336
student table (slide 18)	6850 / 2817 / 206 / 127; IRLS → **−3.50413, 0.40489** (SE 0.07071, 0.11502) in 6 iterations; $-2\ell = 2908.683$; $\hat{P}$ **4.31%** / **2.92%**; OR 1.499
balance + student (slide 20)	$-10.75 + 0.005738x - 0.7149s$; **5.4%** (student) vs **10.5%** at 1500; cut-offs **1998.1** (student), **1873.5**; OR $e^{-0.7149} = 0.489$
deviances	null 2920.6; 1596.5 / 2908.7 / 1571.7; AIC 1600.5 / 2912.7 / 1577.7; LR 1324.1, 11.9, 24.8; McFadden 0.453, 0.004, 0.462
spam `glm` (slides 36–38)	TN 1325, FP 80, FN 93, TP 803: error 7.52%, FNR 10.38%, FPR 5.69%, sensitivity 89.62%, specificity 94.31%, precision 90.94%, **FDR 9.06%** (slide's column 8.20%)
spam, other rows	ridge 8.17 / 14.29 / 4.27 / FDR 7.25; enet **6.52** / 10.49 / **3.99** / 6.53; lasso 6.65 / 10.60 / 4.13 / 6.75; bayes 7.39 / 10.16 / 5.62 / 8.94
admission script	about one third admitted; MLE read off the figure ≈ (−2.7, 0.0027, 0.6) (approximate)

2.5 · Note 05

Case	Result
Bernoulli (slide 43)	$h(0) = \log 2 = 0.693$, $h(1) = \log \frac{2}{3} = -0.405$; ML error 3/8; Bayes rule switches at $\pi_1 = 1/3$ and 3/5
Gaussian, $\mu = (1, 5)$, $\sigma = (6, 2.5)$ (slide 46)	$h = 0.066111x^2 - 0.772222x + 1.110642$; cut-offs **1.680, 10.001**; $h(-20) = 43.0$, $h(20) = 12.1$; min −1.144 at 5.840

Case	Result
misallocation (slide 50)	$p_{21} = 0.3881$, $p_{12} = 0.1148$, total **0.2515**; equal $\sigma = 6$: cut-off 3, $p_{21} = p_{12} =$ **0.3694**
unequal costs, $c_{21}/c_{12} = 2$	threshold $\log \frac{1}{2}$; cut-offs 3.228, 8.453; error 0.2855; expected cost 0.4095 vs 0.4455 for ML; ratio $> e^{1.144} = 3.14$ empties R_2
ISLP `default` LDA, cut-off 0.5 → 0.2	missed 252 → 138 of 333; false alarms 23 → 235; error 2.75% → 3.73%; cost at 4:1, 1031 → 787
iris, sepals (slide 54)	$\lvert\hat{\Sigma}\rvert = 0.021122$; flower 51 → virginica (0.737), flower 101 → versicolor (2.506 vs 2.556), flower 1 → setosa; $\hat{\alpha} = (-7.800, 12.081)$, score 7.738
admission LDA/QDA (slides 58–65)	priors 0.3647, 0.3059, 0.3294; LDA posterior admit 0.5180, QDA 0.9227; apparent error 0.082 vs 0.035
ROC	AUC 0.802 (unequal), $0.681 = \Phi(0.4714)$ (equal); ML points (0.388, 0.885), (0.369, 0.631)

2.6 · Note 06

Case	Result
groups	weekday (Mon–Thu) 16 days, total 1744, mean **109.00**, var 147.73; "weekend" (Fri–Sun) 12 days, total 1000, mean **83.33**, var 130.42
Gaussian	$\hat{\beta}_0 = 109$, $\hat{\beta}_1 = -25.667$ (SE 4.525, $t = -5.67$); $s = 11.85$, $s^2 = 140.4$; $R^2 = 0.553$
Poisson	$\hat{\alpha} = \log 109 =$ **4.69135**, $\hat{\beta} =$ **−0.26850**; SE 0.02395, 0.03967; $z = -6.77$
rate ratio	$e^{-0.26850} = 0.7645$: **−23.5%**; $100\hat{\beta} = -26.85\%$
fit and dispersion	deviance 38.324 on 26 df ($p = 0.056$); null 85.141; LR 46.82; AIC 221.76; Pearson 37.546; $\hat{\sigma}^2 = 1.444$, $\hat{\sigma} = 1.202$
quasi-Poisson	SE 0.0477, $z = -5.63$; LR $46.82/1.444 = 32.4$

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3 · Where these results come back

Result	Next appearance
(21), MSE = variance + bias²	ridge/lasso and shrinkage priors (Classes 3+4); large BVAR (Class 9)
the logit likelihood (9)–(13), IRLS	Bayesian logit (Classes 3+4); probit, ordered and multinomial probit (Class 7)
the posterior predictive, note 04 (21)	Classes 3+4
Bayes rule and loss, note 05 (13)–(15)	Bayes estimators and loss functions (Classes 3+4)
the Poisson GLM and overdispersion	zero-inflated Poisson (Classes 3+4); GLMs in Class 7
confounding (the student sign flip)	omitted-variable bias, Classes 5+6

Links

Map: L02-00-map · Pitfalls: L02-08-pitfalls · Code: L02-09-python · Class 1 formulas: L01-09 · Rules: rules/06.

Class 2 · Pitfalls — fifteen confusions, and a 30-item True/False self-test

Prev: L02-07-formulas · Next: L02-09-python · Map: L02-00-map

Every counterexample uses a number already derived in a section note; the few new ones are checked in `L02_python_checks.py` (section 7, L02-09-python §7).

Part I — the confusions

1 · Training error vs test error

The confusion: the model with the smaller training error predicts better.

Counterexample (tanh world, L02-01-prediction-error-bias-variance §5). Degree 15: expected training MSE **7.47**, expected test error **26.20**. Degree 5: training **13.79**, test **19.81**. Training ranks 15 above 5; test ranks them the other way. Degree 15 even trains *below* the noise floor $\sigma^2 = 16$ — it has reproduced noise.

> **Training error falls with every added term (nested least squares cannot do worse in-sample); test error is U-shaped. The gap is $2\sigma^2(d+1)/n$ on average** (note 01 (33)).

2 · Bias–variance of a learner vs bias of an estimator

The confusion: "OLS is unbiased, so its bias term in (21) is zero."

Counterexample (note 01 §4.1). Fit a straight line to $\psi(x) = 1 - x + 2x^2$. OLS *is* unbiased — for the coefficients of the best **linear** approximation, $\frac{2}{3} + x$. The bias in (21) is measured against ψ itself: $-2(x^2 - x + \frac{1}{6})$, integrated **0.0222**, a thousand times the variance. And the "variance" in (21) is the spread of $\hat{\psi}(x)$ **across training samples**, not $\mathrm{Var}(Y)$ — that is σ^2.

> **"Unbiased" always names a target. Gauss–Markov's target is β of the fitted model; (21)'s target is $E[Y \mid X = x]$.**

3 · LOOCV vs k-fold: which has less bias, which less variance

The confusion: "LOOCV uses $n-1$ points and involves no random split, so it is the best estimate on every count."

Counterexample (note 03 (14)). Bias: yes, LOOCV's fits are the closest to the full fit. Variance: its n fits share $n-2$ of their $n-1$ points, so the errors EQM_i are strongly positively correlated; in the extreme of perfect correlation, (14) collapses to $\mathrm{Var}[CV_{(n)}] = \mathrm{Var}[\mathrm{EQM}_1]$ — **averaging n terms buys nothing.** Deterministic given the data does not mean precise: redraw the data and $CV_{(n)}$ moves.

> **Bias falls hold-out → 10-fold → LOOCV; variance does not** (note 03 §6 table). Cost too: one fit for OLS by (7), n fits for a logit.

4 · CV picks a model, not a truth

The confusion: "10-fold CV chose order 12 in 99 of 100 runs, so the motorcycle curve is a 12th-degree polynomial."

Counterexample (note 03 §8). The stand-in data's mean is a sum of two Gaussian bumps — **no polynomial at all** — and CV picks order 12 there too. CV ranks the *candidates* by estimated test error; if the truth is not among them, it still returns a winner. ISLP (p. 290) warns that $d > 4$ gives "very strange shapes", especially at the boundary — where the order-12 band flares.

5 · Selecting the order, then reporting its in-sample fit

The confusion: "Order 12 has adjusted $R^2 = 0.7798$; that is how well it will predict."

Counterexample. Regress pure noise on a 12-term polynomial with $n = 132$: the expected R^2 is $k/(n-1) = 12/131 = \mathbf{0.0916}$ although the true R^2 is 0 (checked by simulation, 4,000 draws). Pick the best of many candidates and the winner's in-sample fit is biased upward again — adjusted R^2 corrects for the parameters of *one* model, not for the search. The same holds for the smallest of 14 CV estimates (note 03 §9) and for the bootstrap band, which holds $d = 12$ fixed in every resample (§7).

> **The data that chose the model cannot also grade it.** Keep an untouched test set, or nest the CV.

6 · Log points vs percent

The confusion: "log(price) has coefficient 0.306 on rooms: one more room, +30.6%."

Counterexample (note 02 §4.1). Exactly, $100(e^{0.306}-1) = \mathbf{+35.8\%}$; one *fewer* room, $100(e^{-0.306}-1) = \mathbf{-26.4\%}$. The 30.6 is a first-order approximation, five points off here. And a **percentage point** is not a percent: unemployment $8 \to 9$ is one point, **12.5%**, and $\log 9 - \log 8 = \mathbf{0.118}$ log points.

7 · Odds vs probability

The confusion: "An odds ratio of 1.733 means the probability rises by 73%."

Counterexample (note 04 §2.1). At the `default` cut-off, $\hat{\pi} = 0.5$, odds 1. USD 100 more balance multiplies the odds by $e^{0.55} = 1.733$: odds 1.733, probability $1.733/2.733 = \mathbf{0.634}$ — a 27% rise, not 73%. The two coincide only when π is small: on the student table the probability ratio is $4.31/2.92 = 1.478$ against an odds ratio of 1.499.

8 · A logit coefficient vs a marginal effect

The confusion: "$\hat{\beta}_{\text{balance}} = 0.0055$: each dollar adds 0.0055 to the default probability."

Counterexample (note 04 (8)). $\partial\pi/\partial x = \beta\pi(1-\pi)$: **0.001375** per dollar at the cut-off, **0.0000315** at a balance of 1,000 — 44 times apart, and both far below 0.0055. The coefficient is constant on the **log-odds** scale only.

9 · The 0.5 cut-off is not cost-neutral

The confusion: "Classify at 0.5; any other cut-off is arbitrary."

Counterexample (note 05 (15), §10). 0.5 *is* a cost choice: $c_{12}/(c_{12}+c_{21}) = \frac{1}{2}$ means a missed defaulter costs the same as a false alarm. If a missed defaulter costs four times as much, the Bayes cut-off is $1/(1+4) = 0.2$, and ISLP's LDA tables give costs **1031** at 0.5 against **787** at 0.2.

10 · Accuracy under class imbalance

The confusion: "The rule with the lower error rate is the better classifier."

Counterexample (note 05 §10, ISLP `default`). "Nobody defaults" has error **3.33%** (333/10,000) and catches no defaulter. LDA at 0.2 has error **3.73%** — *higher* — and catches 195 of 333. At a 4:1 cost the trivial rule costs $4 \times 333 = 1332$, the LDA rule 787.

> **With 3% positives, the error rate is dominated by the negatives.** Read the confusion matrix (note 04 §9), or the ROC curve (note 05 §11), not one number.

11 · FDR vs FPR

The confusion: "The false-positive rate is the share of flagged e-mails that are not spam."

Counterexample (spam `glm`, note 04 §9). FPR $= FP/(FP+TN) = 80/1405 =$ **5.69%** — of the ham, how much was flagged. FDR $= FP/(FP+TP) = 80/883 =$ **9.06%** — of the flagged, how much was ham. Same numerator, different denominators; FDR > FPR here because fewer e-mails were flagged (883) than were ham (1405). The deck's column, $FP/(FP+TP+FN) = 8.20\%$, is neither.

12 · LDA vs logistic regression

The confusion: "LDA's posterior is a logistic function of x (note 05 §7.1), so LDA and logistic regression are the same method."

Counterexample. Same **form**, different **estimator**: LDA plugs Gaussian MLEs of μ_k, Σ into Bayes' theorem; the logit maximises the conditional likelihood of y. On perfectly separated data the logit MLE **does not exist** — IRLS runs off to infinity (note 04 §3; the check in L02-09-python §4) — while LDA returns a finite rule. With $x =$ the `student` dummy, LDA's Gaussian assumption is plainly false, while the logit is exact (saturated, note 04 §4.1).

13 · ML rule vs Bayes rule

The confusion: "Allocate to the class under which the observation is most likely."

Counterexample (note 05 §4). Bernoulli example with prior $\pi_1 = 0.25$. The ML rule ($x = 0 \to 1$, $x = 1 \to 2$) errs with probability $0.25(\frac{1}{2}) + 0.75(\frac{1}{4}) =$ **0.3125**. Since $\pi_1 < \frac{1}{3}$, the Bayes rule sends **everything** to class 2 and errs only **0.25** of the time. The likelihood ratio must clear the **prior odds** (5), not 1.

14 · Gaussian vs Poisson regression for counts

The confusion: "On the blog data both models give the means 109 and 83.33, so it does not matter which one is fitted."

Counterexample (note 06 §4). The means coincide because the model is saturated in two groups; the **variance** each model assumes does not: constant 140.4 vs equal to the mean. Hence $t = -5.67$ vs $z = -6.77$ for the same effect. With a continuous covariate and small counts the Gaussian line also goes negative (9.6% of ISLP's Bikeshare fits, p. 168); the log link cannot.

15 · Overdispersion

The confusion (two-sided): "Var > mean makes the Poisson coefficients wrong" — or — "Poisson standard errors are fine whatever the variance."

Counterexample (note 06 §5). The blog data have variance/mean 1.36 and 1.57, $\hat{\sigma}^2 = 1.444$. Quasi-Poisson leaves $\hat{\beta} = -0.26850$ **unchanged** (the mean model is still right, the score equations (4) are the same) but multiplies the SE by $\hat{\sigma} = 1.202$: z falls from -6.77 to **-5.63**. Point estimates survive; precision was overstated.

Part II — True/False self-test (original items)

[E] easy · **[M]** medium · **[P]** hard. Scoring as in rules/00: $\max\{0,\ \text{right} - \text{wrong}/2\}$ per block. Numbers marked "illustrative" are invented for the item; all others come from the notes.

Block A — prediction error and bias–variance

1. **[E]** Under squared loss, among all functions of X, the conditional mean $E[Y \mid X]$ has the smallest expected prediction error, whatever the joint distribution of (X, Y) (with finite second moments). **True.** Conditioning on X turns the problem into minimising $g(t) = E[(Y - t)^2 \mid X = x]$ separately at each x; $g'(t) = -2E[Y \mid X = x] + 2t = 0$ at the conditional mean and $g'' = 2 > 0$ (note 01 (2)–(6)). No distributional assumption enters beyond the existence of the moments. Under absolute loss the answer would be the conditional median instead.

2. **[M]** A straight line fitted by OLS to a curved regression function contributes no squared-bias term to (21), because OLS is unbiased. **False.** OLS is unbiased for the coefficients of the best *linear* approximation, not for $\psi(x)$. On $\psi(x) = 1 - x + 2x^2$ with $x \sim U(0, 1)$ the line's bias is $-2(x^2 - x + \frac{1}{6})$ and its integrated square is $1/45 = 0.0222$, more than twice the noise variance 0.01 (note 01 §4.1).

3. **[M]** If method A has a lower expected training MSE than method B, A also has a lower expected test error. **False.** This is the reciprocal the training error invites. In the tanh world the degree-15 polynomial trains at 7.47 against 13.79 for degree 5, yet its test error is 26.20 against 19.81 (note 01 §5). The training error rewards variance; the test error charges for it.

4. **[M]** In the decomposition (21), the variance term measures how much Y varies around $\psi(X)$. **False.** That is the irreducible term σ^2, random over the new ε. The variance term is $\mathrm{Var}[\hat{\psi}(x)]$, random over the **training sample**: how much the learned curve moves when the data are redrawn (note 01 §4 table). A perfectly noise-free world ($\sigma = 0$) can still have a positive variance term if the design is random.

5. **[P]** For OLS with $d + 1$ coefficients on a fixed design and homoskedastic noise, the expected training MSE understates the expected error on new responses at the same design points by exactly $2\sigma^2(d+1)/n$, whatever the true regression function. **True.** The residual vector is $(I - H)(\psi + \varepsilon)$; the ψ part $\|(I-H)\psi\|^2/n$ appears in both expectations and cancels, leaving $\sigma^2(d+1)/n + \sigma^2 - \sigma^2(1-(d+1)/n) = 2\sigma^2(d+1)/n$ (note 01 (30)–(33)). At $d = 5$, $n = 30$, $\sigma = 4$: 6.4. This is the penalty C_p and AIC add back.

6. **[M]** Doubling the training sample lowers the irreducible error. **False.** σ^2 is the variance of the new noise ε, which no sample size touches; the floor of the tanh curves stays at 16 for every n. A larger n lowers the **variance** term (roughly in proportion to $(d + 1)/n$ for polynomial OLS), which is what moves the next item's optimum.

7. **[M]** In the tanh world with $\sigma = 4$, raising the training size from $n = 30$ to $n = 200$ cannot lower the test-optimal polynomial degree. **True.** Larger n shrinks every degree's variance while leaving bias^2 nearly unchanged, so flexibility gets cheaper; the exact computation gives $d^* = 5$ at $n = 30$ and $d^* = 7$ at $n = 60$ and at $n = 200$ (L02-09-python §7). The Flexibility Dial shows the optimum sliding right as n grows.

Block B — validation and cross-validation

8. [**M**] Although LOOCV involves no random split, it can have a larger variance as an estimator of the test error than 10-fold CV. **True.** Given the data LOOCV is deterministic, but as a function of a random sample it has a sampling variance, and its n nearly identical fits make the EQM_i strongly positively correlated, so the covariance term of (14) dominates (note 03 §5; ISLP §5.1.4). 10-fold's fits overlap less.

9. [**E**] For a least-squares fit, LOOCV can be computed from a single fit on all n points. **True.** $CV_{(n)} = \frac{1}{n}\sum(e_i/(1-h_{ii}))^2$ (note 03 (7)), derived by Sherman–Morrison: the left-out residual is the ordinary residual inflated by $1/(1-h_{ii})$ (13).

10. [**M**] The same one-fit shortcut gives the exact LOOCV of a logistic regression fitted by IRLS. **False.** The derivation uses that $\widehat{\beta}$ is linear in y through $(X'X)^{-1}X'y$; a logit's $\widehat{\beta}$ is the fixed point of a re-weighted iteration whose weights change when a point leaves. LOOCV for the logit (or LDA) needs n refits (ISLP p. 205; note 03 §5).

11. [**M**] An 80/20 hold-out estimate tends to overstate the test error of the model refitted on all n observations. **True.** It is computed from a fit on about 80% of the data — 105 points of 132 for the motorcycle — and a fit on fewer points is noisier, so its error is higher than that of the full-sample fit (ISLP p. 208; note 03 §4, §6 table).

12. [**M**] A fintech's analyst picks the polynomial degree that maximises adjusted R^2 among degrees 1–20 and reports that adjusted R^2 as the model's out-of-sample fit; since adjusted R^2 penalises the number of parameters, the report is unbiased. **False.** The penalty corrects for the parameters of one fixed model, not for choosing the best of twenty on the same data. Under pure noise a 12-term polynomial with $n = 132$ already has $E[R^2] = 12/131 = 0.0916$; taking the maximum over candidates pushes the reported figure up further (pitfall 5). An untouched test set or nested CV is needed.

13. [**E**] Since 10-fold CV picked order 12 in 99 of 100 repetitions, the true regression function of the motorcycle data is a polynomial of degree 12. **False.** CV ranks the candidates by estimated test error; it says nothing about whether any candidate is the truth. On the stand-in data, whose mean is a sum of Gaussian bumps, CV also picks 12 (note 03 §8). The 99/100 measures the stability of the choice, not its truth.

14. [**P**] The pairs-bootstrap band drawn around the order-12 fit includes the uncertainty of having chosen the order from the same data. **False.** Every resample refits order 12; the selection step is outside the loop. Putting it inside — re-running CV in each resample — would widen the band (note 03 §7, the "what the band does not cover" box).

15. [**M**] On data whose noise variance changes with x, the pairs-bootstrap band and the formula band $\widehat{f}(x_0) \pm 2\sqrt{\ell_0'\widehat{C}\ell_0}$ will generally differ. **True.** The formula uses one $\widehat{\sigma}^2$ for every x; resampling pairs carries the quiet and the noisy regions in their own proportions. On the stand-in the bootstrap band is wider in the noisy region; on homoskedastic data the two agree to within 10% (L02-09-python §3).

Block C — logistic regression and classification metrics

16. [**E**] In a logit, e^{β_j} is the factor that multiplies the odds when x_j rises by one unit, the same at every value of the covariates. **True.** The odds are e^{η} (note 04 (3)); raising x_j by one adds β_j to η, so the odds go from e^{η} to $e^{\eta+\beta_j}$ — ratio e^{β_j} whatever η was (6).

17. [**M**] In a logit, β_j is the change in $P(y=1)$ per unit of x_j. **False.** The probability effect is $\beta_j\pi(1-\pi)$, largest $(\beta_j/4)$ at $\pi = \frac{1}{2}$ (note 04 (8)). For `default`, 0.001375 per dollar at the cut-off and

0.0000315 at a balance of 1,000, neither equal to 0.0055.

18. [M] A positive coefficient on `student` alone (0.405) and a negative one given `balance` (-0.715) can both be correct. **True.** They answer different questions: students default more *overall* because they carry larger balances; *at the same balance* they default less. Balance confounds the marginal comparison (note 04 §5.3) — the sign flip is evidence of confounding, not of an error.

19. [P] At the MLE of a logit that includes an intercept, the average fitted probability equals the sample share of ones. **True.** The score is $X'(y-\pi)$ (note 04 (12)); its intercept row is $\sum_i (y_i - \hat{\pi}_i) = 0$, so $\frac{1}{n}\sum \hat{\pi}_i = \bar{y}$ — 0.0333 in `default`. It holds for any canonical-link GLM with an intercept (the Poisson version: fitted total = observed total, note 06 (4)).

20. [M] Lowering the probability cut-off of a credit model from 0.5 to 0.2 lowers its overall error rate. **False.** ISLP's LDA on `default`: error 2.75% at 0.5, **3.73%** at 0.2 — 212 more false alarms buy 114 fewer missed defaulters (note 05 §10). The lower cut-off is right when a missed defaulter costs about four times a false alarm (cost 1031 → 787), not because errors fall.

21. [M] A digital bank in São Paulo has 3.33% defaulters in its portfolio. A rule that predicts "no default" for every client has a lower error rate (3.33%) than an LDA rule with cut-off 0.2 (3.73%), so the trivial rule is the better credit tool. **False.** The trivial rule catches none of the defaulters; at any cost ratio above the one that makes the two equal, it loses — at 4:1, $4 \times 333 = 1332$ against 787 (pitfall 10). With 3% positives the error rate is dominated by the negatives and cannot rank credit rules.

22. [M] For the spam `glm`, the false discovery rate exceeds the false-positive rate because fewer e-mails were flagged as spam than were actually ham. **True.** Both have numerator FP = 80; FDR divides by the flagged $FP+TP = 883$, FPR by the ham $FP+TN = 1405$. FDR > FPR exactly when $883 < 1405$: **9.06% vs 5.69%** (note 04 §9).

23. [P] For ungrouped 0/1 data, a residual deviance of 1596.5 on 9998 degrees of freedom shows a good fit, because it is far below its χ^2_{9998} expectation. **False.** With one observation per covariate pattern the deviance does not have a χ^2_{n-p} reference distribution; it equals $-2\ell(\hat{\beta})$ and depends on the $\hat{\pi}_i$ alone (note 04 (23), §7). Only **differences** of deviances between nested models are safely referred to χ^2 — e.g. $1596.5 - 1571.7 = 24.8$ on 1 df for `student`.

Block D — discriminant analysis

24. [E] With equal priors and equal misallocation costs, the Bayes discriminant rule coincides with the ML rule. **True.** The threshold in (14) of note 05 is $\log\{\pi_2 c_{12}/(\pi_1 c_{21})\} = \log 1 = 0$, which is the ML rule $h(x) > 0$.

25. [M] Two univariate Gaussian classes with unequal variances can produce a Bayes rule that allocates **both** tails of the real line to the same class. **True.** h is then quadratic (note 05 (8)); in the deck's example ($\mu = (1, 5)$, $\sigma = (6, 2.5)$) it is a U with roots 1.680 and 10.001, so class 1 — the wider one — takes $(-\infty, 1.680)$ and $(10.001, \infty)$. A single midpoint arises only when $\sigma_1 = \sigma_2$.

26. [M] In that example, an observation at $x = 12$ is allocated to class 2 because it is closer to $\mu_2 = 5$ than to $\mu_1 = 1$. **False.** $h(12) = 0.066111(144) - 0.772222(12) + 1.110642 = 1.364 > 0$: class 1. Far from both means the density with the fatter tails is the larger one; distance to the mean decides only when the variances are equal (note 05 §5.2).

27. [P] A public hospital in Recife screens for a disease (class 1) with the deck's two Gaussians and equal priors (illustrative). Raising the cost of a missed case far enough sends every patient to class 1. **True.** The threshold $\log(c_{12}/c_{21})$ falls as c_{21} rises; once $c_{21}/c_{12} > e^{1.144} = 3.14$ it sits below $\min_x h(x) = -1.144$, so $h(x)$ exceeds it everywhere and R_2 is empty (note 05 §10).

28. [M] Since every Bayes rule is admissible, an admissible rule is a sensible rule. **False** — the reciprocal fails. "Allocate everything to class 2" is admissible in the Bernoulli example (it is the Bayes rule for $\pi_1 < 1/3$) but ignores the data. Admissibility only excludes rules beaten everywhere, such as the reversed rule with $p_{11} = \frac{1}{2}$, $p_{22} = \frac{1}{4}$ (note 05 §9).

Block E — counts

29. [M] Because the Gaussian and Poisson regressions on the weekend dummy give the same fitted means (109 and 83.33), they attach the same statistical significance to the weekend effect. **False.** Same means, different variance functions: constant $s^2 = 140.4$ vs Var $= \lambda$. The Gaussian t is -5.67, the Poisson z is -6.77, and the quasi-Poisson z is -5.63 (note 06 §4–§5).

30. [M] Overdispersion (Var > mean) makes the Poisson estimate of the weekend coefficient inconsistent. **False.** The score equations $X'(y - \lambda) = 0$ only need the **mean** to be right; with $\hat{\sigma}^2 = 1.444$ the quasi-Poisson fit returns the same $\widehat{\beta} = -0.26850$. What overdispersion breaks is the **standard error** — too small by the factor $\hat{\sigma} = 1.202$ (note 06 §5; pitfall 15).

Answer key at a glance: T — 1, 5, 7, 8, 9, 11, 15, 16, 18, 19, 22, 24, 25, 27 (14 items); F — 2, 3, 4, 6, 10, 12, 13, 14, 17, 20, 21, 23, 26, 28, 29, 30 (16 items).

Links

Map: L02-00-map · Formulas: L02-07-formulas · Code: L02-09-python · Class 1 pitfalls: L01-10 · Grading conventions: rules/00 · Rules: rules/06.

Class 2 · Python — the reference implementation, asserted against the sources

> Every routine the class needs, each followed by the `asserts` that tie it to a printed number. numpy, scipy and statsmodels only — **no scikit-learn**. The lecturer's R is the reference, never the code. **The file is the source of truth:** `Notas/L02/L02_python_checks.py`; the blocks below are copied from it verbatim, section by section. Prev: L02-08-pitfalls · Map: L02-00-map · Formulas: L02-07-formulas

Run: `python Notas/L02/L02_python_checks.py` — about ten seconds; it prints one line per section and ends with `ALL L02 CHECKS PASSED` (§8 below shows the run). The first disagreement aborts with an `AssertionError` naming the quantity and the printed value.

Reused, not rewritten.

Existing code	What it supplies	Used in
`figures/make_figures.py` (imported as `mf`)	`psi`, `poly_bias_variance` (exact (28) on a QR/Legendre basis), `mcycle_standin`, `design`, `ols_predict`, `loocv_brute`, `loocv_shortcut`, `kfold_cv`, `h_roots`, the 28 visitor counts `Y`	§1, §3, §5, §6, §7
`companions/check_companions.py`	the Flexibility Dial and the Two-Gaussian Classifier recomputed and compared with the pages' JavaScript under `node` at three control vectors, plus a Monte Carlo of bias² and variance	cited, not imported: it checks the **pages**; this file checks the **notes**

What is new here, and why: the IRLS routine (no existing code fits a GLM by hand), k-fold for an arbitrary fold list with both (15) and (16) (`mf.kfold_cv` implements only the site's layout and (16)), the confusion-matrix rates, and the asserts against the section notes' tables.

---

0 · Set-up: the comparison helper

`check(x, printed, dec)` passes when `x` rounds to the printed value at `dec` decimals, so every assert says *which* printed number it reproduces and to how many places.

```
"""Class 2 (L02) reference implementation with self-checks against the sources'
 ↪  printed numbers.

Run: python Notas/L02/L02_python_checks.py   (numpy, scipy, statsmodels; no
 ↪  scikit-learn)
Every section asserts the numbers of one section note and prints a one-line summary;
 ↪  the first
```

```
mismatch aborts with an AssertionError naming it. Shown section by section in
 ↪ L02-09-python.md.

Reused, not rewritten: figures/make_figures.py (imported as mf) supplies the tanh
 ↪ model psi, the exact
polynomial bias-variance routine, the mcycle stand-in, the OLS helpers, the two LOOCV
 ↪ routines and the
Gaussian discriminant roots h_roots. Importing it does not draw any figure (build()
 ↪ runs only as __main__).
"""
import pathlib
import sys

import numpy as np
import statsmodels.api as sm
from scipy import integrate, stats
from scipy.special import expit, logit

sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent / "figures"))
import make_figures as mf  # noqa: E402


def near(x, printed, dec):
    """True when x rounds to the printed value at `dec` decimals (half a unit in the
     ↪ last place)."""
    return abs(x - printed) <= 0.5 * 10.0 ** -dec + 1e-9


def check(x, printed, dec, what):
    assert near(x, printed, dec), f"{what}: computed {x!r}, source prints {printed}"
```

---

1 · Exact bias–variance on the tanh world → L02-01-prediction-error-bias-variance

The table of note 01 §5 row by row (avg bias², variance, test, training), the optimum $d^* = 5$ with test error 19.81, an independent hat-matrix route for $d = 5$, the optimism (33) as an identity, and the line-on-a-quadratic numbers of §4.1 by quadrature and Monte Carlo.

```
# ============================================================================ 1 ·
 ↪ bias-variance (L02-01)
def section1():
    psi = mf.psi
    check(psi(8.0), 12.339, 3, "psi(8)")
    check(psi(18.0), 101.365, 3, "psi(18)")
    assert abs(psi(13.3) - 57) < 1e-12                        # inflection, (26)
    check(45 / 1.9, 23.68, 2, "slope at the inflection")

    # exact (21) for the degree-d OLS polynomial on 30 equispaced points, sigma = 4
     ↪ (note 01 section 5 table)
    rows = {d: mf.poly_bias_variance(d) for d in range(16)}
```

```
    test = {d: b2 + v + 16 for d, (b2, v, _) in rows.items()}
    dstar = min(test, key=test.get)
    assert dstar == 5, dstar
    table = {1: (74.80, 1.03, 91.83, 93.87), 3: (7.98, 1.97, 25.95, 22.39), 5: (0.94,
↪   2.87, 19.81, 13.79),
             6: (0.70, 3.33, 20.03, 13.01), 7: (0.11, 3.80, 19.91, 11.85), 10:
              ↪  (0.007, 5.37, 21.38, 10.14),
             15: (0.000, 10.20, 26.20, 7.47)}
    for d, (b2, v, te, tr) in table.items():
        cb2, cv, ctr = rows[d]
        check(cb2, b2, 3 if d == 10 or d == 15 else 2, f"bias2 d={d}")
        check(cv, v, 2, f"variance d={d}")
        check(test[d], te, 2, f"test error d={d}")
        check(ctr, tr, 2, f"training MSE d={d}")
    assert max(test[5], test[6], test[7]) - test[5] < 0.25    # "within 0.25 of each
     ↪  other"

    # independent route for d = 5: raw-power design, hat matrix by pinv, direct
     ↪  moments of (28)
    x = np.linspace(8, 18, 30)
    g = np.linspace(8, 18, 400)
    B, B0 = np.vander((x - 13) / 5, 6), np.vander((g - 13) / 5, 6)
    P = B0 @ np.linalg.pinv(B)                                  # row x0: b(x0)'(B'B)^-1
↪   B'
    bias2 = np.mean((P @ psi(x) - psi(g)) ** 2)
    var = 16 * np.mean(np.sum(P ** 2, axis=1))
    assert abs(bias2 + var + 16 - test[5]) < 1e-8

    # optimism (33): E[in-sample test] - E[train] = 2 sigma^2 (d+1)/n, exactly, via
     ↪  trace(I - H) = n - d - 1
    H = B @ np.linalg.pinv(B)
    assert abs(np.trace(np.eye(30) - H) - 24) < 1e-9
    r = (np.eye(30) - H) @ psi(x)
    e_train = r @ r / 30 + 16 * (1 - 6 / 30)                    # (32)
    e_insample = r @ r / 30 + 16 * 6 / 30 + 16                  # bias2 + variance +
↪   noise at the design points
    check(e_insample - e_train, 6.4, 10, "optimism 2 sigma^2 (d+1)/n at d = 5")
    check(e_train, 13.79, 2, "(32) equals the table's training column at d = 5")

    # section 4.1: straight line on psi(x) = 1 - x + 2x^2, x ~ U(0,1), sigma = 0.1
     ↪  (bias-variance-R.txt)
    c = np.linalg.solve([[1, 1 / 2], [1 / 2, 1 / 3]], [1 / 3, 1 / 4])   # projection
↪   of x^2 on {1, x}
    assert np.allclose(c, [-1 / 6, 1])
    ib2 = 4 * integrate.quad(lambda t: (t * t - t + 1 / 6) ** 2, 0, 1)[0]
    assert abs(ib2 - 1 / 45) < 1e-12 and near(ib2, 0.0222, 4)
    rng = np.random.default_rng(2)
    xs = rng.uniform(0, 1, 900)
    Xl = np.column_stack([np.ones(900), xs])
    h = np.sum(Xl * np.linalg.solve(Xl.T @ Xl, Xl.T).T, axis=1)
```

```
    assert abs(h.mean() - 2 / 900) < 1e-12                    # average leverage =
     ↪ (p+1)/n, so (24) is exact at the design
    check(0.1 ** 2 * h.mean(), 0.0000222, 7, "(24) average variance")
    # Monte Carlo, design held fixed (the reading of (24)): redraw only the noise, R
     ↪ times
    R, grid = 4000, np.linspace(0, 1, 201)
    G = np.column_stack([np.ones_like(grid), grid])
    P = G @ np.linalg.solve(Xl.T @ Xl, Xl.T)                   # grid fit = P y
    Y = (1 - xs + 2 * xs ** 2) + rng.normal(0, 0.1, (R, 900))
    fits = Y @ P.T
    mc_b2 = np.mean((fits.mean(0) - (1 - grid + 2 * grid ** 2)) ** 2)
    mc_var = np.mean(fits.var(0))
    assert abs(mc_b2 / ib2 - 1) < 0.05 and abs(mc_var / 2.22e-5 - 1) < 0.1
    assert ib2 / mc_var > 500 and ib2 > 2 * 0.1 ** 2          # bias^2 ~ 1000 x
     ↪ variance, > 2 x noise
    # fresh x in every training set: the misspecification residual 2(x^2 - x + 1/6),
     ↪ variance 1/45,
    # now acts as extra noise, so the variance is ~ 2 (0.01 + 1/45) / 900 or more --
     ↪ still ~ 1/250 of bias^2
    X = rng.uniform(0, 1, (R, 900))
    Y = 1 - X + 2 * X ** 2 + rng.normal(0, 0.1, (R, 900))
    xb, yb = X.mean(1), Y.mean(1)
    slope = ((X - xb[:, None]) * (Y - yb[:, None])).sum(1) / ((X - xb[:, None]) **
↪   2).sum(1)
    fresh_var = np.mean(((yb - slope * xb)[:, None] + slope[:, None] * grid).var(0))
    assert 2 * (0.01 + 1 / 45) / 900 < fresh_var < ib2 / 200
    print(f"1 bias-variance: d* = {dstar}, test error {test[dstar]:.2f}; optimism
     ↪ 6.4; "
          f"line on quadratic: bias2 {ib2:.4f} (MC {mc_b2:.4f}), var 2.22e-5 (MC
           ↪ fixed design {mc_var:.2e}, "
          f"fresh x {fresh_var:.2e})")
```

One nuance the asserts expose. (24) — variance $\sigma^2 \cdot 2/n' = 2.22 \times 10^{-5}$ — is the variance **with the design held fixed** (the Monte Carlo that redraws only the noise gives 2.24×10^{-5}). If every training set also draws fresh x's, the misspecification residual $2(x^2 - x + \frac{1}{6})$, of variance $1/45$, behaves like extra noise and the variance is about 8.6×10^{-5} — four times larger, still some 260 times below the squared bias. Note 01's "1000 times" is the fixed-design figure.

---

2 · Log-level readings, exact vs approximate → L02-02-linear-log-linear-regression

Wooldridge's coefficients through (11) and (12), the log-log case (15)–(16), every printout check of note 02 (Credit, houseprice, Mroz, the quadratics), and a simulation showing that the ratio of medians in a log-level model is $e^{\beta \Delta x}$, not $1 + \beta \Delta x$.

```
# ============================================================================ 2 ·
 ↪ log-level readings (L02-02)
def exact_pct(z):
    """(11): exact % change in y when log y changes by z = beta * dx."""
```

```
    return 100 * np.expm1(z)


def section2():
    for z, approx, exact in [(0.083, 8.3, 8.65), (0.306, 30.6, 35.8), (-0.306, -30.6,
     ↪ -26.4),
                             (-0.052, -5.2, -5.1), (-0.260, -26.0, -22.9)]:
        check(100 * z, approx, 1, f"(12) at {z}")
        check(exact_pct(z), exact, 2 if z == 0.083 else 1, f"(11) at {z}")
    assert -exact_pct(-0.306) < 30.6 < exact_pct(0.306)                  # the arc
     ↪ reading sits between
    assert abs(exact_pct(0.3) - 30 - 100 * 0.3 ** 2 / 2) < 0.5            # error ~
     ↪ 100 z^2 / 2: ~5 points at 0.3
    check(100 * (1.10 ** 0.257 - 1), 2.48, 2, "(16) log-log exact at +10%")
    check(0.257 * 10, 2.57, 2, "(14) log-log approx at +10%")
    check(18.501 * 100, 1850.1, 1, "units: roe -> roedec")
    check(429.6 / 16.61, 25.86, 2, "t Student")
    check(23.18 / 3.639, 6.37, 2, "t Cards")
    check(1 - (1 - 0.9536) * 399 / 395, 0.9531, 4, "(4) adjusted R2, Credit")
    check(0.092903 * 1.886008, 0.17522, 5, "(5) per sqft")
    check(0.092903 * 1.886008 * 1000 / 3.2, 54.75, 2, "dollars per sqft")
    b, n2, br = 1.9374668, -0.3614412, 0.2907379
    for val, want in [(b + br, 2.2282), (b + n2, 1.5760), (b + n2 + br, 1.8668)]:
        check(val, want, 4, "interaction table")
    rss = 735 * 0.4511 ** 2
    check(rss, 149.57, 2, "(19) RSS")
    check(1 - rss / 752, 0.8011, 4, "(19) R2")
    check(1 - (1 - 0.8011) * 753 / 735, 0.7962, 4, "(20) adjusted R2 no intercept")
    check(-0.711872 / 0.032891, -21.64, 2, "t MTR")
    check(np.sqrt(735 / 753), 0.98798, 5, "MLE/OLS SE ratio")
    check(0.029839 * np.sqrt(735 / 753), 0.029480, 6, "LFP MLE SE")
    check(0.298 / (2 * 0.0061), 24.4, 1, "(22) experience turning point")
    check(0.545 / 0.124, 4.4, 1, "(22) rooms turning point")
    check(-54.5 + 12.4 * 5, 7.5, 1, "(23) 5 -> 6 rooms")
    check(-54.5 + 12.4 * 6, 19.9, 1, "(23) 6 -> 7 rooms")
    check(np.log(9) - np.log(8), 0.118, 3, "log points 8 -> 9")
    check(100 * (9 - 8) / 8, 12.5, 1, "percent 8 -> 9")
    # simulation: in a log-level model the fitted ratio y1/y0 is exp(beta dx), not 1
     ↪ + beta dx
    rng = np.random.default_rng(6)
    x = rng.uniform(0, 10, 400_000)
    y = np.exp(1 + 0.306 * x + rng.normal(0, 0.2, x.size))
    bhat = sm.OLS(np.log(y), sm.add_constant(x)).fit().params[1]
    med = [np.median(y[(x > a - 0.05) & (x < a + 0.05)]) for a in (4.0, 5.0)]   #
 ↪ medians of y at x = 4 and 5
    gap = 100 * (med[1] / med[0] - 1)
    assert abs(gap - exact_pct(bhat)) < 1.5 and abs(gap - 100 * bhat) > 3
    print(f"2 log readings: rooms +1 exact {exact_pct(0.306):.1f}% vs 30.6%; -1 exact
     ↪ {exact_pct(-0.306):.1f}%; "
          f"sales +10% exact {100 * (1.1 ** 0.257 - 1):.2f}% vs 2.57%")
```

3 · Validation set, LOOCV, k-fold and the bootstrap band → L02-03-cross-validation-motorcycle

The `mcycle` data are not in the repo, so the data-driven checks run on `make_figures`' **stand-in** (simulated; its numbers are not the site's). The asserts cover: the site's printed counts; LOOCV by brute force (6) = shortcut (7); the Sherman–Morrison step (12)–(13) on one point; $k = n$ folds = LOOCV; (15) ≠ (16) with unequal folds; the whole stand-in table of note 03 §8 with the same seeds and draw order as the figure; and the bootstrap SE (17) against the formula band (18) — equal under homoskedasticity, wider on the heteroskedastic stand-in.

```
# ============================================================================== 3 · validation, CV, bootstrap (L02-03)
def holdout_mse(x, y, d, train_idx, test_idx):
    """(5): validation-set estimate, fit on train_idx only."""
    return np.mean((y[test_idx] - mf.ols_predict(x[train_idx], y[train_idx], x[test_idx], d)) ** 2)


def kfold(x, y, d, folds, weighted=True):
    """k-fold CV for any list of index arrays `folds`: (16) if weighted (SSE / n), else (15) (mean of fold MSEs)."""
    mse, size = [], []
    for te in folds:
        tr = np.setdiff1d(np.arange(len(x)), te)
        mse.append(holdout_mse(x, y, d, tr, te))
        size.append(len(te))
    mse, size = np.array(mse), np.array(size)
    return float(size @ mse / size.sum()) if weighted else float(mse.mean())


def site_folds(perm, k=10):
    """The site's layout: L = trunc(n/k) shuffled positions per fold, the last fold takes the remainder."""
    n, L = len(perm), len(perm) // k
    return [perm[j * L:(j + 1) * L] if j < k - 1 else perm[j * L:] for j in range(k)]


def section3():
    wins = {6: 1, 8: 2, 9: 2, 10: 15, 11: 7, 12: 63, 13: 10}           # slide 7, 100 hold-out splits
    assert sum(wins.values()) == 100 and wins[10] + wins[11] + wins[12] + wins[13] == 95
    check(1 - 0.7798, 0.2202, 4, "(4) sigma2 at d = 12")
    check(np.sqrt(1 - 0.7798), 0.4693, 4, "(4) sigma at d = 12")
    check(13 / 132, 0.0985, 4, "mean leverage d = 12")
    check(4 / 132, 0.0303, 4, "mean leverage d = 3")
    assert [len(f) for f in site_folds(np.arange(132))] == [13] * 9 + [15]
    rng = np.random.default_rng(0)
```

```
    z = rng.normal(size=20)
    s = lambda v: (v - v.mean()) / v.std(ddof=1)
    assert np.allclose(s(3.7 * z - 2), s(z))                              # (2): affine
     ↪  maps are undone

    x, y = mf.mcycle_standin()
    ds = np.arange(1, 15)
    # LOOCV: brute force (6) vs the one-fit shortcut (7), both from make_figures
    brute = np.array([mf.loocv_brute(x, y, d) for d in ds])
    short = np.array([mf.loocv_shortcut(x, y, d)[0] for d in ds])
    assert np.allclose(brute, short, rtol=1e-8)
    # Sherman-Morrison step (12) on one point, with an orthonormal basis to keep it
     ↪  well conditioned
    Q, _ = np.linalg.qr(mf.design(x, 5))
    i = 7
    bfull = Q.T @ y
    e = y - Q @ bfull
    hii = Q[i] @ Q[i]
    keep = np.arange(132) != i
    b_loo = np.linalg.lstsq(Q[keep], y[keep], rcond=None)[0]
    assert np.allclose(b_loo, bfull - Q[i] * e[i] / (1 - hii))            # (12) with A
     ↪  = Q'Q = I
    assert abs((y[i] - Q[i] @ b_loo) - e[i] / (1 - hii)) < 1e-12          # (13)
    # (15) vs (16) differ when fold sizes differ; (16) reproduces make_figures'
     ↪  site-layout k-fold
    perm = np.random.default_rng(1).permutation(132)
    f = site_folds(perm)
    assert abs(kfold(x, y, 12, f) - mf.kfold_cv(x, y, 12, perm)) < 1e-12
    assert kfold(x, y, 12, f) != kfold(x, y, 12, f, weighted=False)
    assert abs(kfold(x, y, 3, [np.array([j]) for j in range(132)]) - brute[2]) <
     ↪  1e-12   # k = n is LOOCV

    # the stand-in table of note 03 section 8, same seeds and order of draws as
     ↪  make_figures.fig_cv
    train = np.array([np.mean((y - mf.ols_predict(x, y, x, d)) ** 2) for d in ds])
    rng = np.random.default_rng(56789)
    hold = np.array([[holdout_mse(x, y, d, p[:105], p[105:]) for d in ds]
                     for p in (rng.permutation(132) for _ in range(100))])
    kf = np.array([[kfold(x, y, d, site_folds(p)) for d in ds] for p in
↪   (rng.permutation(132) for _ in range(100))])
    want = {1: (0.9252, 0.9550, 0.9461, 0.9481), 3: (0.7376, 0.7954, 0.7781, 0.7810),
            5: (0.5388, 0.6192, 0.5900, 0.5917), 7: (0.3219, 0.3746, 0.3567, 0.3577),
            10: (0.1847, 0.2262, 0.2174, 0.2197), 12: (0.1737, 0.2220, 0.2082,
             ↪  0.2117),
            13: (0.1736, 0.2357, 0.2109, 0.2164), 14: (0.1722, 0.2501, 0.2107,
             ↪  0.2179)}
    for d, row in want.items():
        for got, w, lab in zip((train[d - 1], hold.mean(0)[d - 1], short[d - 1],
         ↪  kf.mean(0)[d - 1]), row,
                               ("train", "hold-out", "LOOCV", "10-fold")):
```

```
        check(got, w, 4, f"stand-in {lab} d={d}")
for est in (hold.mean(0), short, kf.mean(0)):
    assert ds[np.argmin(est)] == 12
assert np.all(train < short) and np.all(short < kf.mean(0)) and
 ↪ np.all(kf.mean(0)[4:] < hold.mean(0)[4:])
pick_h = np.bincount(ds[hold.argmin(1)], minlength=15)
pick_k = np.bincount(ds[kf.argmin(1)], minlength=15)
assert (pick_h[14], pick_h[12], pick_h[10], pick_h[13], 100 - pick_h10, 12, 13,
 ↪ 14.sum()) == (31, 30, 23, 10, 6)
assert pick_k[12] == 82

# pairs bootstrap (17) vs the formula band (18): equal on homoskedastic data,
 ↪ bootstrap wider here
def bands(xx, yy, d, B, seed):
    g = np.linspace(xx.min(), xx.max(), 200)
    rb = np.random.default_rng(seed)
    boot = np.array([mf.ols_predict(xx[k], yy[k], g, d) for k in rb.integers(0,
↪ len(xx), (B, len(xx)))])
    X = mf.design(xx, d)
    beta, rss = np.linalg.lstsq(X, yy, rcond=None)[:2]
    G = mf.design(g, d)
    Qx, Rx = np.linalg.qr(X)
    se = np.sqrt(rss[0] / (len(xx) - d - 1)) *
↪ np.linalg.norm(np.linalg.solve(Rx.T, G.T), axis=0)
    return g, boot.std(0, ddof=1), se, np.percentile(boot, [2.5, 97.5], axis=0)
xh = np.random.default_rng(4).uniform(-1, 1, 200)
yh = np.sin(2 * xh) + np.random.default_rng(5).normal(0, 0.3, 200)
_, seb, sef, _ = bands(xh, yh, 3, 2000, 7)
assert abs(np.mean(seb / sef) - 1) < 0.1                              #
 ↪ homoskedastic: the two agree
g, seb, sef, (lo, hi) = bands(x, y, 12, 1000, 31415)
noisy = g > -0.5
assert np.mean(hi[noisy] - lo[noisy]) > np.mean(4 * sef[noisy])      #
 ↪ heteroskedastic stand-in: bootstrap wider
print(f"3 CV: stand-in argmin 12 for hold-out/LOOCV/10-fold; 10-fold picks 12
 ↪ {pick_k[12]}/100, "
      f"hold-out {pick_h[12]}/100; LOOCV shortcut = brute force; bootstrap
       ↪ SE/formula SE "
      f"(homoskedastic) = 1 +- 0.1")
```

---

4 · Logistic regression by IRLS, against statsmodels → L02-04-logistic-regression

`irls` is note 04 (16)–(17) as code, with frequency weights so that the four cells of the student table stand for 10,000 rows. It reproduces the iteration table of note 04 §4.1 to five decimals, the closed form, the slide's SEs and deviance, and `statsmodels`' GLM. The `default` and balance + student numbers come from the **printed coefficients** (the data are not on disk); a simulated default-like sample checks `irls` against `sm.Logit` on a continuous covariate; perfectly separated data make it fail, as they must.

```
# ============================================================================ 4 · logistic regression (L02-04)
def irls(X, y, link, f=None, beta0=None, tol=1e-10, maxit=100):
    """Newton-Raphson as iteratively reweighted least squares (note 04 (16)-(17); note 06 section 3).

    X: n x (p+1) design with a leading column of ones; y: responses (0/1 for "logit", counts for "log");
    link: "logit" (Bernoulli) or "log" (Poisson), both canonical; f: frequency weights (rows repeated
    f_i times; None = 1); beta0: start (default 0). Returns (beta, standard errors from (X'WX)^-1,
    history of (beta, -2 loglik up to a constant) after each iteration). Raises RuntimeError when it
    does not converge, e.g. under perfect separation (no finite logit MLE).
    """
    f = np.ones(len(y)) if f is None else np.asarray(f, float)
    mean = expit if link == "logit" else np.exp
    vfun = (lambda m: m * (1 - m)) if link == "logit" else (lambda m: m)
    m2ll = ((lambda m: -2 * np.sum(f * (y * np.log(m) + (1 - y) * np.log1p(-m)))) if link == "logit"
            else (lambda m: -2 * np.sum(f * (y * np.log(m) - m))))
    beta = np.zeros(X.shape[1]) if beta0 is None else np.asarray(beta0, float)
    hist = []
    for _ in range(maxit):
        eta = X @ beta
        mu = mean(eta)
        w = f * vfun(mu)
        z = eta + (y - mu) / vfun(mu)                                   # adjusted response (17)
        new = np.linalg.solve(X.T @ (w[:, None] * X), X.T @ (w * z))    # weighted LS (16)
        hist.append((new, m2ll(mean(X @ new))))
        done = np.max(np.abs(new - beta)) < tol
        beta = new
        if done:
            break
    else:
        raise RuntimeError("IRLS did not converge")
    mu = mean(X @ beta)
    cov = np.linalg.inv(X.T @ ((f * vfun(mu))[:, None] * X))
    return beta, np.sqrt(np.diag(cov)), hist


def confusion_metrics(tn, fp, fn, tp):
    """Rates of note 04 section 9, 'positive' = spam; 'slide_fd' is the deck's FP / (FP + all actual positives)."""
    return dict(error=(fp + fn) / (tn + fp + fn + tp), fnr=fn / (fn + tp), fpr=fp / (fp + tn),
```

```
                sens=tp / (tp + fn), spec=tn / (tn + fp), prec=tp / (tp + fp), fdr=fp
↪ / (fp + tp),
                slide_fd=fp / (fp + tp + fn))


def section4():
    # the student-only model from the 2x2 table of slide 18, as four weighted rows
    s = np.array([0, 0, 1, 1.0])
    yv = np.array([0, 1, 0, 1.0])
    cnt = np.array([6850, 206, 2817, 127])
    X = sm.add_constant(s)
    beta, se, hist = irls(X, yv, "logit", f=cnt)
    table = [(-1.88322, 0.05577, 4116.20), (-2.78054, 0.15372, 3072.81), (-3.31150,
↪ 0.29836, 2917.50),
             (-3.48780, 0.39184, 2908.74), (-3.50400, 0.40477, 2908.683), (-3.50413,
↪ 0.40489, 2908.683)]
    for k, (b0, b1, dev) in enumerate(table):
        (c0, c1), cdev = hist[k]
        check(c0, b0, 5, f"IRLS iteration {k + 1} beta0")
        check(c1, b1, 5, f"IRLS iteration {k + 1} beta1")
        check(cdev, dev, 3 if k >= 4 else 2, f"IRLS iteration {k + 1} -2loglik")
    check(beta[0], np.log(206 / 6850), 9, "closed form beta0")
    check(beta[1], np.log((127 / 2817) / (206 / 6850)), 9, "closed form beta1")
    check(beta[0], -3.50413, 5, "slide 18 beta0")
    check(beta[1], 0.40489, 5, "slide 18 beta1")
    check(se[0], 0.07071, 5, "slide 18 SE beta0")
    check(se[1], 0.11502, 5, "slide 18 SE beta1")
    assert abs(se[1] - np.sqrt(1 / 206 + 1 / 6850 + 1 / 127 + 1 / 2817)) < 1e-9
    glm = sm.GLM(yv, X, family=sm.families.Binomial(), freq_weights=cnt).fit()
    assert np.allclose(glm.params, beta, atol=1e-8) and np.allclose(glm.bse, se,
↪ atol=1e-8)
    check(glm.deviance, 2908.7, 1, "slide 18 residual deviance")
    check(glm.null_deviance, 2920.6, 1, "null deviance")
    check(-2 * (333 * np.log(0.0333) + 9667 * np.log(0.9667)), 2920.6, 1, "null
↪ deviance by hand")
    check(100 * expit(beta.sum()), 4.31, 2, "P(default | student)")       # the deck
↪ prints 0.43%
    check(100 * expit(beta[0]), 2.92, 2, "P(default | non-student)")     # the deck
↪ prints 0.29%
    check(np.exp(beta[1]), 1.499, 3, "odds ratio student")
    assert abs(np.sum(cnt * (yv - expit(X @ beta)))) < 1e-6               # score row
↪ 1: mean fitted = share of 1s

    # the balance-only model from its printed coefficients (slides 13, 15, 17)
    b0, b1 = -10.6513, 0.0055
    check(100 * expit(b0 + b1 * 1000), 0.58, 2, "P at 1000")
    check(100 * expit(b0 + b1 * 2000), 58.6, 1, "P at 2000")
    check(-b0 / b1, 1936.6, 1, "(19) 50% cut-off")
    for c, x1, x2 in [(0.1, 1537.1, 1537.4), (0.2, 1684.5, 1684.9), (0.3, 1782.5,
↪ 1782.9), (0.4, 1862.9, 1863.2),
```

```
                    (0.6, 2010.3, 2010.7), (0.7, 2090.7, 2091.0), (0.8, 2188.7,
↪ 2189.1), (0.9, 2336.1, 2336.5)]:
        check((logit(c) - b0) / 0.0055, x1, 1, f"cut-off c={c}, slope 0.0055")
        check((logit(c) - b0) / 0.005499, x2, 1, f"cut-off c={c}, slope 0.005499")
    check(0.07519 / 0.0001299, 578.8, 1, "LPM zero crossing")
    check(-0.07519 + 0.0001299 * 2000, 0.185, 3, "LPM at 2000")
    check(np.exp(0.0055 * 100), 1.733, 3, "odds ratio per USD 100")
    pe_cut, pe_1000 = 0.0055 * 0.25, 0.0055 * expit(b0 + b1 * 1000) * (1 - expit(b0
↪ + b1 * 1000))
    check(pe_cut, 0.001375, 6, "(8) at the cut-off")
    check(pe_1000, 0.0000315, 7, "(8) at 1000")
    assert round(pe_cut / pe_1000) == 44
    check(-10.65 / 0.3612, -29.49, 2, "z intercept")
    check(0.005499 / 0.0002204, 24.95, 2, "z balance")
    for dev, k, aic in [(1596.5, 2, 1600.5), (2908.7, 2, 2912.7), (1571.7, 3,
     ↪ 1577.7)]:
        check(dev + 2 * k, aic, 1, "AIC = deviance + 2k")
    for lr, want in [(2920.6 - 1596.5, 1324.1), (2920.6 - 2908.7, 11.9), (1596.5 -
     ↪ 1571.7, 24.8)]:
        check(lr, want, 1, "LR test")
        assert lr > stats.chi2.ppf(0.95, 1)
    for dev, want in [(1596.5, 0.453), (1571.7, 0.462), (2908.7, 0.004)]:
        check(1 - dev / 2920.6, want, 3, "McFadden pseudo-R2")
    # balance + student (slide 20) and the group cut-offs (slide 22 swaps them)
    c0, c1, c2 = -10.75, 0.005738, -0.7149
    check(100 * expit(c0 + c1 * 1500 + c2), 5.4, 1, "student at 1500")
    check(100 * expit(c0 + c1 * 1500), 10.5, 1, "non-student at 1500")
    check(-(c0 + c2) / c1, 1998.1, 1, "50% cut-off, student")
    check(-c0 / c1, 1873.5, 1, "50% cut-off, non-student")
    check(np.exp(c2), 0.489, 3, "odds ratio student | balance")

    # IRLS on simulated default-like data vs statsmodels' Logit, and recovery of the
     ↪ true coefficients
    rng = np.random.default_rng(10)
    bal = rng.gamma(4.0, 210.0, 10_000)
    yy = rng.binomial(1, expit(b0 + b1 * bal)).astype(float)
    Xs = sm.add_constant(bal)
    bh, seh, _ = irls(Xs, yy, "logit")
    ref = sm.Logit(yy, Xs).fit(disp=0)
    assert np.allclose(bh, ref.params, rtol=1e-8) and np.allclose(seh, ref.bse,
     ↪ rtol=1e-6)
    assert np.all(np.abs(bh - [b0, b1]) < 3.5 * seh)
    try:                                                          # perfect
     ↪ separation: no finite MLE
        with np.errstate(all="ignore"):
            irls(sm.add_constant(np.arange(10.0)), (np.arange(10) > 4).astype(float),
↪ "logit", maxit=30)
        raise AssertionError("separated data should not converge")
    except (RuntimeError, np.linalg.LinAlgError):
        pass
```

```
    # the spam confusion matrices (slides 36-38)
    spam = {"glm": (1325, 80, 93, 803, 7.52, 10.38, 5.69, 9.06, 8.20),
            "ridge": (1345, 60, 128, 768, 8.17, 14.29, 4.27, 7.25, 6.28),
            "enet": (1349, 56, 94, 802, 6.52, 10.49, 3.99, 6.53, 5.88),
            "lasso": (1347, 58, 95, 801, 6.65, 10.60, 4.13, 6.75, 6.08),
            "bayes": (1326, 79, 91, 805, 7.39, 10.16, 5.62, 8.94, 8.10)}
    for name, (tn, fp, fn, tp, err, fnr, fpr, fdr, sfd) in spam.items():
        m = confusion_metrics(tn, fp, fn, tp)
        assert tn + fp + fn + tp == 2301 and fn + tp == 896
        for key, want in zip(("error", "fnr", "fpr", "fdr", "slide_fd"), (err, fnr,
         ↪ fpr, fdr, sfd)):
            check(100 * m[key], want, 2, f"spam {name} {key}")
    m = confusion_metrics(*spam["glm"][:4])
    check(100 * m["sens"], 89.62, 2, "sensitivity")
    check(100 * m["spec"], 94.31, 2, "specificity")
    check(100 * m["prec"], 90.94, 2, "precision")
    assert min(spam, key=lambda k: spam[k][4]) == "enet" and max(spam, key=lambda k:
     ↪ spam[k][5]) == "ridge"
    print(f"4 logistic: IRLS student model {beta.round(5)} (SE {se.round(5)}) =
     ↪ statsmodels; "
          f"P 0.58%/58.6%, cut-off {-b0 / b1:.1f}; spam FDR 9.06% (slide 8.20%)")
```

The student model is the one `default` fit that can be reproduced exactly, because slide 18 prints the whole 2×2 table: `irls` returns $(-3.50413,\ 0.40489)$ with SEs $(0.07071,\ 0.11502)$, and the fitted probabilities are 2.92% and 4.31% — not the slide's 0.29% and 0.43%.

---

5 · Discriminant analysis: Gaussian cut-offs, costs, ROC, LDA/QDA on iris → L02-05-discriminant-analysis

The Bernoulli example and its prior thresholds; the quadratic h of (7) with roots 1.680 and 10.001 (and `mf.h_roots` agreeing); the misallocation probabilities 0.3881 / 0.1148 / 0.3694; a scan showing that **no single cut-off beats the two-cut-off Bayes rule** (the content of (3)); the cost rule; ISLP's `default` tables; both AUCs, the unequal one by direct integration (independent of the figure's trapezoid); the LDA posterior as a logistic function; and iris LDA from the slide's summaries, with QDA computed on the same summaries.

```
# ============================================================================ 5 ·
 ↪ discriminant analysis (L02-05)
def h_coef(m1, m2, s1, s2):
    """(7) as a x^2 + b x + c."""
    a = -0.5 * (1 / s1 ** 2 - 1 / s2 ** 2)
    b = m1 / s1 ** 2 - m2 / s2 ** 2
    c = -0.5 * (m1 ** 2 / s1 ** 2 - m2 ** 2 / s2 ** 2) + np.log(s2 / s1)
    return a, b, c


def misallocation(lo, hi, m1=1.0, m2=5.0, s1=6.0, s2=2.5):
```

```
    """p21, p12 when R2 = (lo, hi) and R1 is its complement (section 6)."""
    p21 = stats.norm.cdf(hi, m1, s1) - stats.norm.cdf(lo, m1, s1)
    p12 = 1 - (stats.norm.cdf(hi, m2, s2) - stats.norm.cdf(lo, m2, s2))
    return p21, p12


def section5():
    # Bernoulli example (slide 43) and the priors that move it
    p1, p2 = {0: 0.5, 1: 0.5}, {0: 0.25, 1: 0.75}
    check(np.log(p1[0] / p2[0]), 0.693, 3, "h(0)")
    check(np.log(p1[1] / p2[1]), -0.405, 3, "h(1)")
    assert 0.5 * (0.5 + 0.25) == 3 / 8
    rules = [(1, 2), (2, 2), (1, 1), (2, 1)]                                    #
    ↪ allocation of x = 0, x = 1
    pkk = {r: (sum(p1[x] for x in (0, 1) if r[x] == 1), sum(p2[x] for x in (0, 1) if
    ↪ r[x] == 2)) for r in rules}
    assert pkk[(1, 2)] == (0.5, 0.75) and pkk[(2, 1)] == (0.5, 0.25)      # (2,1)
     ↪ dominated by ML: inadmissible
    for pi1, want in [(0.30, (2, 2)), (0.34, (1, 2)), (0.59, (1, 2)), (0.61, (1,
     ↪ 1))]:
        bayes = tuple(1 if pi1 * p1[x] > (1 - pi1) * p2[x] else 2 for x in (0, 1))
        assert bayes == want, (pi1, bayes)
    assert np.log(0.5 / 0.25) > 0                                               #
     ↪ multinomial x = (1,0): class 1, slide 44 says "< 0"

    # univariate Gaussians, mu1 = 1, mu2 = 5, sigma1 = 6, sigma2 = 2.5 (slides 45-47)
    a, b, c = h_coef(1, 5, 6, 2.5)
    for v, want in [(a, 0.066111), (b, -0.772222), (c, 1.110642)]:
        check(v, want, 6, "h coefficients")
    roots = np.sort(np.roots([a, b, c]).real)
    assert np.allclose(roots, mf.h_roots(0.0))
    check(roots[0], 1.680, 3, "x_L")
    check(roots[1], 10.001, 3, "x_U")
    hx = lambda x: a * x * x + b * x + c
    assert abs(hx(3.0) - (stats.norm.logpdf(3, 1, 6) - stats.norm.logpdf(3, 5, 2.5)))
     ↪ < 1e-12
    check(hx(-20), 43.0, 1, "h(-20)")
    check(hx(20), 12.1, 1, "h(20)")
    check(-b / (2 * a), 5.840, 3, "argmin h")
    check(hx(-b / (2 * a)), -1.144, 3, "min h")
    p21, p12 = misallocation(*roots)
    check(p21, 0.3881, 4, "p21")
    check(p12, 0.1148, 4, "p12")
    check(0.5 * (p21 + p12), 0.2515, 4, "total error, unequal variances")
    pe = stats.norm.sf(3, 1, 6)
    check(pe, 0.3694, 4, "p21 = p12, equal variances")
    check(-(-20 - 3) / 9, 2.556, 3, "equal-variance h(-20)")
    # Bayes optimality (3): no single cut-off (a linear rule) beats the quadratic
     ↪ Bayes rule
    t = np.linspace(-10, 15, 5001)
```

```
    single = 0.5 * (stats.norm.sf(t, 1, 6) + stats.norm.cdf(t, 5, 2.5))   # allocate
↪   to 1 iff x < t
    assert single.min() > 0.5 * (p21 + p12)
    # unequal costs: c21/c12 = 2 -> threshold log(1/2) (14)
    rc = mf.h_roots(np.log(0.5))
    check(rc[0], 3.228, 3, "cost-rule x_L")
    check(rc[1], 8.453, 3, "cost-rule x_U")
    q21, q12 = misallocation(*rc)
    for v, want in [(q21, 0.2481), (q12, 0.3228), (0.5 * (q21 + q12), 0.2855),
                    (0.5 * (2 * p21) + 0.5 * p12, 0.4455), (0.5 * (2 * q21) + 0.5 *
↪   q12, 0.4095)]:
        check(v, want, 4, "cost table")
    check(np.exp(-hx(-b / (2 * a))), 3.14, 2, "ratio that empties R2")
    assert mf.h_roots(-np.log(3.2)) is None and mf.h_roots(-np.log(3.1)) is not None
    # posterior threshold (15) and I's default tables
    assert abs(1 / (1 + 4) - 0.2) < 1e-15                                    #
     ↪  c12/(c12+c21) = 0.2 <=> c21 = 4 c12
    for miss, fa, err, rate in [(252, 23, 2.75, 75.7), (138, 235, 3.73, 41.4)]:
        check(100 * (miss + fa) / 10_000, err, 2, "I default error rate")
        check(100 * miss / 333, rate, 1, "I missed share")
    assert 4 * 252 + 23 == 1031 and 4 * 138 + 235 == 787
    # AUCs (section 11): equal variance in closed form; unequal by direct
     ↪  integration
    check(stats.norm.cdf(4 / np.sqrt(72)), 0.681, 3, "AUC equal variances")

    def p_h1_above(k):                                                  # P(h(X1) >
     ↪  k), X1 ~ N(1, 36)
        r = mf.h_roots(k)
        return 1.0 if r is None else 1 - (stats.norm.cdf(r[1], 1, 6) -
         ↪  stats.norm.cdf(r[0], 1, 6))
    auc = integrate.quad(lambda u: p_h1_above(hx(u)) * stats.norm.pdf(u, 5, 2.5),
↪   -15, 25, limit=200)[0]
    check(auc, 0.802, 3, "AUC unequal variances")
    # LDA posterior is a logistic function of x (7.1), here p = 1, sigma = 6, pi1 =
     ↪  0.3
    xg = np.linspace(-10, 15, 7)
    post = 0.3 * stats.norm.pdf(xg, 1, 6) / (0.3 * stats.norm.pdf(xg, 1, 6) + 0.7 *
↪   stats.norm.pdf(xg, 5, 6))
    alpha = (1 - 5) / 36
    assert np.allclose(post, expit(np.log(0.3 / 0.7) - alpha * 3 + alpha * xg))

    # iris, the two sepal measurements (slide 54), LDA by Mahalanobis distance (10)
    mu = np.array([[5.006, 3.428], [5.936, 2.770], [6.588, 2.974]])
    S = np.array([[[0.122, 0.097], [0.097, 0.141]], [[0.261, 0.083], [0.083, 0.096]],
                  [[0.396, 0.092], [0.092, 0.102]]])
    Sp = S.mean(0)
    assert np.allclose(Sp, [[0.25967, 0.09067], [0.09067, 0.11300]], atol=5e-6)
    check(np.linalg.det(Sp), 0.021122, 6, "|Sigma pooled|")
    Si = np.linalg.inv(Sp)
    assert np.allclose(Si, [[5.3499, -4.2925], [-4.2925, 12.2937]], atol=5e-4)
```

```
    flowers = np.array([[5.1, 3.5], [7.0, 3.2], [6.3, 3.3]])
    D = np.array([[(f - m) @ Si @ (f - m) for m in mu] for f in flowers])
    want = [[0.053, 15.530, 21.966], [25.814, 4.402, 0.737], [10.581, 2.506, 2.556]]
    assert np.all(np.abs(D - want) <= 0.0005), D                          # every
     ↪ distance to 3 decimals
    assert list(D.argmin(1)) == [0, 2, 1]                                  # setosa,
     ↪ virginica (wrong), versicolor (wrong)
    al = Si @ (mu[0] - mu[1])
    assert np.allclose(al, [-7.800, 12.081], atol=2e-3)
    check(al @ (flowers[0] - mu[:2].mean(0)), 7.738, 2, "flower 1, setosa vs
↪   versicolor")
    # QDA (12) with class covariances; with every Sigma_k set to the pooled one it
     ↪ must reproduce LDA
    def qda_scores(xv, covs):
        return np.array([-0.5 * np.log(np.linalg.det(Sk)) - 0.5 * (xv - m) @
         ↪ np.linalg.solve(Sk, xv - m)
                         for m, Sk in zip(mu, covs)])
    assert all(qda_scores(f, [Sp] * 3).argmax() == D[i].argmin() for i, f in
     ↪ enumerate(flowers))
    qda = [int(qda_scores(f, S).argmax()) for f in flowers]
    for k, want in [(31 / 85, 0.3647), (26 / 85, 0.3059), (28 / 85, 0.3294)]:
        check(k, want, 4, "admission priors")
    names = ["setosa", "versicolor", "virginica"]
    print(f"5 discriminant: cut-offs {roots.round(3)}, errors {p21:.4f}/{p12:.4f}
     ↪ (total {0.5 * (p21 + p12):.4f}); "
          f"cost rule {rc.round(3)}; AUC {auc:.3f}; iris LDA -> {[names[k] for k in
           ↪ D.argmin(1)]}, "
          f"QDA (computed here, no source value) -> {[names[k] for k in qda]}")
```

QDA on the three printed iris flowers (class covariances from slide 54, equal priors): setosa, virginica, virginica — flower 101 moves from versicolor (LDA) to its true class. **The deck prints no QDA allocation for these flowers, so this is a computed value with no source to check against**; the assert that does bind is that QDA with every Σ_k set to the pooled matrix reproduces LDA.

---

6 · Poisson regression by IRLS, against statsmodels → L02-06-gaussian-vs-poisson-regression

The same `irls` with the log link; every number of note 06 — the closed form, the SEs from the group totals, rate ratio, deviances, Pearson dispersion and the quasi-Poisson correction (also through `statsmodels`' `scale="X2"`), and the Gaussian fit beside it.

```
# ============================================================================ 6 ·
 ↪ Gaussian vs Poisson (L02-06)
def section6():
    y = mf.Y.astype(float)
    w = (np.tile(np.arange(1, 8), 4) <= 3).astype(float)                  # fri, sat,
 ↪ sun = weekend on the page
    assert (w.sum(), y[w == 0].sum(), y[w == 1].sum()) == (12, 1744, 1000)
```

```
    X = sm.add_constant(w)
    beta, se, hist = irls(X, y, "log", beta0=[np.log(y.mean()), 0.0])
    ref = sm.GLM(y, X, family=sm.families.Poisson()).fit()
    assert np.allclose(beta, ref.params, atol=1e-9) and np.allclose(se, ref.bse,
     ↪  atol=1e-9)
    check(beta[0], 4.69135, 5, "alpha")
    check(beta[1], -0.26850, 5, "beta")
    assert abs(beta[0] - np.log(109)) < 1e-9 and abs(beta[1] - np.log((1000 / 12) /
     ↪  109)) < 1e-9
    check(se[0], 0.02395, 5, "SE alpha")
    check(se[1], 0.03967, 5, "SE beta")
    assert abs(se[0] - 1 / np.sqrt(1744)) < 1e-9 and abs(se[1] - np.sqrt(1 / 1744 + 1
     ↪  / 1000)) < 1e-9
    check(beta[1] / se[1], -6.77, 2, "z")
    check(np.exp(beta[1]), 0.7645, 4, "rate ratio")
    check(100 * np.expm1(beta[1]), -23.5, 1, "exact % change")
    check(100 * beta[1], -26.85, 2, "100 beta shortcut (note 06 section 4)")
    check(100 * beta[1] - 100 * np.expm1(beta[1]), -3.30, 2, "gap (note 06 says 3.4:
↪   difference of rounded values)")
    check(ref.deviance, 38.324, 3, "residual deviance")
    check(ref.null_deviance, 85.141, 3, "null deviance")
    check(ref.null_deviance - ref.deviance, 46.82, 2, "LR weekend")
    check(ref.aic, 221.76, 2, "AIC")
    check(stats.chi2.sf(ref.deviance, 26), 0.056, 3, "P(chi2_26 > deviance)")
    lam = ref.fittedvalues
    pearson = np.sum((y - lam) ** 2 / lam)
    check(pearson, 37.546, 3, "Pearson chi2")
    s2q = pearson / 26
    check(s2q, 1.444, 3, "dispersion")
    check(np.sqrt(s2q), 1.202, 3, "sigma hat")
    check(np.sqrt(s2q) * se[1], 0.0477, 4, "quasi-Poisson SE")
    check(beta[1] / (np.sqrt(s2q) * se[1]), -5.63, 2, "quasi-Poisson z")
    check((ref.null_deviance - ref.deviance) / s2q, 32.4, 1, "LR / dispersion")
    qp = sm.GLM(y, X, family=sm.families.Poisson()).fit(scale="X2")
    assert abs(qp.bse[1] - np.sqrt(s2q) * se[1]) < 1e-12
    # the Gaussian regression on the same dummy
    ols = sm.OLS(y, X).fit()
    check(ols.params[0], 109.000, 3, "beta0 OLS")
    check(ols.params[1], -25.667, 3, "beta1 OLS")
    s = np.sqrt(ols.scale)
    check(s, 11.85, 2, "residual SE")
    check(s ** 2, 140.4, 1, "s^2")
    check(ols.bse[1], 4.525, 3, "SE beta1")
    check(ols.bse[0], 2.962, 3, "SE beta0")
    check(ols.tvalues[1], -5.67, 2, "t")
    check(ols.rsquared, 0.553, 3, "R2")
    assert abs(ols.pvalues[1] - 5.8e-6) < 0.05e-6
    for grp, mean_, sd_, var_, ratio in [(0, 109.00, 12.15, 147.73, 1.36), (1, 83.33,
     ↪  11.42, 130.42, 1.57)]:
        v = y[w == grp]
```

```
        check(v.mean(), mean_, 2, "group mean")
        check(v.std(ddof=1), sd_, 2, "group sd")
        check(v.var(ddof=1), var_, 2, "group variance")
        check(v.var(ddof=1) / v.mean(), ratio, 2, "variance / mean")
    check(1.96 * np.sqrt(109), 20.5, 1, "Poisson band weekday")
    check(1.96 * np.sqrt(1000 / 12), 17.9, 1, "Poisson band weekend")
    outside = np.sum(np.abs(y - lam) > 1.96 * np.sqrt(lam))
    assert outside == 3 and abs(28 * 0.05 - 1.4) < 1e-12
    print(f"6 Poisson: IRLS {beta.round(5)} (SE {se.round(5)}) = statsmodels in
     ↪ {len(hist)} iterations; "
          f"rate ratio {np.exp(beta[1]):.4f}; dispersion {s2q:.3f}; quasi z {beta[1]
           ↪ / (np.sqrt(s2q) * se[1]):.2f}")
```

A rounding remark on note 06 §4. From the printed $\hat{\beta} = -0.26850$, the log reading is $100\hat{\beta} = -26.85\%$ and the exact one -23.55%: they differ by **3.30** points. Rounding the first to -26.9% before subtracting would give 3.4, the difference of two rounded figures.

---

7 · The counterexamples of the pitfalls note → L02-08-pitfalls

The numbers Part I and Part II use that are not already in a section note: the optimum moving from $d^* = 5$ to 7 as n grows, the pure-noise $E[R^2] = k/(n-1)$, the probability after an odds ratio, the trivial credit rule, FDR vs FPR, ML vs Bayes error, $h(12)$, and quasi-Poisson leaving $\hat{\beta}$ unchanged.

```
# ============================================================================== 7 ·
 ↪ counterexamples of L02-08-pitfalls
def section7():
    # training error ranks the degrees backwards (pitfall 1): d = 15 beats d = 5 in
     ↪ training, loses in test
    tr5, tr15 = mf.poly_bias_variance(5)[2], mf.poly_bias_variance(15)[2]
    te = lambda d, n=30: sum(mf.poly_bias_variance(d, n=n)[:2]) + 16
    assert tr15 < tr5 and te(15) > te(5) and tr15 < 16                  # d = 15 even
     ↪ trains below sigma^2
    # more data moves the optimum right (V/F 7): d* = 5 at n = 30, 7 at n = 60 and
     ↪ 200
    assert [int(np.argmin([te(d, n) for d in range(16)])) for n in (30, 60, 200)] ==
     ↪ [5, 7, 7]
    # selection optimism (pitfall 5): pure noise, k = 12 polynomial terms, n = 132 ->
     ↪ E[R2] = k / (n - 1)
    rng = np.random.default_rng(0)
    r2 = []
    for _ in range(4000):
        x, y = rng.normal(size=132), rng.normal(size=132)
        e = y - mf.ols_predict(x, y, x, 12)
        r2.append(1 - e @ e / np.sum((y - y.mean()) ** 2))
    check(12 / 131, 0.0916, 4, "E[R2] under pure noise")
    assert abs(np.mean(r2) - 12 / 131) < 0.003
    # odds ratio is not a probability ratio (pitfall 7): odds 1 -> 1.733 moves p from
     ↪ 0.5 to 0.634
```

```
    check(np.exp(0.55) / (1 + np.exp(0.55)), 0.634, 3, "p after an odds ratio of
↪   1.733 at p = 0.5")
    check((127 / 2944) / (206 / 7056), 1.478, 3, "probability ratio, student table")
    # accuracy under imbalance (pitfall 10): the all-"no" rule vs LDA at 0.2, I's
     ↪  default tables
    assert 333 / 10_000 < (138 + 235) / 10_000 and 4 * 333 > 4 * 138 + 235
    # FDR > FPR iff fewer flagged than actual negatives (pitfall 11)
    tn, fp, fn, tp = 1325, 80, 93, 803
    assert (fp / (fp + tp) > fp / (fp + tn)) == (fp + tp < fp + tn)
    # ML vs Bayes rule (pitfall 13): Bernoulli example with pi1 = 0.25
    ml_err = 0.25 * 0.5 + 0.75 * 0.25                                    # ML rule (1,
↪   2): p21 = 1/2, p12 = 1/4
    bayes_err = 0.25                                                     # pi1 < 1/3:
↪   everything to class 2
    assert (ml_err, bayes_err) == (0.3125, 0.25)
    # a far observation goes to the wide class (V/F 23): h(12) > 0 in the deck's
     ↪  example
    a, b, c = h_coef(1, 5, 6, 2.5)
    check(a * 144 + b * 12 + c, 1.364, 3, "h(12)")
    # overdispersion leaves the Poisson point estimates alone (pitfall 15)
    y = mf.Y.astype(float)
    X = sm.add_constant((np.tile(np.arange(1, 8), 4) <= 3).astype(float))
    p, q = sm.GLM(y, X, family=sm.families.Poisson()).fit(), sm.GLM(y, X,
↪   family=sm.families.Poisson()).fit(scale="X2")
    assert np.allclose(p.params, q.params) and np.all(q.bse > p.bse)
    print("7 pitfalls: training ranks d = 15 over d = 5; d* 5 -> 7 as n grows;
     ↪  pure-noise E[R2] 0.0916; "
          "OR 1.733 at p = 0.5 gives 0.634; ML 0.3125 vs Bayes 0.25 at pi1 = 0.25")


if __name__ == "__main__":
    section1()
    section2()
    section3()
    section4()
    section5()
    section6()
    section7()
    print("ALL LO2 CHECKS PASSED")
```

---

8 · The run

```
1 bias-variance: d* = 5, test error 19.81; optimism 6.4; line on quadratic: bias2
 ↪  0.0222 (MC 0.0227), var 2.22e-5 (MC fixed design 2.24e-05, fresh x 8.56e-05)
2 log readings: rooms +1 exact 35.8% vs 30.6%; -1 exact -26.4%; sales +10% exact
 ↪  2.48% vs 2.57%
3 CV: stand-in argmin 12 for hold-out/LOOCV/10-fold; 10-fold picks 12 82/100,
 ↪  hold-out 30/100; LOOCV shortcut = brute force; bootstrap SE/formula SE
 ↪  (homoskedastic) = 1 +- 0.1
```

```
4 logistic: IRLS student model [-3.50413  0.40489] (SE [0.07071 0.11502]) =
 ↪  statsmodels; P 0.58%/58.6%, cut-off 1936.6; spam FDR 9.06% (slide 8.20%)
5 discriminant: cut-offs [ 1.68  10.001], errors 0.3881/0.1148 (total 0.2515); cost
 ↪  rule [3.228 8.453]; AUC 0.802; iris LDA -> ['setosa', 'virginica', 'versicolor'],
 ↪  QDA (computed here, no source value) -> ['setosa', 'virginica', 'virginica']
6 Poisson: IRLS [ 4.69135 -0.2685 ] (SE [0.02395 0.03967]) = statsmodels in 5
 ↪  iterations; rate ratio 0.7645; dispersion 1.444; quasi z -5.63
7 pitfalls: training ranks d = 15 over d = 5; d* 5 -> 7 as n grows; pure-noise E[R2]
 ↪  0.0916; OR 1.733 at p = 0.5 gives 0.634; ML 0.3125 vs Bayes 0.25 at pi1 = 0.25
ALL L02 CHECKS PASSED
```

---

9 · What is deliberately absent

Not here	Why
scikit-learn (`cross_val_score`, `LogisticRegression`, `LinearDiscriminantAnalysis`)	not installed; every routine is a few lines of numpy, which also shows the formula
the `mcycle`, `default`, spam, iris and admission **data**	not on disk: the checks use printed tables, printed coefficients and, for CV, a labelled stand-in
the smoothing spline of Fundamentos	a black box on purpose (note 01 §7); polynomials stand in
ridge, lasso, elastic net; every Bayesian fit (logit, Poisson grid, SIR)	Classes 3+4
plotting	`figures/make_figures.py` draws the figures and asserts their plotted numbers

---

Links

Map: L02-00-map · Formulas: L02-07-formulas · Pitfalls: L02-08-pitfalls · Class 1 code: L01-11 · Figures: `figures/make_figures.py` · Companion checks: `companions/check_companions.py` · Rules: rules/06.